IN PRAISE OF UNCERTAINTY

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UNCERTAINTY RULES

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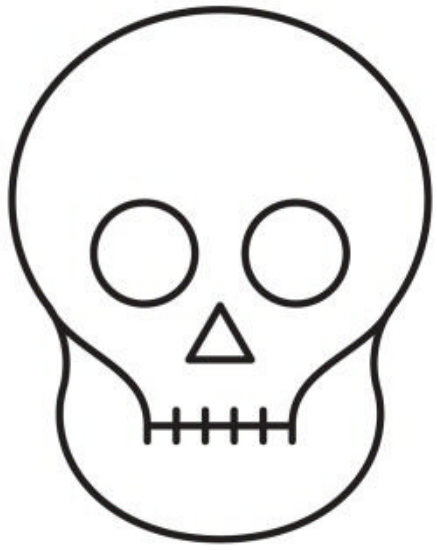
The correct use of beasts

PATRICK STOKES
Wounded stories

STEPHEN LAW
Cast-iron certainty



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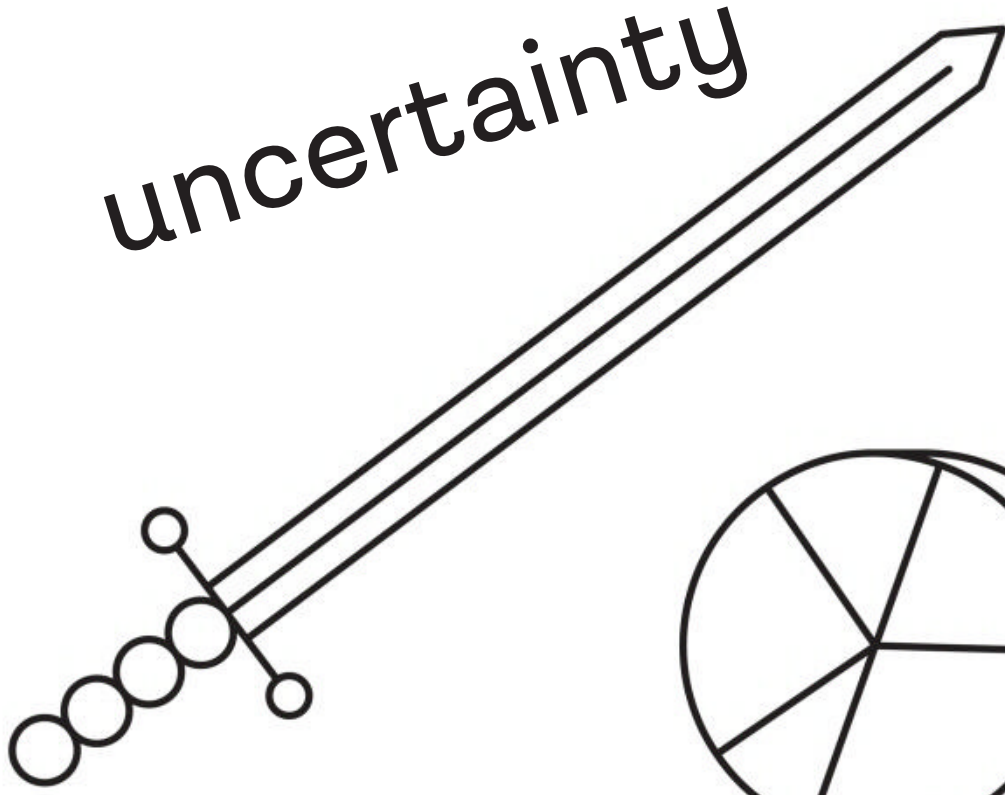

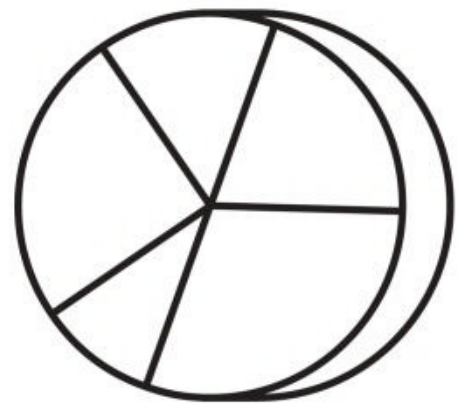

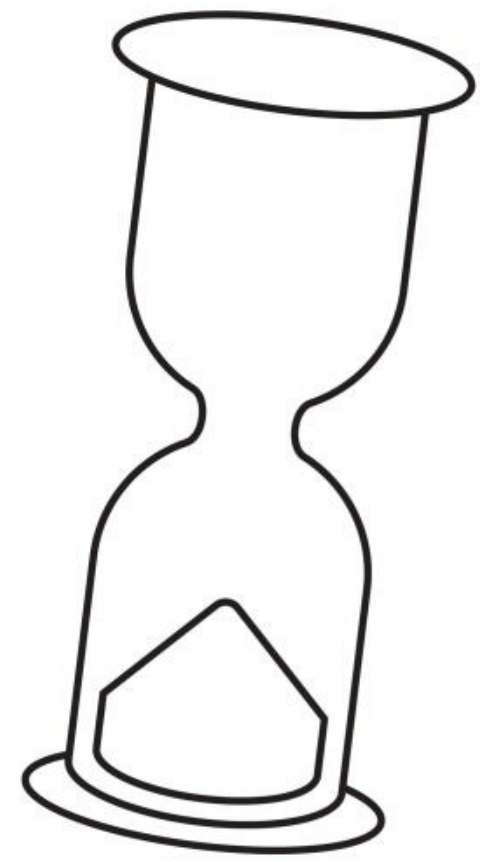

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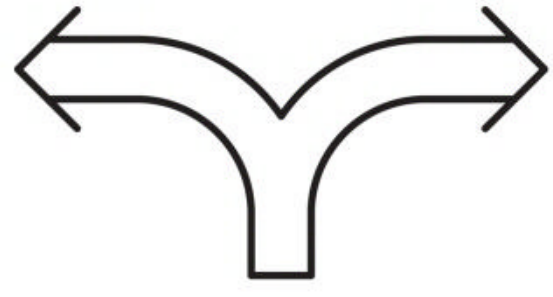

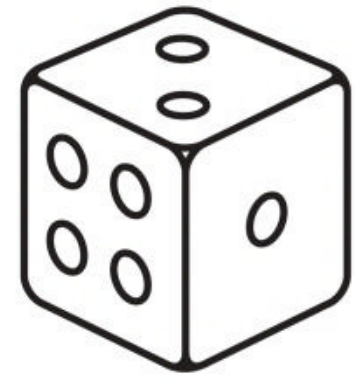

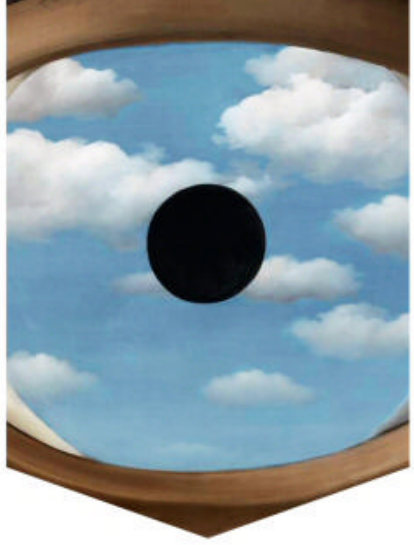

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"Life is to be lived, not controlled; and humanity is won by continuing to play in face of certain defeat."

Ralph Ellison

Uncertainty

The Fates – *Moirai* to the Greeks and *Parcae* to the Romans – were three sisters in Greek and Roman mythology who handled a length of thread that represented each and every life: the 'thread of destiny'. Through each thread they controlled the destinies of mortals and gods alike, ensuring that lives were led according to the laws of the universe. Clotho would spin the thread; Lachesis would determine the length of and amount of luck in a life; and Atropos would cut it with her shears to bring it to an end.

We humans are quite fond the idea of certainty, be it of our past, present, or future; of knowing *what comes next*. But, in the end, the only real certainty we have is that Atropos will wield her shears and cut the thread. (And, as Benjamin Franklin quipped, that someone will ensure we pay our dues.)

Rather than causing anguish or fear, the certainty of death in an uncertain life should galvanise us into action to live life to the fullest. We mightn't have much control over our fate – when we are born, the amount of luck in our lives, and when it will end – but we could do worse than to accept this, and take Ralph Ellison's advice that "life is to be lived, not controlled".

Zan Boag
Editor,
New Philosopher

NewPhilosopher

Editor
Zan Boag
Editorial Director
Antonia Case
Art Directors
Carlos Egan, Aida Novoa
Cover Design
Genís Carreras
Editor-at-large
Nigel Warburton
Contributors
Mariana Alessandri, John Armstrong, Antonia Case, Tom Chatfield, André Dao, Stephen Law, DBC Pierre, Clarissa Sebag-Montefiore, Sir David Spiegelhalter, Patrick Stokes, Kate Sweeny, Stuart Vyse, Nigel Warburton
Illustrators / Artists
Genís Carreras, Carlos Egan, Russel Herneman, Corey Mohler, Aida Novoa
Photographers
Library of Congress, NASA, Thomas Pesquet
Administration
Marnie Anderson, Claudio Faerman

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Contact
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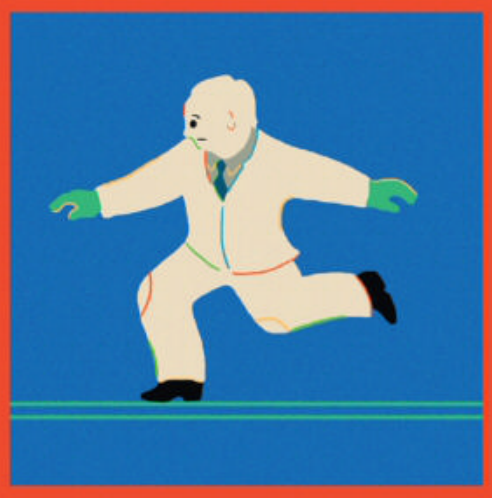

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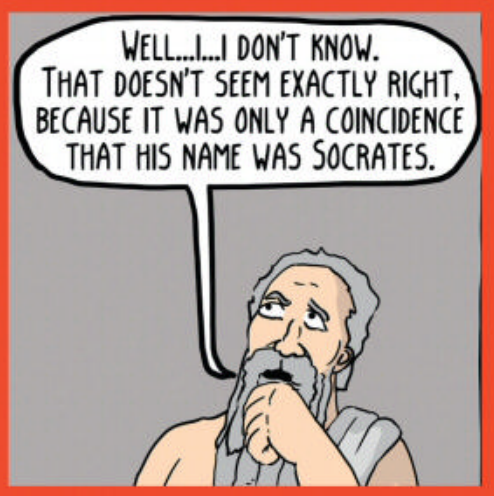

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The fly that rouses

Socrates described his job as the "office of such a fly; and all day long I never cease to settle here, there, and everywhere, rousing, persuading, reproving every one of you."

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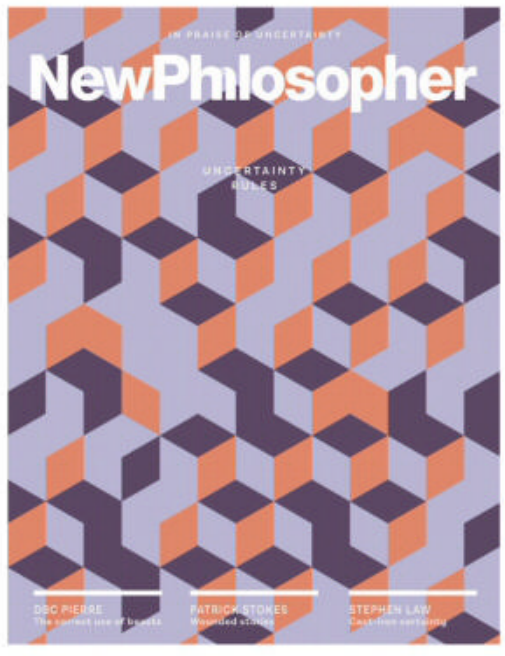

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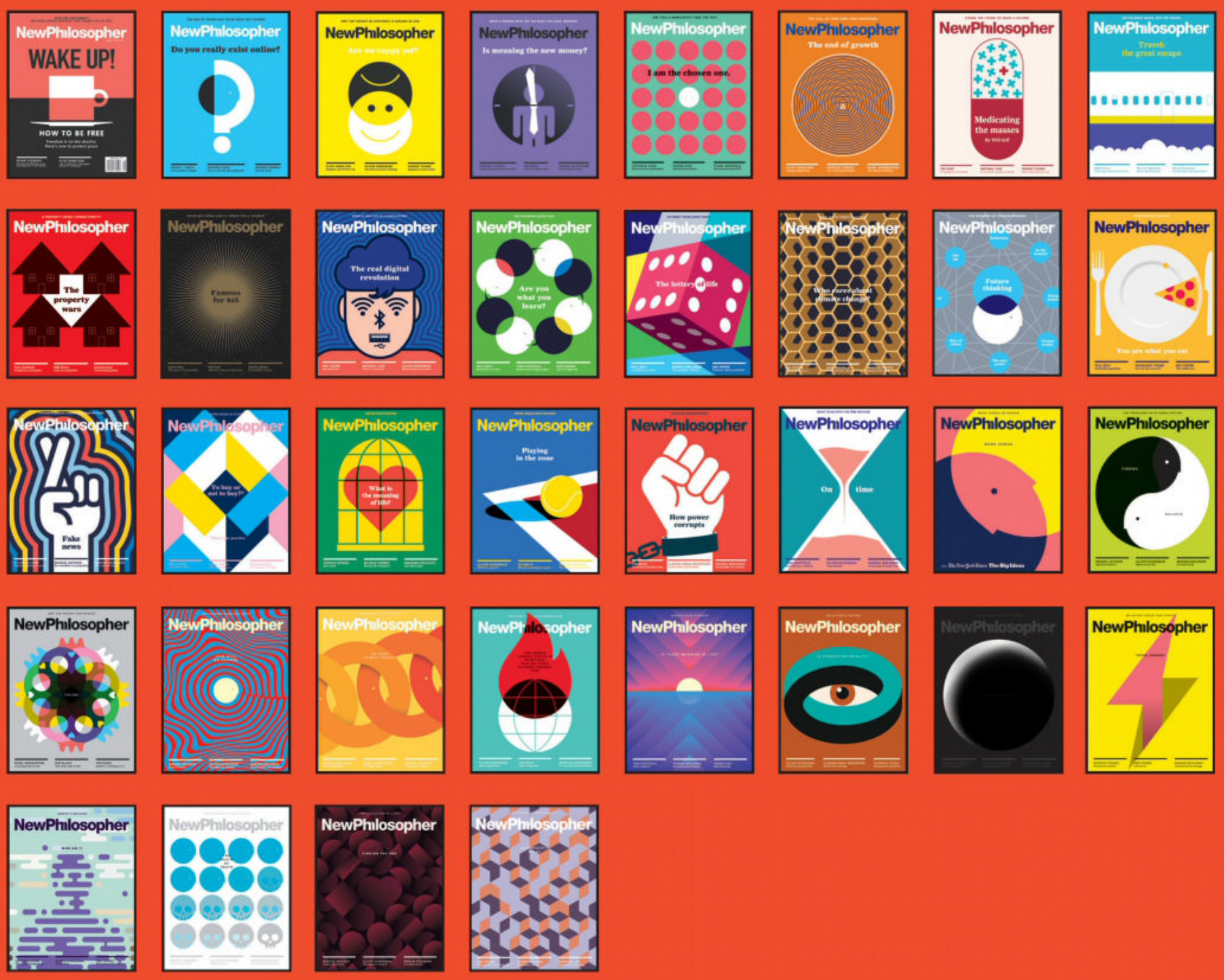

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Contributors

DBC Pierre

DBC Pierre won the Booker Prize for his debut novel *Vernon God Little*, which was also awarded the Whitbread First Novel Award in 2003 – the first time the two awards had been granted to the same book. Pierre is also the author of *Ludmilla's Broken English*, *Lights out in Wonderlan*d, a book of short stories, and a novella, *Breakfast with the Borgias*. He was awarded the James Joyce Award from the Literary and Historical Society of University College Dublin.

Mariana Alessandri

Mariana Alessandri is Associate Professor of Philosophy at the University of Texas Rio Grande Valley. She has written for *The New York Times, Philosophy Today, Womankind magazine, Times Higher Education, Chronicle of Higher Education,* and many academic journals. Alessandri is the author of the forthcoming book, *Night Vision: Seeing Wisdom in our Darker Moods*, and her teaching interests include Existentialism and Mexican-American Philosophy.

Patrick Stokes

Patrick Stokes is a lecturer in philosophy at Deakin University, Melbourne. He specialises in 19th and 20th century European philosophy, personal identity, narrative selfhood, moral psychology, and death and remembrance. A particular focus is bringing Kierkegaard into dialogue with contemporary analytic philosophy of personal identity and moral psychology. Stokes was awarded the 2014 AAP media prize.

Sir David Spiegelhalter

Sir David Spiegelhalter has been Winton Professor of the Public Understanding of Risk at the University of Cambridge since October 2007. He is Chairman at The Winton Centre for Risk and Evidence Communication, a Fellow at Churchill College, and Associate Fellow of the Centre for Science and Policy. He was elected FRS in 2005 and awarded an OBE in 2006 for services to medical statistics. He received a knighthood in the Queen's Birthday Honours List in 2014 for services to statistics.

Stephen Law

Stephen Law is a philosopher and author. Currently Director of Philosophy and Cert HE at Oxford University Department of Continuing Education, Law has published many popular books including *The Philosophy Gym, What Am I Doing With My Life?*, *The Great Philosophers: The Lives and Ideas of History's Greatest Thinkers, The Complete Philosophy Files*, and *Believing Bullshit.* He is a Fellow of The Royal Society of Arts and Commerce and in 2008 became the provost of the Centre for Inquiry UK.

Kate Sweeny

Kate Sweeny is a Professor of Psychology and Teresa and Byron Pollitt Endowed Term Chair at the University of California, Riverside, where she runs the Life Events Lab. Sweeny received her PhD in 2008 from the University of Florida where she studied social psychology. Her research examines two broad questions. First, how do people cope with acute and stressful moments of uncertainty, like the wait for important news? Second, how should doctors talk to their patients?

Stuart Vyse

Stuart Vyse is a behavioural scientist, teacher, and writer. He is a columnist with *Skeptical Inquirer* magazine and has written for the *Observer*, *The Atlantic*, and *Time*. He is the author of *Believing in Magic*, which won the William James Book Award; *Going Broke*; *Superstition*; and *The Uses of Delusion*. As an expert on superstition and irrational behaviour, he has been quoted in *The New York Times, the Washington Post, the Los Angeles Times*, and has appeared on CBS, CNN, and PBS NewsHour.

Clarissa Sebag-Montefiore

Clarissa Sebag-Montefiore lived in China from 2009 to 2014 during which time she worked as the associate editor for *Time Out Beijing*, the art editor for *Time Out Shanghai*, and as an oped columnist for the *International New York Times*, reporting from China for the blog Latitude: Views From Around the World. She writes for *The Guardian, The Economist, Financial Times, The New York Times, Womankind, Wall Street Journal, New Statesman, New Internationalist, The Huffington Post*, and Time magazine.

John Armstrong

John Armstrong is a scholar, writer, philosopher, and speaker on art and culture. He was formerly Associate Professor of Philosophy at the University of Melbourne and Philosopher-in-Residence at Melbourne Business School. Armstrong is the author of several internationally acclaimed books, including *The Intimate Philosophy of Art*, *The Secret Power of Beauty*, *Conditions of Love*, and *Love, Life, Goethe*.

Zan Boag

Zan Boag is Editor of *New Philosopher,* Editorial Director of the international newsstand magazine *Womankind*, and Director of *poet* bookstore. In 2017 he won the Australasian Association of Philosophy Media Professionals Award and was shortlisted for Editor of the Year in the international Stack Awards. Boag speaks regularly on philosophy, technology, the media, and ethics, and is the co-founder and host of the monthly philosophical discussion series *Bright Thinking*. He is a Fellow of the Royal Society of the Arts.

André Dao

André Dao is a writer and editor who is co-founder of *Behind the Wire,* an oral history project documenting people's experience of immigration detention, and a producer of the Walkley-award winning podcast, *The Messenger.* His work has appeared in *The Monthly, SBS True Stories, Meanjin*, and *Al Jazeera English*. Formerly the editor-in-chief of human rights publication *Right Now*, Dao was a finalist for the Australian Human Rights Commission's Young People's Medal in 2011.

Russel Herneman

Russel Herneman is an award-winning cartoonist whose work has appeared in *The Times of London, Private Eye, Prospect, The Spectator,* and many others. In 2018 he won Pocket Cartoon of the Year 2018 in the Political Cartoon Awards, European Newspaper Design award for illustration, and Society of News Design Award of excellence for Illustration. He was an exhibitor at the Society of Graphic Fine Art Draw 18 at Mennier Gallery, London, and in 2016 and 2017 he was a finalist in Best Use of Illustration in the Stack Awards.

Antonia Case

Antonia Case is the literary editor of *New Philosopher*, the editor of *Womankind* magazine, and an award-winning writer and journalist. Case was selected as 'philosopher in residence' for the 2016 Brisbane Writers' Festival and is currently writing a book on personal identity and change. She was the winner of the 2013 Australasian Association of Philosophy Media Professionals' Award and in 2016 was shortlisted for editor of the year in the Stack Awards.

Oliver Burkeman

Oliver Burkeman is the winner of the Foreign Press Association's Young Journalist of the Year and was shortlisted for the Orwell Prize in 2006. His books include *HELP! How to Become Slightly Happier and Get a Bit More Done* and *The Antidote: Happiness for People Who Can't Stand Positive Thinking* – which explores the upsides of negativity, uncertainty, failure, and imperfection. His new book, *Four Thousand Weeks*, is about making the most of our finite lives.

Genís Carreras

Genís Carreras is the designer of every cover of *New Philosopher* magazine and the creator of *Philographics: Big Ideas in Simple Shapes*. Carreras's work has been recognised in the AOI World Illustration Awards, the Laus Awards, and the Stocks Taylor Benson Awards, and his work has been featured in the books *MIN: New Simplicity in Graphic Design, Playing with Type, Geometry Makes Me Happy*, and *Geo/Graphics.*

Nigel Warburton

Nigel Warburton is a freelance philosopher, podcaster, writer, and the Editor-at-large of *New Philosopher*. Described as "one of the most-read popular philosophers of our time", his books include *A Little History of Philosophy*, *Thinking from A to Z*, and *Philosophy: The Classics*. The interviewer for the Philosophy Bites podcast, Warburton was previously Senior Lecturer in Philosophy at the Open University and Lecturer in Philosophy at Nottingham University.

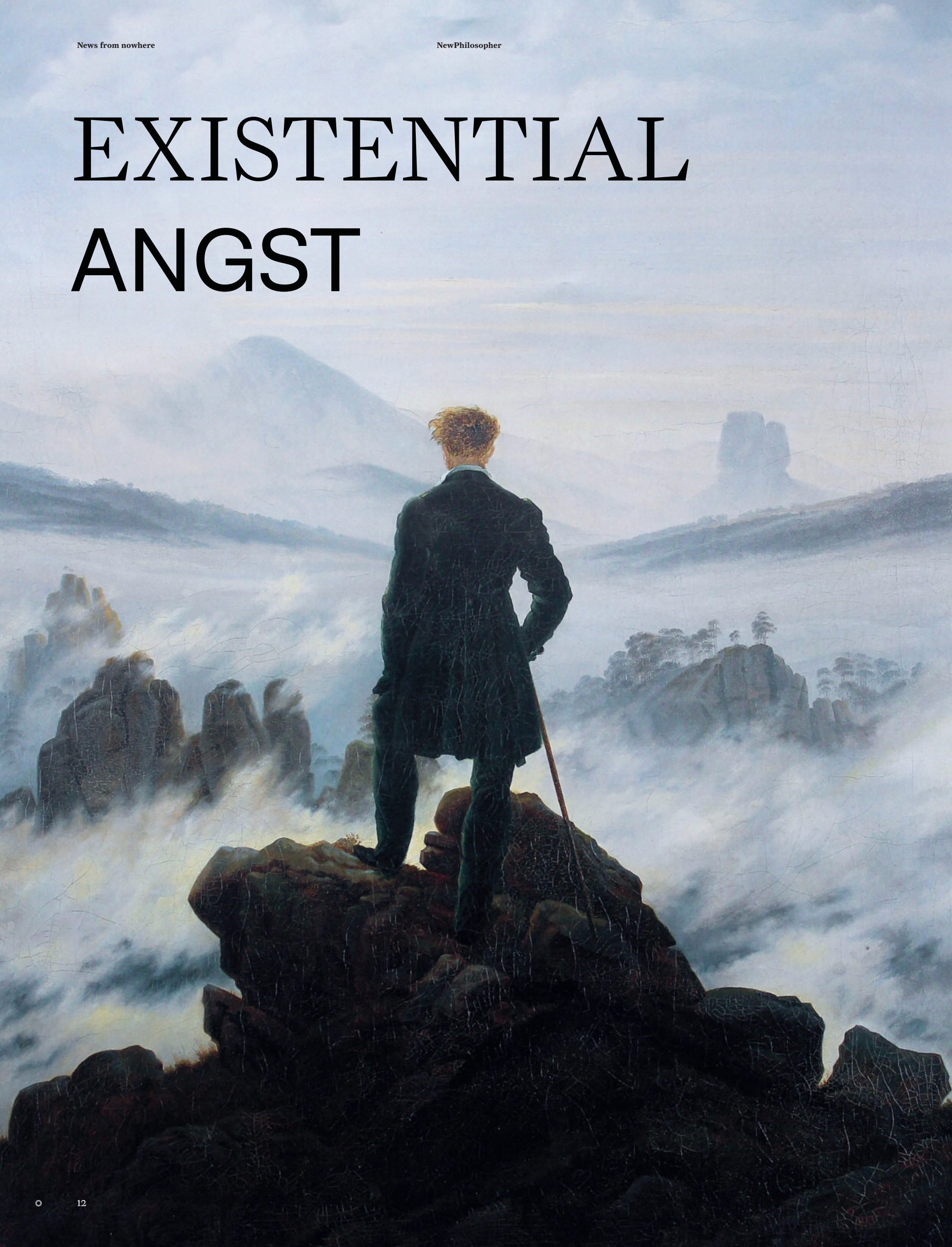
Tom Chatfield

Tom Chatfield is a British writer, broadcaster, and tech philosopher. He is the author of six books, including *Netymology, Live This Book*!, and *How to Thrive in the Digital Age*, and speaks around the world on technology, the arts, and media. Chatfield was launch columnist for the BBC's worldwide technology site, BBC Future, is a Visiting Associate at the Oxford Internet Institute, and is a senior expert at the Global Governance Institute.

Aida Novoa & Carlos Egan

Aida Novoa & Carlos Egan are the art directors of *New Philosopher* and *Womankind* magazine, as well as for *poet* tea, which is produced by the magazines' publishers. Their work for the publications has been recognised by AIGA, the oldest and largest organisation for design in the United States, as well as by *Computer Arts* magazine, *Desktop Mag*, and *Creative Journal.*

EXISTENTIAL ANGST



> “Vertigo announces it-self through fear; I am on a narrow path - without a guardrail - which goes along a precipice. The precipice presents itself to me as to be avoided; it represents a danger of death.”
>
> **— Jean-Paul Sartre**

Despite being one of the most popular philosophical movements of the last century, Existentialism is famously hard to define as a theory. In popular imagination, Existentialism has become associated more with a series of philosophical moods and attitudes, such as ‘despair’, ‘bad faith’, ‘anticipatory resoluteness’, and, most commonly, ‘angst’.

That last concept passes through a number of hands and languages, from Kierkegaard’s *angest* to Heidegger’s *angst* and Sartre’s *l’angoisse*. It’s often translated ‘anxiety’, ‘dread’, or ‘anguish’, but ‘angst’ avoids any unhelpful medical connotations.

All these thinkers, despite their significant philosophical and religious differences, see angst as linked not just to the uncertainty of the future, but to the uncertainty of human behaviour. Existential angst is not simply anxiety about what could happen, but anxiety over what we might do ourselves. Angst is, quite simply, the dizzying, even nauseating feeling of freedom.

Sartre famously likened existential angst to the sort of vertigo you might experience while walking along the edge of a cliff. Angst is not simply fear of what might happen to you – the fear that you might lose your footing, say, or that the cliff face may give way, or a sudden gust of wind might blow you over the edge. Rather, angst is the awareness that nothing, literally *nothing*, stops you from throwing yourself off the cliff or behaving recklessly. Perhaps you make a decision: you resolve to stop cheating on your partner or exercise more this year or watch your step until you’re safely away from the cliff edge. But in the very next moment, your resolution retreats into the past, and you’re confronted by freedom of choice all over again.

For the Existentialists, we can never really escape angst. It is, quite simply, the price we pay for freedom – not that we ever had a choice.

Knowledge vs. opinion

For many philosophers, both ancient and modern, knowledge is something certain. Plato, for instance, distinguished knowledge (*epistêmê*) from mere opinion (*doxa*). You might strongly suspect that something is true, but that is not the same as *knowing* it. Later, Stoic philosophers declared knowledge was 'kataleptic' – that is, some impressions are experienced as self-certifying in a way others are not. Because the mind has this capacity for telling true impressions from false ones, we can have certain knowledge about the world.

By this time, however, the institution Plato had founded, the Academy, had passed to a school of Sceptic philosophers who were opponents of the Stoics. These academic Sceptics argued that true impressions and false impressions are in fact often indistinguishable. That means that knowledge never wears its certainty on its face, and so while impressions *might* correspond to reality (maybe even most of the time), we can never be *certain* that they do. As the Sceptics were denying *katalepsis*, this doctrine became known as 'acatalepsy'.

Scepticism can be an unsettling and even distressing form of philosophy. (Centuries later, David Hume found sceptical reflections so disturbing he had to go play backgammon to calm himself down again.) After all, the idea that we can never be certain our sense impressions of the world correspond to reality leaves us seemingly helpless and knowledgeless. But it is important to be clear about what Sceptics do and don't claim. The doctrine of acatalepsy didn't mean that we know nothing. Rather, it means that our beliefs, instead of being either knowledge or non-knowledge (whether mere opinion or error) can only ever have degrees of plausibility. We can never have absolute certainty. But some things are still much more likely to be true than others – and that might be enough to say that we know what's what.

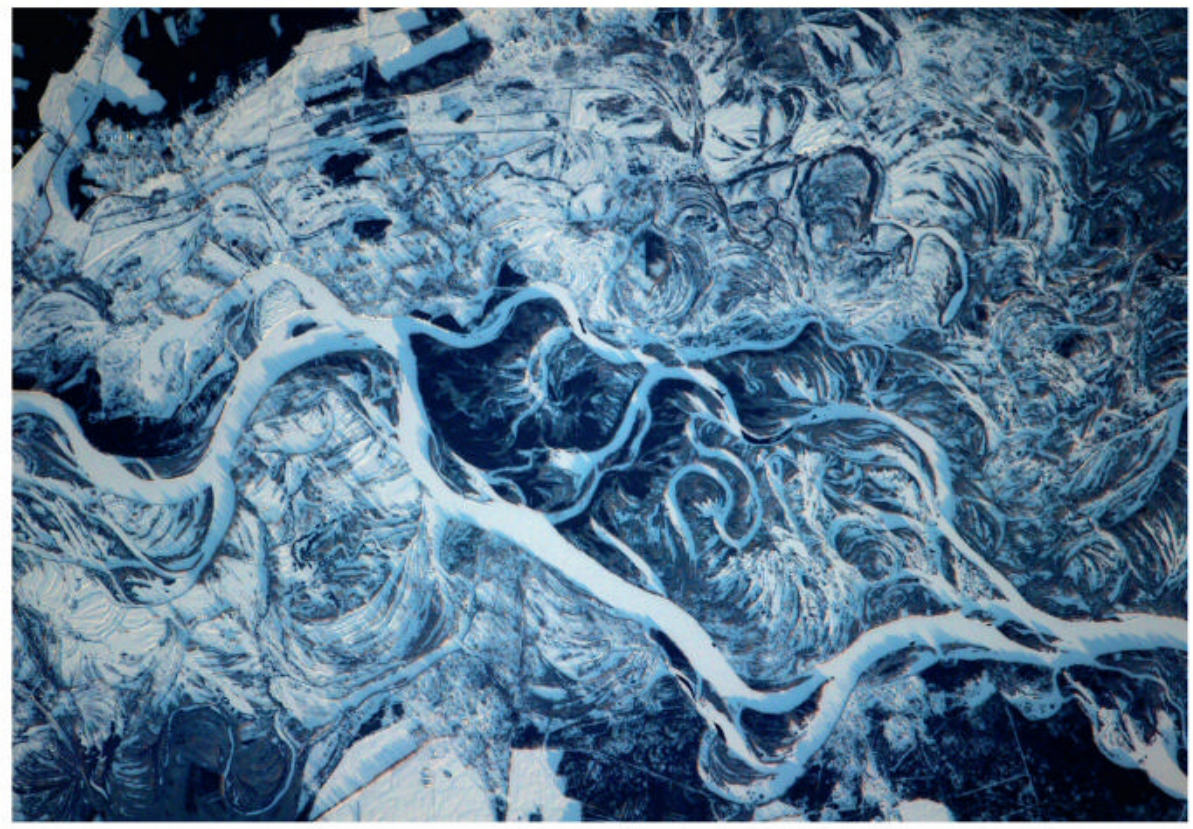
The Death of Seneca, by Manuel Domínguez Sánchez, 1871

"We may not be able to get certainty, but we can get probability, and half a loaf is better than no bread."

— C.S. Lewis

A VUCA WORLD

The Frozen Wild Dnieper River, photo by NASA/ESA/Thomas Pesquet



It seems that we live in times that are increasingly difficult to navigate. The pace of technological change is an obvious starting point: it took humankind untold millennia to learn to fly, but once the Wright brothers managed the first ever flight (travelling the grand distance of 36 metres) it was only another 66 years until the Moon landing – less than a lifetime. Now, of course, your smartphone puts enough computing power in your pocket to guide millions of simultaneous Apollo 11s to the Moon.

Another case in point is the waning of religion. Where once religious faith gave unshakeable certainty to the vast majority of human lives – and anchored the world's major philosophies – the same cannot be said today. Add to the mix the collapse of the old geopolitical uncertainties after the end of the Cold War, and the decline of American hegemony, not to mention climate change, and novel viruses crossing over from animals to humans… it's little wonder that people in the leadership and strategy industry keep insisting that we live in a VUCA world.

That's VUCA as in Volatility, Uncertainty, Complexity, and Ambiguity: an acronym first coined by the US Army to describe a world that is constantly changing (volatility), increasingly difficult to predict (uncertainty), hard to analyse (complexity), and resistant to interpretation (ambiguity). From military circles, where it was used to describe the turbulence of the battlefield, VUCA made its way to the world of business, where CEOs and corporate strategists thought it captured something of the flux of 21st century markets.

Then again, there's nothing particularly new about the observation that everything is changing all the time. Just ask Heraclitus of Ephesus – he'd tell you that the only certain thing is that nothing remains the same. As he was apparently fond of saying: *panta rhei*. Everything flows.

"At last – opportunity knocks!"

Schrödinger's cat

Erwin Schrödinger is best remembered today, at least outside physics departments, for his notorious thought experiment about a cat. Imagine we've put a cat in a box with a vial of poisonous prussic acid linked to a device with a tiny amount of radioactive material in it. Over the course of one hour, the radioactive material might decay and set off the device, releasing the poison and killing the cat. Or it might not. An hour passes. Before we open the box to check, is the cat alive, or dead? Or does the cat somehow exist in a third state – is it both alive *and* dead?

The example, of course, isn't really about cats, but about fundamental particles and 'quantum superposition'. The famous 'slit' experiment shows that light can behave like either a wave or a particle. When we make an observation, light will appear to be one but not the other – but *before* we make the observation, both options seem to exist.

While Schrödinger's Cat has become a cliché, it's easy to miss the point Schrödinger was actually trying to make with his poor feline. Schrödinger wasn't offering a helpful illustration of this 'both alive and dead' understanding of quantum superposition, but attacking it. His cat is meant as a *reductio ad absurdum* of the both/and 'Copenhagen Interpretation'. Schrödinger is mocking the idea that there might be no fact of the matter until an observer arrives to check, because it is self-evidently ludicrous to say the cat is both alive and dead until we open the box.

As Schrödinger says, "There is a difference between a shaky or out-of-focus photograph and a snapshot of clouds and fog banks." Some things are genuinely blurry, but often things are in fact clear enough – it's the camera that's the problem.

Box for animal mummy, 664–30 BCE, The Met

"'Tis much better to do a little with certainty, & leave the rest for others that come after you, than to explain all things by conjecture without making sure of any thing."

— **Sir Isaac Newton**

HEART OF HEARTS

"I just knew in my heart of hearts it was the right thing to do." "Deep down, I knew it was wrong." It's the kind of thing we say all the time – as if within us there is some fundamental self, and as if that fundamental self somehow has a direct line to capital T Truth. Someone with a religious bent – or an Ancient Greek philosopher – might put it this way: it is my soul that knows for certain.

For the American pragmatist philosopher Richard Rorty, such notions are simply wrongheaded. According to Rorty, all knowledge – whether moral or more straightforwardly factual – is a matter of language and social context. While the conventional view is that we access the raw data of the world through our perceptions, Rorty argues that we never have direct access to the world – our perceptions are always already entangled with concepts we have learned through language. Thus, as Rorty says, "there is nothing deep down inside us except what we have put there ourselves".

This means that whatever your heart of hearts knows about right and wrong is not natural or pre-given, and it is not true in the absolutist sense of the word. Instead, your soul's convictions about moral truth were – and are still being – formed by your peers, your community's traditions, and your own responses to others. Some have seen in Rorty's argument a kind of moral nihilism – the death of any sense of right or wrong that runs deeper than 'what society lets us say'. But the social construction of morality might not be cause for despair. There remains the possibility of solidarity with those others with whom we are constantly creating worlds of right and wrong. As Rorty has put it, we still have "our loyalty to other human beings clinging together against the dark".

Diogenes, by Jean-Léon Gérôme, 1860

"There is no such thing as absolute certainty, but there is assurance sufficient for the purposes of human life."

J.S. Mill

Artwork: *This is Not a Pipe*, by René Magritte, 1929

Uncertain times

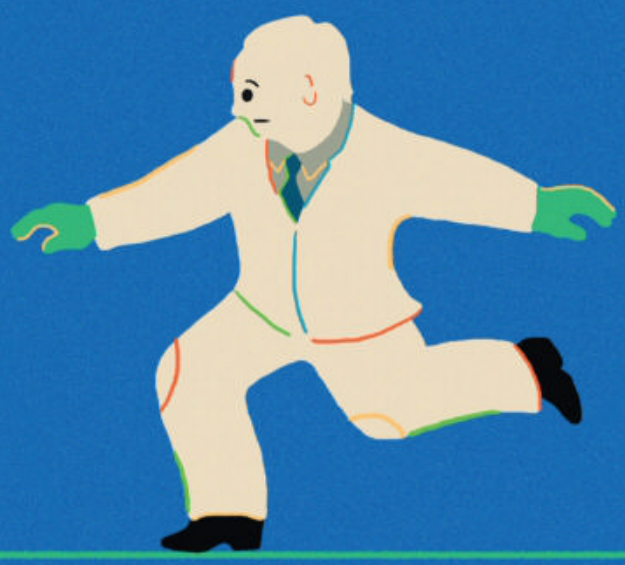
When Vladimir Putin ordered Russian nuclear forces onto high alert, days after he invaded Ukraine, people far from the warzone began to report bone-deep feelings of uncertainty. Media accounts suggested a sudden spike in insomnia, and an epidemic of compulsive social media 'doomscrolling'. Meanwhile, anxious parents wondered how to discuss the unfolding crisis with their children, and the phrase "in these uncertain times" started cropping up in more than a handful of news stories and politicians' speeches.

Search Google's digitised archive of scholarly and historical literature, though, and you'd find that phrase recurring in hundreds of journals and books, in almost every decade, back to the 17th century. The belief that things are unprecedentedly uncertain right now, it turns out, is one that people almost always hold about the era in which they happen to live. "There is… the feeling that we live in a time of unusual insecurity," wrote the spiritual teacher Alan Watts, in his book *The Wisdom of Insecurity*. "As the years go by, there seem to be fewer and fewer rocks to which we can hold, fewer things which we can regard as absolutely right and true, and fixed for all time." Who today could disagree? But Watts was writing in 1951, a time generally remembered as one of peace and prosperity.

At the very least, this ought to give us pause. Of course, it *might* be the case that the world is genuinely becoming a more uncertain place with every successive year, thus vindicating each generation's conviction that the present is more uncertain than the past.

Illustration by Aida Novoa & Carlos Egan

by Oliver Burkeman



But another possibility is the one the writer C. S. Lewis articulated in a sermon delivered in London in the early days of World War II: that extreme uncertainty is humanity's default state, and that circumstances just occasionally converge so as to allow us to forget it for a while. "The war creates no absolutely new situation," Lewis told his audience. "It simply aggravates the permanent situation so that we can no longer ignore it. Human life has always been lived on the edge of a precipice."

The truth, after all, is that we're all going to die – at a time we're unlikely to be able to predict – and that in the meantime anything at all could happen, at any moment, to ourselves or to anyone we care about. We are all inescapably vulnerable to events. You could walk out of a high-rise building tomorrow afternoon, and a grand piano could fall on your head from an apartment above. Is that likely? Not remotely. But you can't rule it out. And that tiny sliver of possibility is all that's required for uncertainty to wreak its emotional distress. You desperately want to be sure about how things will turn out – and you definitely never can be.

One way of defining the phenomenon of worry, in fact, might be as the experience of frustration encountered in repeatedly trying, yet failing, to achieve a sense of certainty about the future. A trivial case in point: at the moment, my family and I are house-hunting, hoping to find an appropriate home in the rural area to which we moved last year, before our current temporary arrangement comes to an end. Sometimes, I catch myself worrying

about it: yearning to know that it'll all work out and feeling the discomfort of the fact that I don't know yet know if it will. And yet, by definition, I won't know that it'll work out until it already *has* worked out – whereupon, I suspect, I'll find something else that hasn't happened yet to fret about instead. The future is annoying like that: it can never be known with certainty from the vantage-point of the present, simply because it's the future.

This is why respite from the distress of uncertainty can only be reached via a counterintuitive route: not by successfully eliminating the uncertainty, but by coming to see that the struggle to eliminate it is so completely futile that you might as well give it up entirely. We're all in the same boat, being carried forward on the river of time, unable to know what's around the next bend – and to treat *that* as a problem in need of a solution is to treat life itself as a problem, which is a recipe for nothing but anxiety. This helps explain the insight usually associated with Buddhist philosophy that the cause of our suffering isn't to be found in the way life is, but in our deep emotional investment in wishing it were otherwise. Or as the American Zen teacher Charlotte Joko Beck liked to say: "What makes it unbearable is your mistaken belief that it can be cured."

But now take one step further. What if learning to embrace uncertainty wasn't just an unfortunate necessity – a tactic for reconciling oneself to the dispiriting truth of the human condition – but something actively positive, the precondition for a fulfilling life? The certainty-seeking mindset has a way of causing those trapped in it to miss out on their lives: it renders them allergic to the present, where uncertainty reigns, and intently focused on the future – where certainty might, they hope, prevail at last.

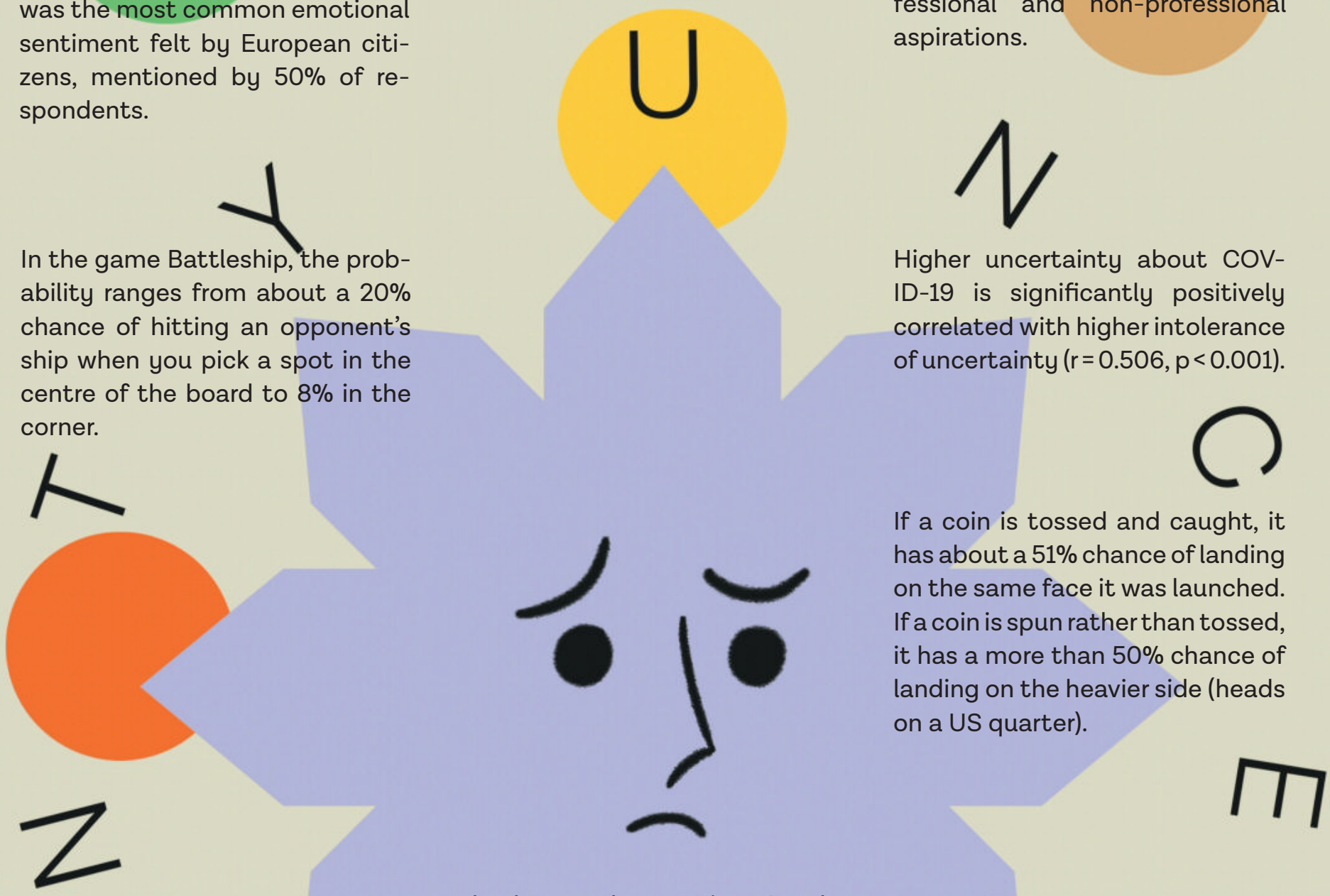
It was becoming a father that finally drove this home to me. At first, I agonised about what I *should* be doing – in terms of nutrition, or reading books, or sleep training, or screen time, or a thousand other variables endlessly dissected in the literature on parenting – until it finally dawned on me that approaching fatherhood in this manner would mean missing out on fatherhood entirely, to my son's detriment and my own. Even if there were a 'right way' to approach any of these matters, I would only be able to evaluate the choices I'd made – and thus figure out if I'd chosen correctly – when my son was much older. My craving for a feeling of certainty about whether or not I was on the right track thus meant focusing my gaze on the future, instead of attending to my actual relationship with my son, which unfolded exclusively in the present. I came to realise that an authentic experience for both of us would only be found by engaging with what was happening *now*, and that seeking certainty got in the way of that – underscoring the insight of the psychoanalyst and philosopher Erich Fromm that "the quest for certainty blocks the search for meaning".

The self-help writer Susan Jeffers once suggested an exercise for learning to embrace uncertainty: write down a list of the things you hope will come true in your lifetime – in your career, your relationships, or in the wider world – and then try rewriting them, this time replacing the phrase "I hope that…" with "I wonder if…" Hope, at least in this sense of the word, is a deeply conditional state of mind: it puts you in the position of anxiously waiting to see if your preferred outcomes will arrive, while intensifying your feelings of disappointment if they don't. Wondering, by contrast, leaves you curious about what's coming next, and at least potentially gratified or enriched by whatever ends up happening. You get to stop fighting for a sense of closure you'll never attain, and relax into the inescapable, exhilarating openness of life as it actually is.

The certainty-seeking mindset has a way of causing those trapped in it to miss out on their lives: it renders them allergic to the present, where uncertainty reigns.

In 2020, when asked, “What feeling best describe your current emotional status?”, ‘uncertainty’ was the most common emotional sentiment felt by European citizens, mentioned by 50% of respondents.

Young people with uncertain career ambitions earn significantly lower hourly wages in young adulthood than those with professional and non-professional aspirations.



In the game Battleship, the probability ranges from about a 20% chance of hitting an opponent’s ship when you pick a spot in the centre of the board to 8% in the corner.

Higher uncertainty about COVID-19 is significantly positively correlated with higher intolerance of uncertainty ($r = 0.506$, $p < 0.001$).

If a coin is tossed and caught, it has about a 51% chance of landing on the same face it was launched. If a coin is spun rather than tossed, it has a more than 50% chance of landing on the heavier side (heads on a US quarter).

In chess, aside from Kings, which can’t be taken, the end Pawns have the highest survival rate at 66%-73%. Rooks also tend to be hardy, with a 55% survival rate.

The house always wins. On the games with the lowest house edge, a casino might be generating anywhere from around 0.5% to a bit over 2% profit. On other games, it may make profits of anywhere from 15% to 40%.

Between 8% and 25% of Americans over the age of thirty-five have experienced a mid-life crisis.

The country with the highest life expectancy in the world is Monaco, at 86.8 years (up from 62.2 years in 1950). The lowest is the Central African Republic, at 53.3 years (up from 31.8 years in 1950). In the US it is 78.9, the same as in Lebanon and Curacao.

Since 2000, the estimated level of human-induced warming has been equal to the level of observed warming with a likely range of ±20% accounting for uncertainty due to contributions from solar and volcanic activity over the historical period.

If you get the first move in Connect Four, you can win 100% of the time within 41 moves.

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STEPHEN LAW
Cast Iron certainty

by Patrick Stokes

Wounded stories

In some corners of the internet, you'll find people sharing and comparing photos of their 'bug-out bags'. These are backpacks or duffle bags stuffed full of things their owners believe they'll need in an emergency, from water and bandages to flares and gas masks. Keeping such a bag packed sounds quite prudent, especially for people who live in areas prone to flood or wildfire, or who may need to escape an unsafe environment. Disaster can strike anytime, so it's good to be prepared.

However, many of these bags are clearly built around survivalist fantasies of rugged self-reliance, rather than real-life emergencies. (The suspiciously high prevalence of weapons is a giveaway. You rarely need to shoot a forest fire). As one online wit pointed out, when the time comes that you really do need a bug-out bag, it won't be because of zombie invasion or a sudden breakdown of society. It'll be a phone call to say your parent is in hospital in another city. You won't need a bowie knife and the know-how to skin a moose, but you will need clean underwear and loose cash for vending machines.

No doubt there's an element of self-aggrandisement in many bug-out bags. But there's also a very human vulnerability: our inability to foresee the sorts of abrupt events that change our whole lives instantly. When we imagine fleeing into the woods to live by our wits or hide out in pre-stocked survival bunkers, we're imagining still being *us* doing these things. We would be the same person, just living in a world that had suddenly changed, for everyone.

But imagine what would happen if, tomorrow, you suffered an incapacitating stroke, or your child died, or you lost your home. The rest of the world would remain more or less untouched, but *your* life is fundamentally and irreparably altered. Perhaps that's the reason these sorts of sudden transformations, despite being so common, are so hard to anticipate. Doing so asks us to envisage becoming someone other than who we are.

This language sounds metaphorical. But for many philosophers, it's more than that. Beginning in the 1970s, theorists in a range of disciplines – philosophy, history, psychology, and others – made what became known as the 'narrative turn.' (Humanities scholarship never merely develops, for some reason, but 'turns.'). These researchers began to see storytelling not just as something

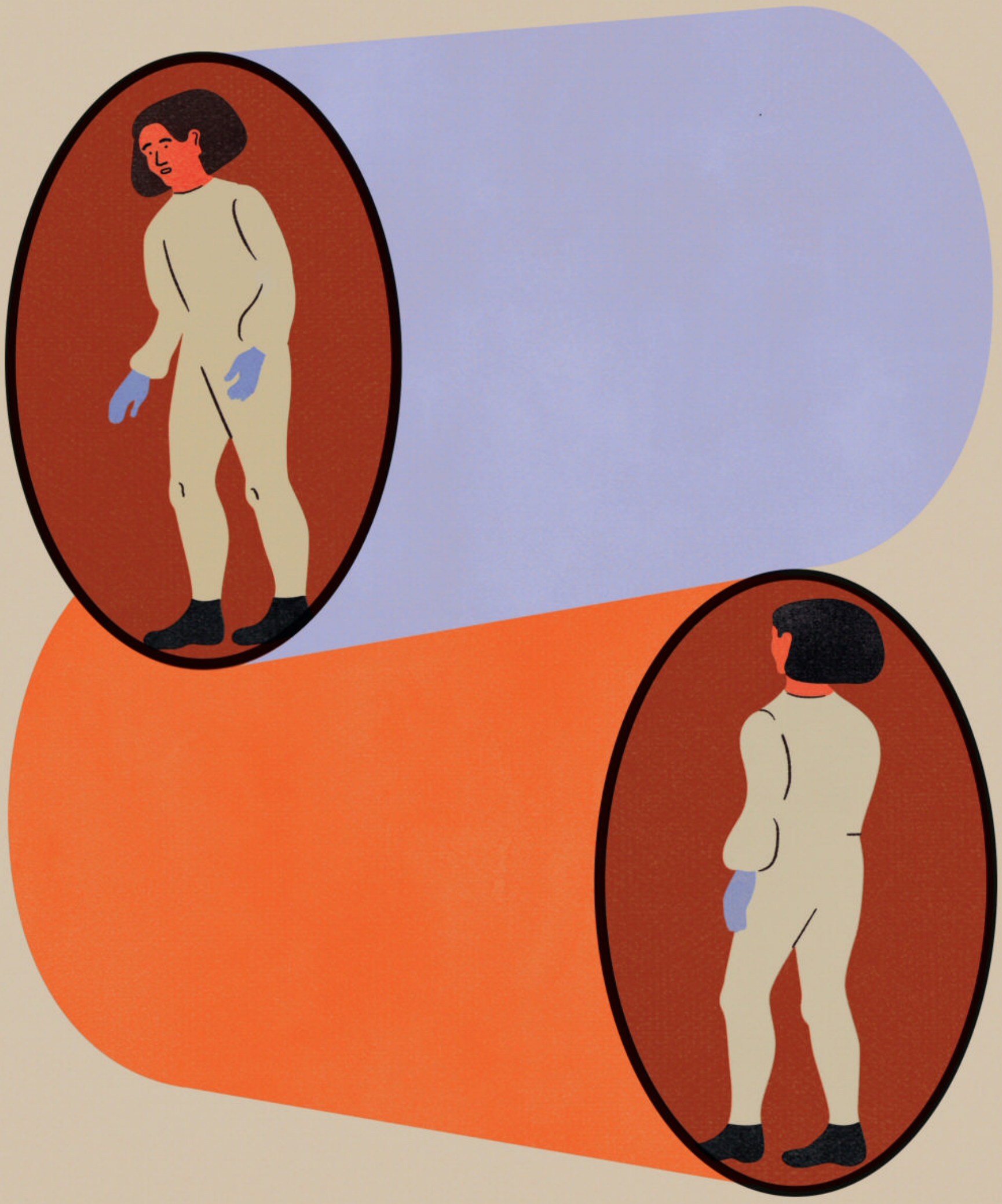


Illustration by Aida Novoa & Carlos Egan

humans do once events have happened, but as a precondition for being able to experience events in the first place. As the philosopher Alasdair MacIntyre argued, even something as simple as understanding that someone is making a cup of coffee requires us to be able to interpret their movements as part of a narratively structured, meaningful sequence.

That applies not just to individual acts of coffee-making, but to entire human lives. According to this view, *the things that make you who you are* have that status because they are narratively selected and organised. If I ask you to tell me the story of your life, you'll pick out the specific chain of events you think I need to hear to get a sense of who you are, to explain how you came to this moment – and where you think you're headed next.

The problem is that, as Kierkegaard memorably put it, life can only be understood backwards but must be lived forwards. Or, as the philosopher of history Louis O. Mink pointed out nearly fifty years ago, there is a fundamental difference between *following* a story and *telling* a story. When you tell a story, there's a kind of inevitability to it – the story is always going to end the way it does, even if this is only obvious in retrospect – which is reflected in the details you include and the ones you leave out.

But following a story is different. You don't really know how things will end until they do. Think of watching a tense sporting match versus recounting the match afterwards. The retelling is structured by the knowledge of who will ultimately win; the watching is suffused with the anxiety of not knowing exactly that.

Wounds can heal, and that is true of narrative injury as well.

"Uh oh. Mr Nostradamus is building a bunker in his garden."

In other words, the stories we rely on to make sense of ourselves and the world are in tension with the radical openness of the future. The uncertainty of life makes any story we can tell about our life so far provisional, always open to sudden and dramatic reversal. At any moment it can turn out that we're living in a very different story to the one we'd assumed.

Nothing shows that disjunction up more clearly than what's been called 'narrative injury': those sudden, unplanned for events like illness, divorce, bereavement, war, or disaster, that completely knock a life off its trajectory. The tragedy of such sudden reversals of fortune isn't just the good things we lose when they happen, bad enough though that is. It is not just that we used to have health, or a marriage, or a career, and now we do not. It's that we used to be a certain sort of person, with a more-or-less clear path laid out ahead. We do not simply lose parts of ourselves when these things happen. We lose the capacity to make sense of the parts that remain. Our ability to keep narrating ourselves into existence is wounded.

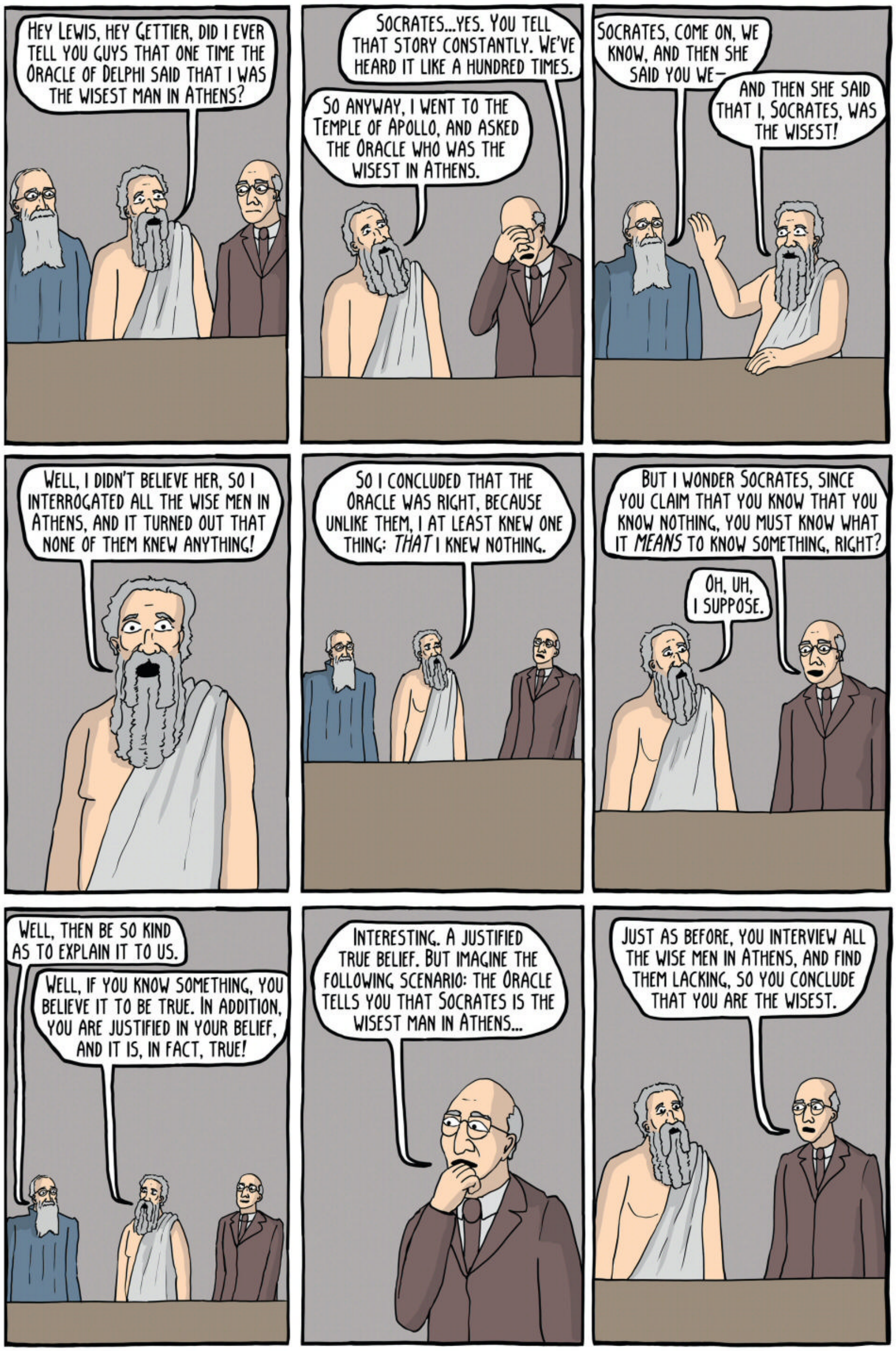
When terrible, life-shattering things happen, whether to ourselves, those we love, or even those whose stories we only encounter at a distance and in passing, there is often a sense that what's happened isn't just a great pity, but is also deeply unfair. The sense of a tragedy as unjust, and not just cruel or lamentable, goes to some very deep, inchoate assumptions about how the world should be arranged. And those assumptions are often diachronic – that is, they're intrinsically spread out across time. If you work hard, you should be rewarded with material security; if you love your partner you should be able to expect many more years together; you should outlive your children. None of that, of course, is how the world necessarily works. Nothing is guaranteed. So, when it suddenly turns out the story you're living isn't the one you thought it was, when you find out your life up until now has been the first act of a tragedy and not a rom-com, there can be a sense not just of loss, but of narrative indignation.

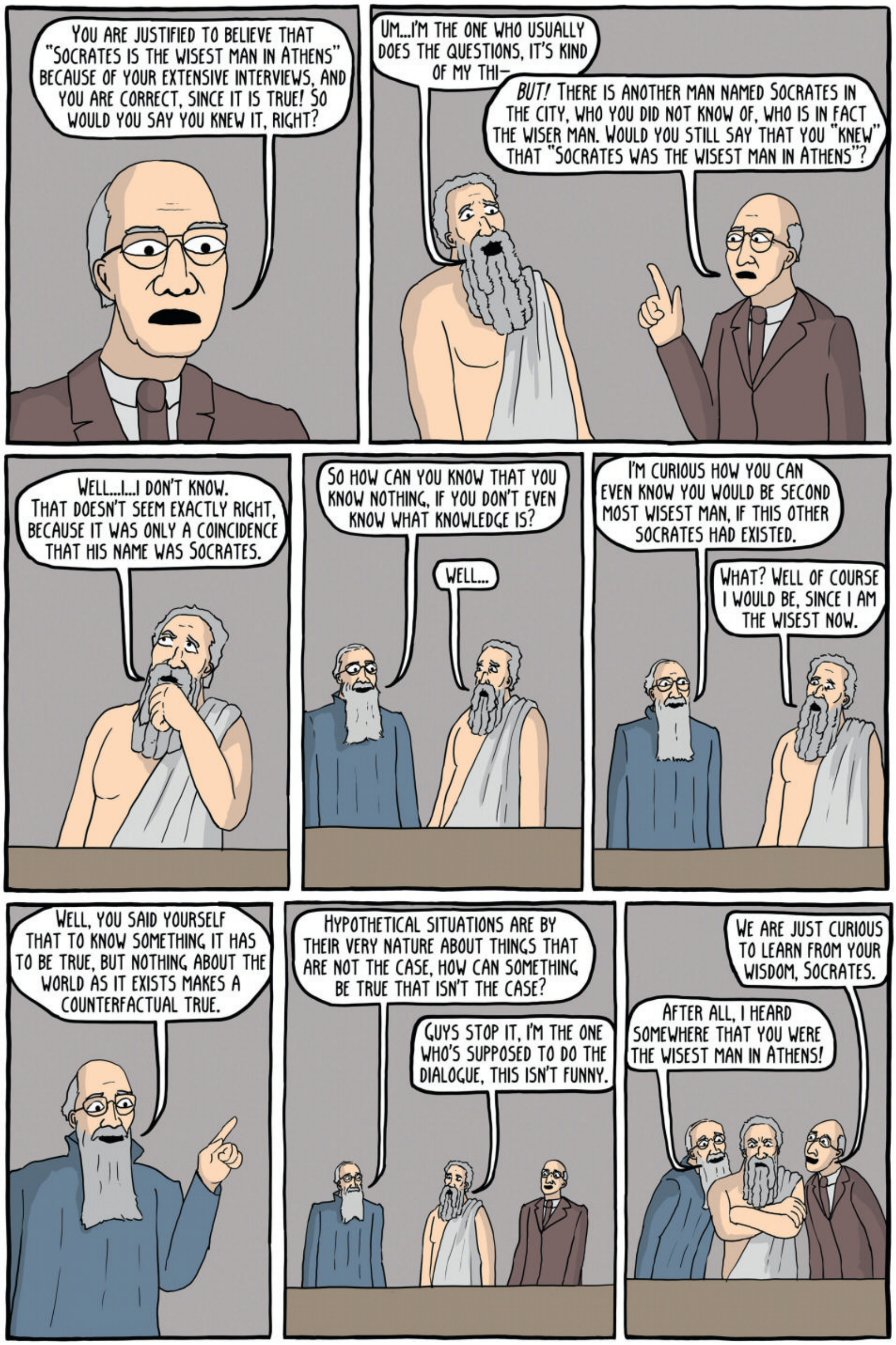
Wounds can heal, and that is true of narrative injury as well. With both kinds of injuries, though, they don't always heal back to how things were before. The bones may knit into a different shape, the connective tissues never quite regain their old suppleness or capacity to bear weight. When people who have suffered illness, bereavement, or career loss talk about adjusting to a 'new normal', part of what they mean is getting used to a revised sense of what they can expect to happen next. And that in turn means putting together a new story, or at least a story that's different to the previous one, something that makes sense in that distinctively narrative way. It's not so much 'putting the pieces back together', a metaphor that suggests something static like a puzzle, but something temporal. A narrative reconfiguration.

The stories we tell about ourselves are fragile things. They're rope bridges woven from the most tenuous thread, slung over a vast abyssal drop of contingency and loss. Most of the time we simply keep our head up and don't notice the constant danger of falling.

Perhaps that is the real promise of 'bug-out bags'. It's not that having such a bag ready to go is a symbol of survival per se, but of narrative continuity. They offer the illusion that you've already thought of all the contingencies, planned for them, spun them into your story. You've got it all planned out. Until, suddenly, the phone rings. ◘

Socrates gets Socrates'd





Author/illustrator: Corey Mohler, Existential Comics. For more comics visit existentialcomics.com

Illustration by Aida Novoa & Carlos Egan

by Clarissa Sebag-Montefiore

Denying death

Carroll Gardens, the neighbourhood where I live in Brooklyn, is famous for two things: Italian food and annual Halloween decorations.

One historic brownstone sticks out from the others: each October its owners adorn the exterior of the building with a plethora of lurid, fantastical furnishings. There's a skeleton dangling in a cage. A bloody baby, its head pulled off, in a pram. A corpse hanging by its neck from an upstairs window. A zombie girl on a swing. And, most gruesome of all, a 'prisoner' in an orange jumpsuit in an electric chair, his eyes squeezed shut in pain, his mouth set in a tight grimace. A mechanism makes his limbs and body shake.

Halloween has its origins in paganism: it was believed that during Samhain, the Celtic new year, the dead would walk among the living. Today it has become a caricature of death itself. Kids gleefully look up at the hanging man on their way to get candy; adults point out the zombie girl and laugh as they walk home from the local pizzeria. If there is anything in life that is certain, it is that we will die. So why do we treat Halloween, a festival of death, as a joke?

Death, in western culture, has become the last remaining taboo: we avoid talking about it or preparing for it physically, mentally, and spiritually. "Imaginary death – such as the cartoon-like violence portrayed on television and in movies – has replaced reality for most people. When death strikes, as it must, and especially if it strikes suddenly and unexpectedly, we respond with discomfort, distress, and dismay," writes Kenneth V. Iserson, Professor Emeritus of emergency medicine at The University of Arizona, in his book *Grave Words: Notifying Survivors About Sudden, Unexpected Deaths*.

Or, as Anita Hannig, an associate professor of anthropology at Brandeis University and author of *The Day I Die: The Untold Story of Assisted Dying in America*, tells me, we live in a "culture of denial" when it comes to death: "Nobody wants to face mortality, because it is this one thing that is the end for all of us. We want to push it as far away from us as possible."

An easier, and less emotionally taxing, route is to treat death as something that happens elsewhere to other people – to steer clear of its implications, even if our collective denial creates worse outcomes for the bereaved and for the dying. The fact that death, with the advancement of medical science, has become a calamity largely reserved for the old, increases the urge to ignore it altogether. Child mortality rates have plummeted in the last 200 years. In the United States, roughly three-quarters of all deaths occur at 65 and older, usually tied to illnesses such as cancer or heart disease. Even COVID-19 overwhelmingly targets the elderly.

Is it any surprise, then, that the young want to live as if they're immortal? Or that, in western culture, any discussion of death is seen as either morbid or awkward, or both? One

British people avoid using the words "death" and "dying" when discussing a recent loss.

2011 survey commissioned by the Dying Matters Coalition in the UK found that death, as a topic, is strenuously avoided. Indeed, most British people avoid using the words "death" and "dying" when discussing a recent loss – instead euphemisms like "passed away" (57%), "deceased" (23%), "kicked the bucket" (20%) and "popped their clogs" (17%) are more common.

"Although someone in Britain dies every minute, our research has found that many people do all they can to avoid talking about dying," Eve Richardson, then chief executive of the National Council for Palliative Care and the Dying Matters Coalition, noted in an introduction. This stretches to practical considerations, too – just 33 per cent of people surveyed had discussed with their partner what kind of funeral they wanted and whether they had a will.

While festivals like Halloween exaggerate the idea of death, thereby neutering its effect, the reality of death – with its inevitable physicality, smells, hardship, and pain – has become all but completely obscured. Most of us go our entire lives without ever seeing a dead body or witnessing a death. Sixty per cent of all Americans die in hospital and 20 per cent in nursing homes. When death does occur, the body is covered and removed.

"Like so many aspects of American life, [death] has become outsourced and commercialised," says Hannig. "Death has become invisible, because it's sequestered in hospitals and mortuaries, and it's muzzled in everyday conversations."

In centuries past, death was an everyday occurrence, for the young and old alike – and religious cultures strived to remind its citizens that the Grim Reaper could come knocking at any time. In art, jewellery, and architecture, *momento mori* (literally: "remember that you die") served as reminders of our fragility on earth. In paintings, a skeleton might sit in the corner of a portrait; or an hourglass would represent life ticking away. Cadaver tombs were carved with decaying corpses. Mary Queen of Scots – who lost her head at the orders of her cousin and rival Elizabeth I – carried a large silver watch carved in the shape of a skull.

Today we have this "obsession with longevity for longevity's sake", says Hannig. "The end of life has become so medicalised, that death is seen as a failure… We have this unshakeable trust in the power of medicine, so death has become almost this 'enemy' we need to defeat." In the secular world, too, in which people no longer believe in the afterlife, death is not a journey to another world, but a final ending; and that is frightening.

Perhaps, though, radical acceptance and calmer planning might fill us with less fear. There is something to be learned from other cultures, too. For the Hyolmo Buddhists in Nepal, as Hannig writes in an article for *Undark*, "dying is regarded as an intricate art to be learned – a project undertaken with foresight and self-awareness to ensure a smooth passage into the next life as well as a successful rebirth". The Korowai in West Papua, Indonesia, meanwhile, frequently discuss death; elderly men sing about their upcoming demise.

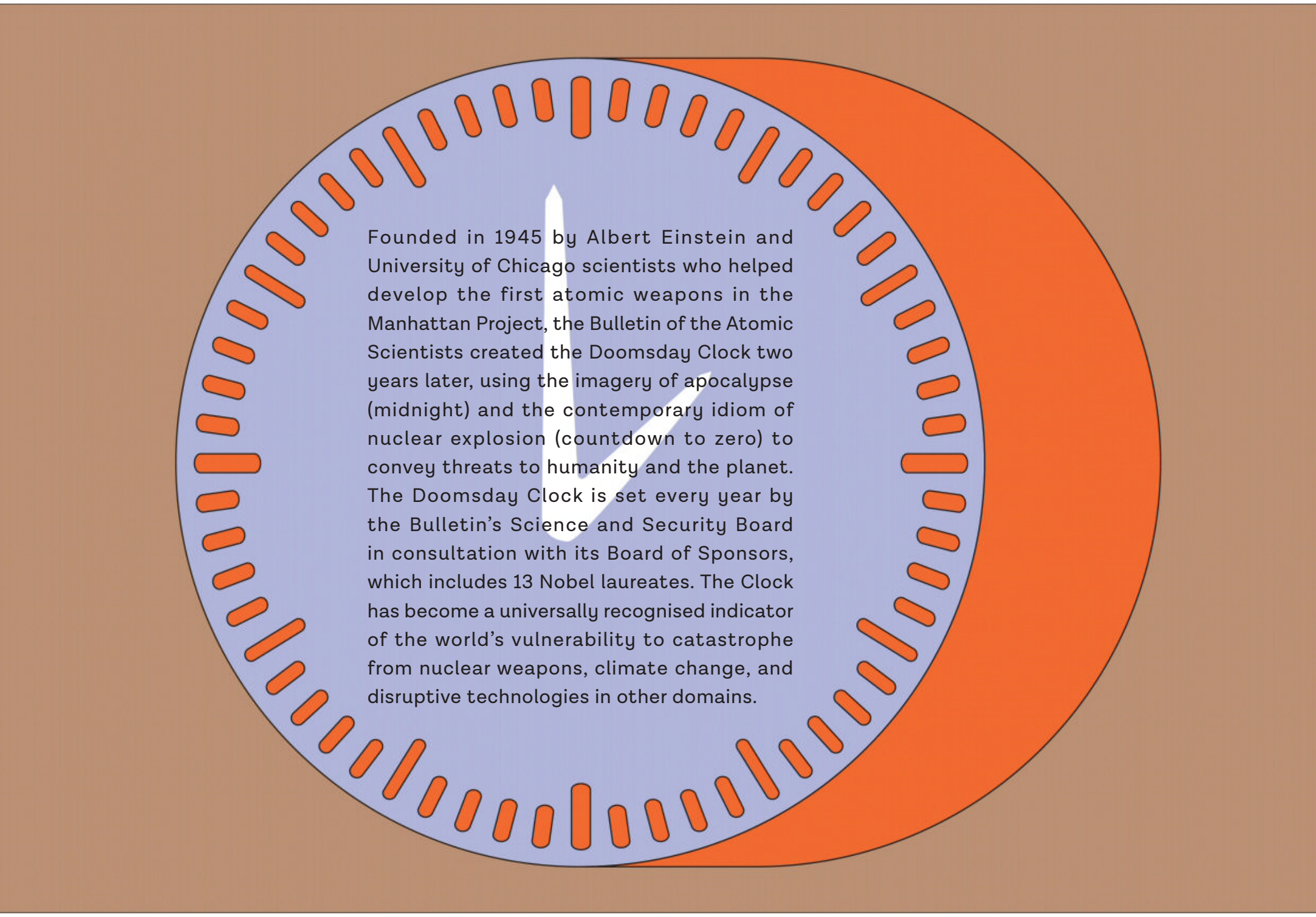
Skeletons need not only be associated with dress-up, decorations, and trick-or-treating. In Capela dos Ossos in Evora, Portugal, the bones and skulls of human remains are stuck in intricate patterns onto the walls and ceilings. Roughly 5,000 corpses were dug up from Evora's medieval cemeteries by Franciscan monks to decorate this 'chapel of bones'. They served a different purpose altogether from that of our now sanitised Halloween: to remind visitors that we will all die, whoever we are, wherever we live, in whatever era, on whatever soil. As a saying carved over the entrance goes: "We bones, lying here bare, await yours."

"Since we can never know anything for sure, it is simply not worth searching for certainty; but it is well worth searching for truth; and we do this chiefly by searching for mistakes, so that we have to correct them."

Sir Karl Popper

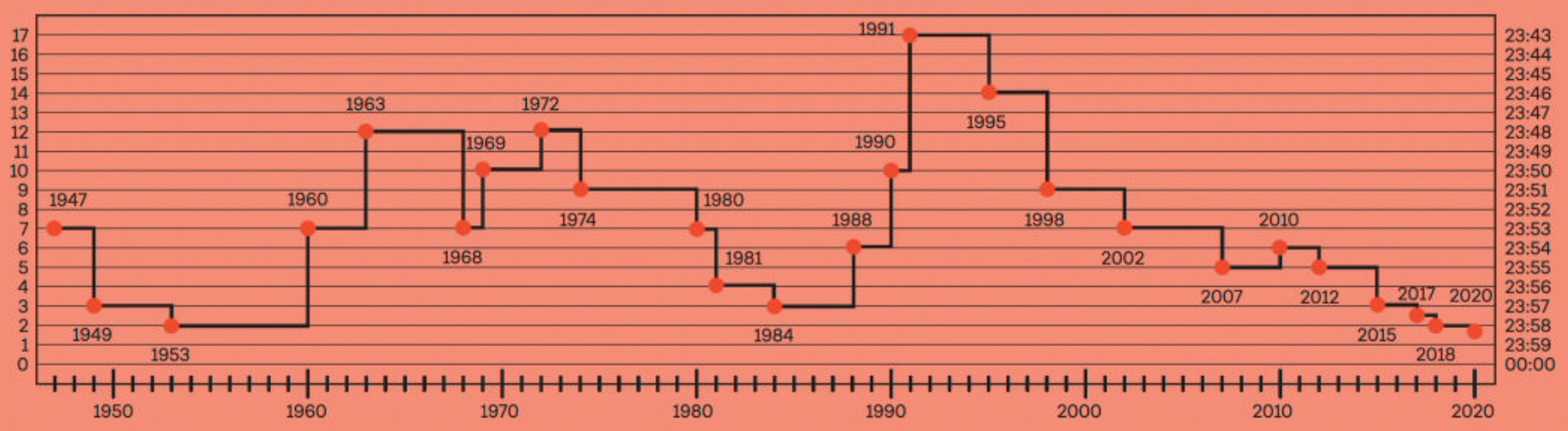
Artwork: *Dux et Comes I*, Edward Wadsworth. 1932

Minutes to midnight



Doomsday clock: minutes to midnight, 1974-2020

Source: Bulletin of Atomic Scientists



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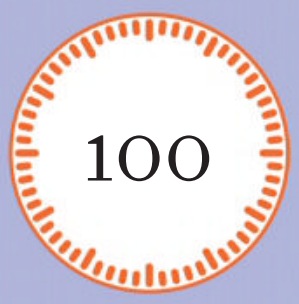

SECONDS
TO MIDNIGHT

2018-19:

MINUTES
TO MIDNIGHT

2017:

MINUTES
TO MIDNIGHT

2015-16:

MINUTES
TO MIDNIGHT

2012-14:

MINUTES
TO MIDNIGHT

2010-11:

MINUTES
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2007-09:

MINUTES
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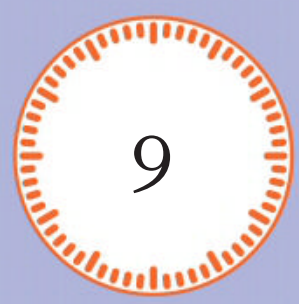
2002-06:

MINUTES
TO MIDNIGHT

1998-2001:



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MINUTES
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1991-94:

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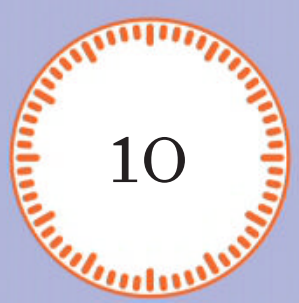
1990:



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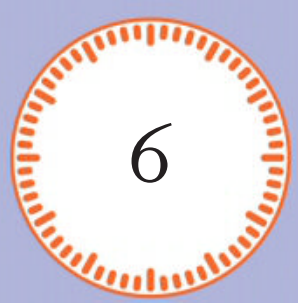

MINUTES
TO MIDNIGHT

1984-87:

MINUTES
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1981-83:

MINUTES
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1980:

MINUTES
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1974-79:

MINUTES
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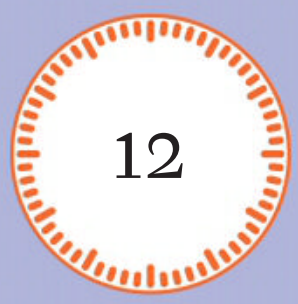
1972-73:



MINUTES
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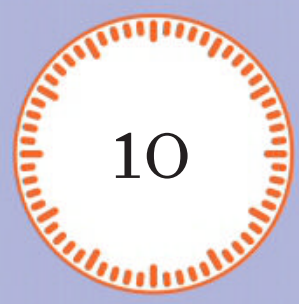

MINUTES
TO MIDNIGHT

1968:

MINUTES
TO MIDNIGHT

1963-67:

MINUTES
TO MIDNIGHT

1960-62:

MINUTES
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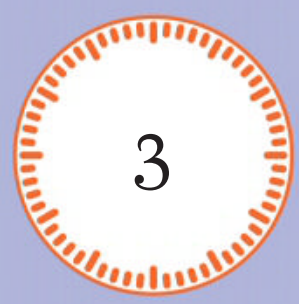
1953-59:

MINUTES
TO MIDNIGHT

1949-52:



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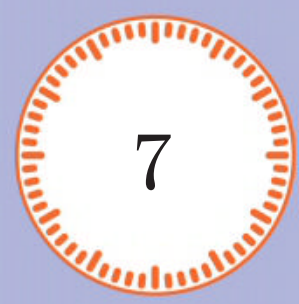


MINUTES
TO MIDNIGHT

by André Dao

A dance floor for divine accidents

"Now, what I want is, Facts." So begins Charles Dickens' *Hard Times*, with the words of Thomas Gradgrind, a semi-retired merchant and would-be politician. Gradgrind, who is also a school reformer, is speaking to the young pupils at his school about his pedagogical philosophy, which amounts to Facts, not Nonsense, where the latter is rather all-inclusive: folk tales, Twinkle Twinkle Little Star, free will, a morality based on something other than self-interest – all Nonsense.

Published in 1854, *Hard Times* was Dickens' biting satire of those who used "figures and averages, and nothing else". Those people were the representatives of what Dickens called "the wickedest and most enormous vice of this time" – namely, the utilitarian philosophy of the late Jeremy Bentham and his followers, which was at that time ascendant in Victorian Britain. Dickens' satire was memorable enough that we now use Gradgrindism to refer to particularly hardline strains of utilitarianism – the kind that might introduce itself, as Thomas Gradgrind does, with a rather pompous monologue in the third person: "With a rule and a pair of scales, and the multiplication table always in his pocket, sir, ready to weigh and measure any parcel of human nature, and tell you exactly what it comes to. It is a mere question of figures, a case of simple arithmetic."

Dickens, of course, does not leave Gradgrind on his high horse (a metaphor that Gradgrind himself would dismiss as Nonsense: a horse is not a moral position but, as one of the model students at Gradgrind's school defines it: "Quadruped. Graminivorous. Forty teeth, namely twenty-four grinders, four eye-teeth, twelve incisive."). Gradgrind is brought low when his son, Tom Jr, is caught thieving from the bank where he works. The father pronounces himself shocked, but the son is ready with a retort: "So many people are employed in situations of trust; so many people, out of so many, will be dishonest. I have heard you talk, a hundred times, of its being a law. How can I help laws?"

Coming from the callous thief – Tom Jr had previously tried to frame a saintly coal miner for the robbery – the reply appears both self-serving and ridiculous. Yet Dickens was parodying a view that was taken seriously at the time – indeed, it dominated intellectual conversations not only in Britain but across 19th century Europe. The catalyst for this heated debate was what the Canadian philosopher of science Ian Hacking calls the "avalanche of printed numbers" – the publication, from around 1820 onwards and for the first time in history,

Illustration by Aida Novoa & Carlos Egan

of statistics about such vital matters as marriages, births, deaths, criminality, and suicide. These numbers revealed something rather startling: that in a given population of a sufficient size, the rates of major life and death events would remain constant.

For some, this regularity of numbers could only mean one thing: that as sure as there are laws of physics determining the course of individual atoms, so too are there social 'laws' determining the course of individuals within a society. According to those 'laws', events as seemingly capricious as murders and suicides are not radically beyond the normal order of things, but rather form part of a probabilistic world. To take the morbid example that fascinated a series of the 19th century's greatest European minds: in such and such a society, X number of people will kill themselves in any given year. Thus, in any given year, a person in that society will have 'X/overall population' chance of committing suicide.

This was an astounding Fact indeed – for it implied that if someone happened to fall within X for that year, that was hardly their (or anyone else's) fault. After all, someone, had to do it. Or, more precisely, X number of someones had to do it. As Dickens' Tom Jr might have put it, in the face of such laws, what could an individual do?

For another literary figure of the era, such thinking was not only ridiculous, but risked, in the name of science, eliminating from humans their very humanity. Fyodor Dostoevsky, in his 1864 notebook, wrote that "since all volition and all reasoning may be tabulated, because the laws of our so-called free will may indeed be discovered, it follows, quite seriously, that some sort of table may be drawn up and that we shall exercise our wills in accordance to that table". But what those "statisticians and experts and lovers of humanity" left out of their tables was the very essence of humanity: "one's own caprice, however wild, one's own fancy, inflamed sometimes to the point of madness". We are not atoms guided by laws but free-wheeling creatures bordering on madness – dancing, in Nietzsche's memorable phrase, on a "dance floor for divine accidents". The stuff then, of 19th century novels.

Indeed, Dickens and Dostoevsky are both towering figures of European literature. If you're looking for complex, representative individuals rather than the abstraction of statistical averages, you could do a lot worse than dipping into *Bleak House* or *The Brothers Karamazov.* Yet for all their undoubted success as novelists – and as savage as their satires of statistical thinking could be – we seem to live in a statistician's world, rather than a humanist's.

Hacking said in his classic book *The Taming of Chance* that probabilistic thinking is the most successful philosophy of recent times. The testament to its success is its ubiquity and its invisibility as a style of reasoning. We mostly don't notice it, but we are constantly couching our decisions in the language of probability. What policy is made today without some calculation of impact? What financial decision is made without some assessment of risk? As Hacking puts it, "probability cannot dictate values, but it now lies at the basis of all reasonable choice made by officials… By covering opinion with a veneer of objectivity, we replace judgement by computation."

How was this victory achieved – and against such formidable literary foes? The secret, counter-intuitively, was to agree with the opponents of statistical laws. Let's not be Gradgrinds, said the new probabilists. Let us concede that statistical laws aren't really laws after all: the regularity of X number of suicides each year doesn't *cause* a single suicide, it forces no one's hand. No law made Tom Jr steal from the bank vault. Laws produce certainty, and all we have here is uncertainty.

So far, so good, Dickens and Dostoevsky might say. If statistical laws are really only 'laws' then that does leave room for humanity, capricious will and all. But this is precisely where the

No law made Tom Jr steal from the bank vault. Laws produce certainty, and all we have here is uncertainty.

new probabilists pulled off their great coup: free from the shackles of determinism and the search for causes, the new probabilists discovered correlation. Certain Facts *correlate* to higher rates of murder and suicide, to lower birth rates or shorter marriages. Now, what if some of those correlative Facts were changed, through government action? Lo and behold: the murder rate, the suicide rate, the birth rate, the divorce rate – they changed too.

Thus was born a whole new species of social control: one which turned uncertainty on its head. Long the bane of rulers, the fact that life and society appear to be random (or stochastic, to use a more technical term) – that the killer's actions stem from no discernible cause – became the basis for a kind of knowledge, and in turn governance, that was agnostic as to causes. As Hacking put it, chance had been tamed. Today, we see this legacy playing out in the work of behaviouralists and so-called government 'nudge' units. There, the descendants of the new probabilists play the odds, designing systems that work on statistically significant biases of populations – nudging us into healthier eating by labelling healthy food green in the canteen, or into saving for retirement through superannuation schemes. Funnily enough, such programs tend to re-run old debates about free will, but with a twist. More freedom than ever for the governors, but determinism for the governed, who will, on the whole, regress to the mean.

"What made you decide to write a book on determinism?"

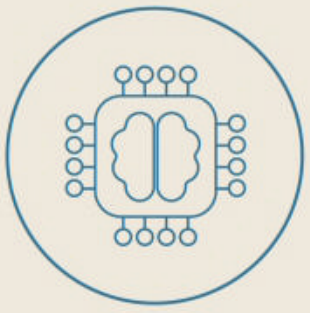
AN UNCERTAIN FUTURE WITH AI

Uncertainty is seen in emerging public views of AI and human enhancement applications. For instance, 42% of Americans are not sure how the widespread use of robotic exoskeletons in manual labour jobs would impact society. Similarly, 39% say they are not sure about the potential implications for society if gene editing is widely used to change the DNA of embryos to greatly reduce a baby's risk of developing serious diseases or health conditions over their lifetime.

Source: Pew Research Center

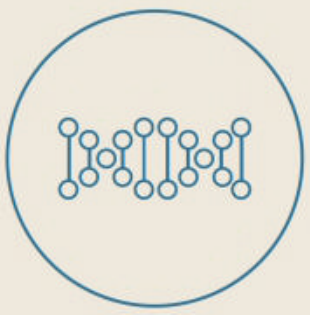
87%:

of Americans say that higher standards for testing driverless cars should be in place.



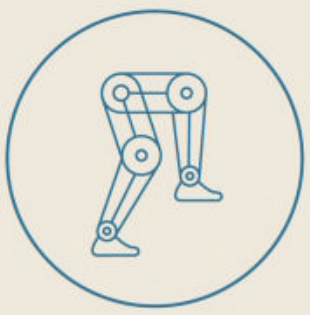
83%:

believe the testing of brain chip implants should meet a higher standard than is currently in use to test medical devices.



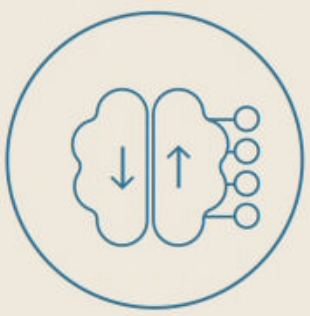
80%:

say that the testing regime for gene editing to greatly reduce a baby's risk of serious diseases should be higher.



72%:

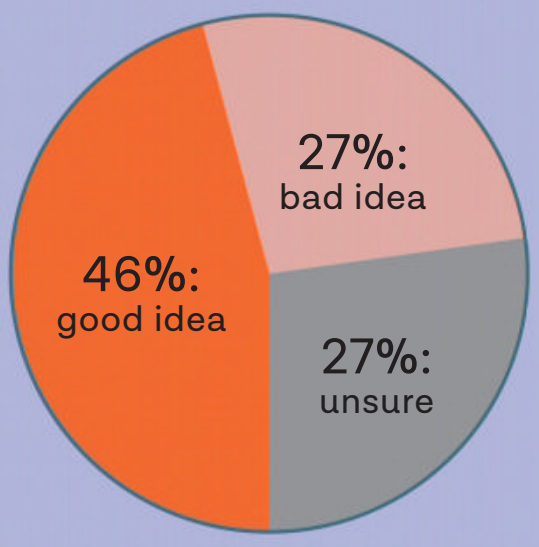
think the testing of robotic exoskeletons for manual labour should use higher standards.

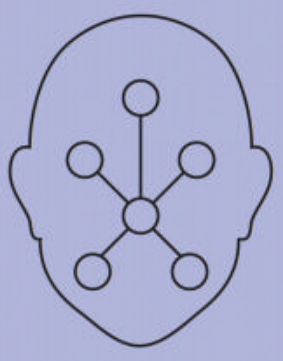


57%:

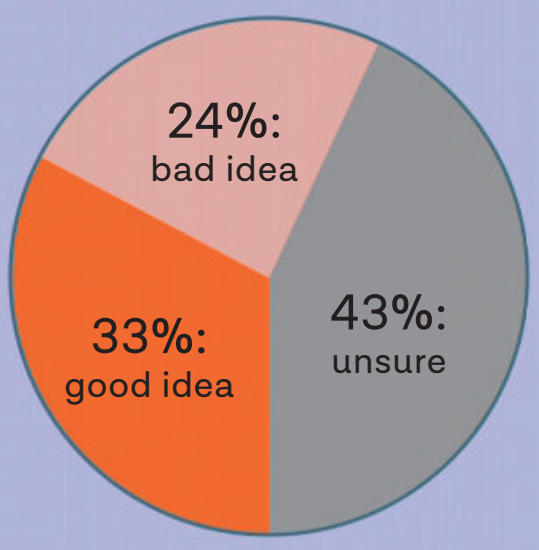
say the widespread use of brain chips for enhanced cognitive function would increase the gap between higher- and lower-income Americans.

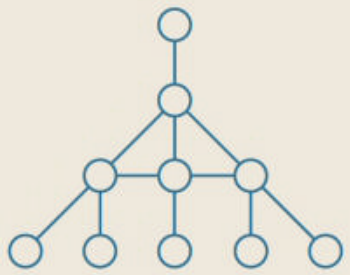
% of US adults who think the use of facial recognition technology by police to monitor crowds and look for people who may have committed a crime is a good or bad idea:



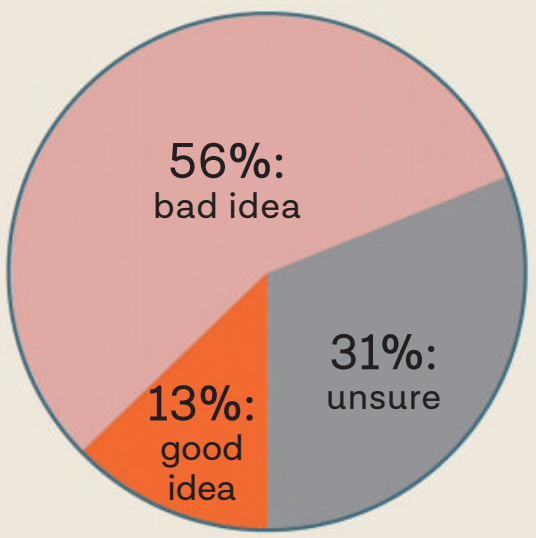


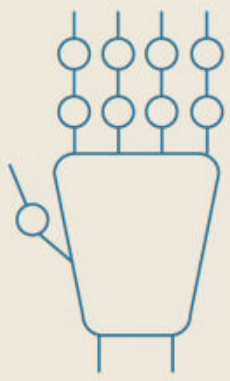
% of US adults who think the use of robotic exoskeletons with a built-in AI system to increase strength for manual labour jobs is a good or bad idea:



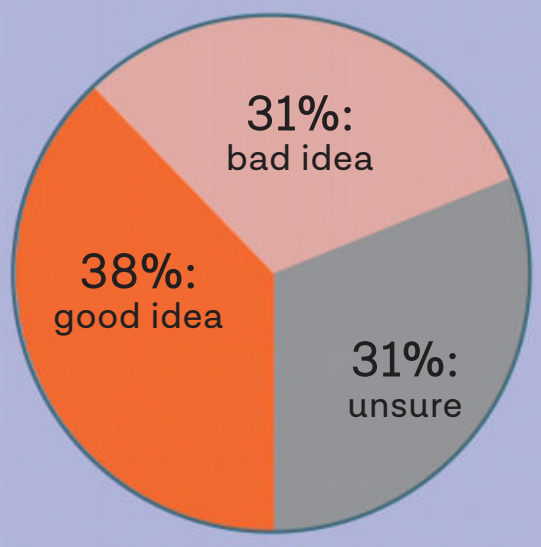


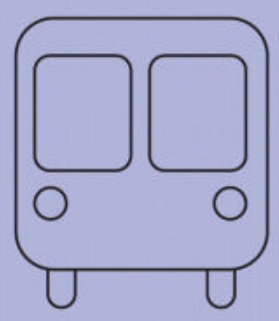
% of US adults who think the widespread use of computer chip implants in the brain to allow people to far more quickly and accurately process information is a good or bad idea:



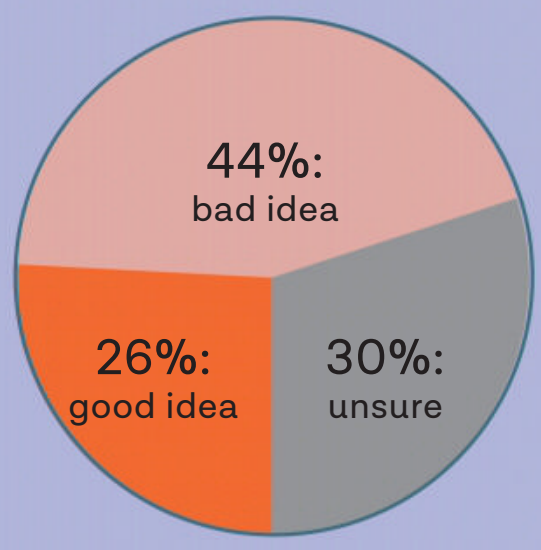


% of US adults who think the use of computer algorithms by social media companies to find false information on their sites is a good or bad idea:





% of US adults who think that autonomous passenger vehicles in widespread use is a good or bad idea:



by Francine Russo,
Scientific American

Uncertainty intolerance

Anxiety disorders are the most common type of mental disorders in the US. But a simple diagnosis of generalised anxiety disorder may not capture the specifics of what bothers some people. A subgroup of anxiety sufferers may experience extreme distress when confronting the inherent uncertainties that turn up in any life circumstance, whether helping a relative with a severe illness or choosing a new route to work that is safe and easy to navigate.

Such people may respond to their feelings by leaving nothing to chance. At work, they appear polished and prepared when giving a presentation because they consider every question that could be posed by colleagues beforehand and memorise possible answers in the days leading up to a meeting. Still, all of the prep work may do little to relieve feelings of painful anxiety.

Psychologists, though, are learning new ways to treat these sufferers, who can be identified through their answers to a psychological measurement known as the intolerance of uncertainty scale (IUS). Patients can then benefit from specialised treatment.

These patients are typical of the ones psychologist Keith Bredemeier sees at the University of Pennsylvania's Center for the Treatment and Study of Anxiety. When encountering patients with a high score on the IUS, Bredemeier may explicitly target enhancing their comfort levels with life's unpredictability as a treatment goal. Since therapy is adapted to patients' salient personality traits, treatment for uncertainty distress may be added to their treatment plans just as treatments for eating disorders, for example, may focus on clients' perfectionism. And therapy for obsessive-compulsive disorder may concentrate on patients' inflated sense of responsibility.

Difficulty dealing with uncertainty was first identified in the early 1990s as a distinguishable personality trait. It is one we all have to some degree, but many of us have it at an elevated level. Like perfectionism or rigidity, it is really a predisposition, not a diagnosable disorder. People who have high levels of perfectionism, for example, are at greater risk of developing anorexia. And someone with a high level of intolerance of uncertainty (IU) may end up developing an anxiety disorder.

To cope, a person may live along a narrow, predictable track. Everything may seem fine to them on the surface, but their risk for developing anxiety, depression or emotional distress is higher – especially if something such as, say, a pandemic wreaks havoc on a carefully planned life.

In the decades since the IUS was developed, researchers have found it useful for understanding why some

Illustration by Aida Novoa & Carlos Egan

patients are anxious, devising targeted therapies and assessing a patient's progress. In psychology labs, investigators have probed how uncertainty affects people's brain and behaviour. Clinicians have used the concept to develop programs that teach parents of autistic children strategies to help their children better tolerate uncertainty. And tourism specialists have delved into which clients might prefer guided tours instead of unguided ones. Most of all, because the pandemic blew away a sense of predictability in everyday life, the concept has sparked an explosion of research globally to find out how those who typically feel this type of distress are faring in these uncertain times and what might be done to help them.

In the 1990s Mark Freeston and Michel Dugas were part of a team at Laval University in Quebec looking into generalised anxiety disorder. The focus in the field was on worry. The thinking at the time "was that worry was some form of problem-solving gone wrong or a problem-solving deficit," notes Freeston, who is now at Newcastle University in England. But Freeston and his colleagues saw something different in their patients' behaviour. "They knew how to solve problems, but something was getting in the way," he says. The team hypothesised that anxious people might have a negative reaction to uncertainty. Following a "clinical hunch," he says, the group developed the IUS and tested it out.

When filling out this questionnaire, people indicated how they might react to statements such as "Unforeseen events upset me greatly" and "It frustrates me not having all the information I need." The scale accurately assessed subjects' reactions – cognitive, emotional, and behavioural – to uncertain situations in which they felt they did not know vital information or could not control the future.

For humans and other animals, fear of uncertainty is a default state, says health psychologist Jos F. Brosschot of Leiden University in the Netherlands. Carry your cat into an unfamiliar empty room, he suggests. No fierce dog or hostile human is there, but there is also no obvious place to hide. Without a safe spot, the animal might be apt to claw at your chest in terror. "In the wild," Brosschot and his colleagues wrote in a 2018 paper, "organisms have survived not by waiting for more evidence of threat but instead by erring on the side of caution… Those who fled at the first sign of unsafety continued to live and pass their genes."

From birth, humans have to carefully learn the cues that signal safety. For a toddler, safety is wherever their caregiver is. Over time, if we grow up in a secure environment with supportive parents,

In psychology labs, investigators have probed how uncertainty affects people's brain and behaviour.

we learn to feel safe when there is no obvious danger. We become better able to tolerate uncertainty. But for those of us who grew up abused, neglected, or otherwise deprived of feeling safe, uncertainty is threatening.

At the University of Reading in England, psychologist Jayne Morriss has subjected volunteers in her laboratory to unpleasant experiences that they may or may not be able to predict, from a mild electric shock to a terrifying scream. Before doing so, she already assessed them with the usual psychological tests, including the IUS. She also ascertained their thresholds for receiving a mild electric shock. Then, as they sat in front of a computer that told them whether, when or if they would receive a shock (or hear a scream), she measured their physiological responses with a magnetic resonance imaging scanner, skin conductance sensors or registers of startle response, depending on the study. Invariably, people high in IU show greater physiological distress.

One of her studies, published last year and entitled, in part, "I Told You It Was Safe," produced intriguing results. The subjects were told that they would receive a shock at a certain time, at an unknown time, or not at all – or, more emphatically, that they definitely would not get shocked. The surprising result is that those with high IU scores showed the greatest physiological distress during the certain-to-be-safe period.

This seems counterintuitive, yet it is easy to explain. Those of us who grew up with unpredictable danger or inconsistent safety are ever vigilant for harm. "People consistently high in IU have trouble recognising when they are safe," Freeston says. "Just telling them it's safe isn't enough."

In fact, Morriss has shown in other studies that subjects with high IU need a greater number of exposures to a now safe scenario that was previously unsafe before they have no distress response. In other words, they have more difficulty updating their perception of safety.

As intolerance of uncertainty has begun to be studied as a separate trait from a tendency to worry, psychologists have identified typical behaviours – often unconstructive ones – that people use to tamp down their distress at not knowing. They overprepare, perhaps memorising answers to all the possible questions anticipated for an upcoming presentation. Some collect every bit of information they can find online and offline. They constantly seek reassurance from others and look for media sources they can trust. Or they avoid situations with unknown outcomes. If they do not know how they will do on a test, they may avoid looking at their textbooks or procrastinate. They may impulsively make a choice, even a bad one, in order to resolve their uncertainty about something. They may also decide that they know an answer and close their minds to new information.

Bredemeier adapts treatments for these patients at his clinic. To help a client feel more comfortable with

For those of us who grew up abused, neglected or otherwise deprived of feeling safe, uncertainty is threatening.

Therapists have begun to target their patients' uncertainty intolerance to help them through the pandemic.

uncertainty, he first has them list situations that feel mildly to intensely uncomfortable. Let's say someone dislikes ordering from a new restaurant without first reading reviews or avoids taking a different route to work. The worst situation of all would be delegating an important task at work. The person just cannot do it.

In response, the therapist asks a client to do each task, starting with the easiest one first. A feeling of uncertainty grows, but the person stays aware of thoughts and feelings and reflects on the outcome. Maybe the client wishes they had ordered the usual takeout. Maybe they have found a favourite new restaurant. The outcome is less important than becoming more comfortable with some unpredictability. After a series of these experiments, the client is likely to score lower on IU and eventually reduce inflexible behaviours.

When the pandemic hit, none of us knew from day to day what would happen. No one knew how infections – or the threat of them – would affect work, school, or travel. Nobody liked the unpredictability. Dozens of studies worldwide showed that people high in IU were at greatest risk of emotional problems.

A study published in January 2022 is especially noteworthy. Two years before COVID emerged, a team at the University of Illinois at Chicago used two methods to measure volunteers' uncertainty intolerance during predictable and unpredictable shocks. The researchers administered the IUS self-report questionnaire before the experiment. And during it, they monitored subjects' neural response with functional magnetic resonance imaging.

Two years later, at the height of the pandemic, the researchers asked these same volunteers detailed questions about their emotional reactions to the crisis. Having higher self-reported IU or greater activity in a brain area called the anterior insula during imaging independently predicted an increase in anxiety, depression, or emotional distress.

Therapists have begun to target their patients' uncertainty intolerance to help them through the pandemic because COVID's unpredictability exacerbates classic IU behaviours in some people. Some read everything they can find on the Internet, wash their hands incessantly and rarely leave home. Others refuse to alter their routines even if they are putting themselves or others at risk. There are endless permutations of dealing with this continuing state of just not knowing.

As Bredemeier says, "All of us like to feel certain. It's just a question of how much distress it causes us. Intolerance of uncertainty is a matter of degree."

“It is the customary fate of new truths to begin as heresies and to end as superstitions.”

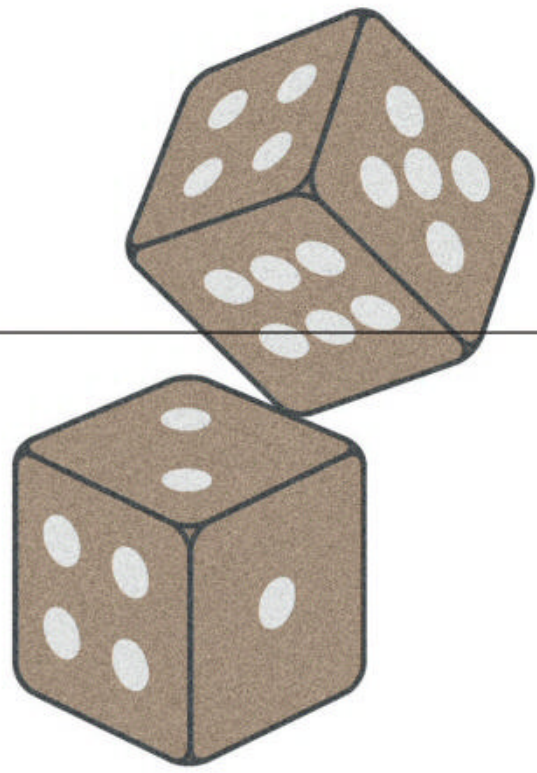
Thomas Henry Huxley

Artwork: *Senecio*, Paul Klee, 1922

Illustration by Aida Novoa & Carlos Egan

by Tom Chatfield

Landscape of uncertainty



Noughts and crosses – also known as tic-tac-toe – isn't a very interesting game. To play it, you draw a three-by-three grid, then take it in turns with another player to fill in a square with your mark: **X** or **O**. The winner is the first person to get three in a row. If you know what you're doing, however, this should never happen. So long as nobody makes a mistake, the result will always be a draw.

I know this because noughts and crosses is a 'solved game': one whose outcome can correctly be predicted at every stage, so long as both players play perfectly. It's simple enough that most children can do so by the age of six. Indeed, the lesson that it can't be won is part of the point. By contrast, achieving perfect play in another solved game – draughts, also known as checkers – is so complex that it's beyond any human. It's easy enough for computers, however. As was first shown in 2007, by a team lead by the Canadian computer scientist Jonathan Schaeffer, a draw is guaranteed if both players play perfectly.

A solved game is one denuded of *epistemic uncertainty*. As the word epistemic (which means 'related to the nature and scope of knowledge') suggests, this describes uncertainty about things that you *could* know in principle, but don't know in practice. It has always been the case that a best move exists, in principle, for every situation in checkers. But until recently it was impossible to know what this move was. Now it's easy to find out. By contrast, *aleatory uncertainty* describes things that you *cannot* be certain about, even in principle, like the result of rolling some dice (aleatory uncertainty takes its name from the Latin for dice, *alea*). In checkers, the end result of every possible game can be calculated with absolute certainty, because these results are a logical consequence of the rules. By contrast, the mechanics governing the result of a dice roll are not deterministic. Over time, statistically predictable patterns emerge from them. But this doesn't mean that even the fastest computer can predict any particular result.

What does it mean to remove uncertainty from a game? In one sense, it stops it from being a game at all. Games have been described in countless ways over the centuries, but perhaps my favourite definition captures the importance of uncertainty to them. It comes from the philosopher Bernard Suits who, in his 1978 book *The Grasshopper*, describes playing a game as a "voluntary attempt to overcome unnecessary obstacles". The words *voluntary* and *attempt* are vital here. A game in the sense that interests Suits isn't just a set of rules or possibilities. It's also an experience entered into for the purpose of pleasure: one in which we try (and may fail) to overcome an arbitrary test of skill or luck. Take away the uncertainty and you hollow out the

satisfactions both of trying and of succeeding. You're left with little more than a demonstration: evidence that a known proposition does indeed play out in a particular way.

One crucial ingredient in the predictability of a solved game is the requirement that each side plays perfectly. This brings with it the further requirement that each side has access to a perfect strategy and chooses to use it – something that shouldn't be taken for granted when it comes to people. When I play checkers with my eight-year-old son, we don't consult a computer to achieve perfect play because (*pace* Suits) this would turn a fun contest into mindless mathematics. When I play noughts and crosses with my four-year-old niece, I let her win, because repeatedly beating small children at a simple game isn't my idea of fun.

Restoring uncertainty to these games goes hand-in-hand with the insight that winning need not be their sole purpose. Indeed, so far as human players are concerned, the 'point' of any game invariably entails external as well as internal elements. Unlike machines – which can no more derive satisfaction from winning than they can know they're playing a game in the first place – we are constantly conferring meaning upon our experiences based on external factors and values. Winning means nothing if there is nothing but the game.

All this comes into intriguing focus when you consider the field summarised by Nobel laureate Robert Aumann as "a mathematical formalisation of interactive decision-making", otherwise known as game theory. Game theory is interested in what strategies people *ought* to follow in a given scenario to ensure they gain as much as possible from it. This means that it's also about what happens when you assume that everyone else is calibrating their self-interest against yours – and thus that self-interest is the only thing you should treat as certain.

Two provably 'perfect' individual choices in this Nash Equilibrium result in a collective loss.

Perhaps the most famous such scenario is known as the Prisoner's Dilemma. This sees two players assigned the roles of prisoners facing interrogation. Each is offered the following deal. If you confess to a crime while your accomplice remains silent, you'll be set free while they get a long sentence. But if they confess while you stay silent, they'll be set free while you get a long sentence. If you both confess, you'll both get a moderate sentence. And if you both stay silent, you'll both get a very short sentence.

Here's the dilemma. The safest option is to confess, because this removes the worst possible result: the possibility of a long sentence based on you staying silent while your counterpart confesses. Confessing, however, also removes the best possible mutual outcome: silence on both sides. If only you could read each other's minds, this would be the ideal outcome! But because you cannot know what the other player is doing, confession is the only optimal strategy.

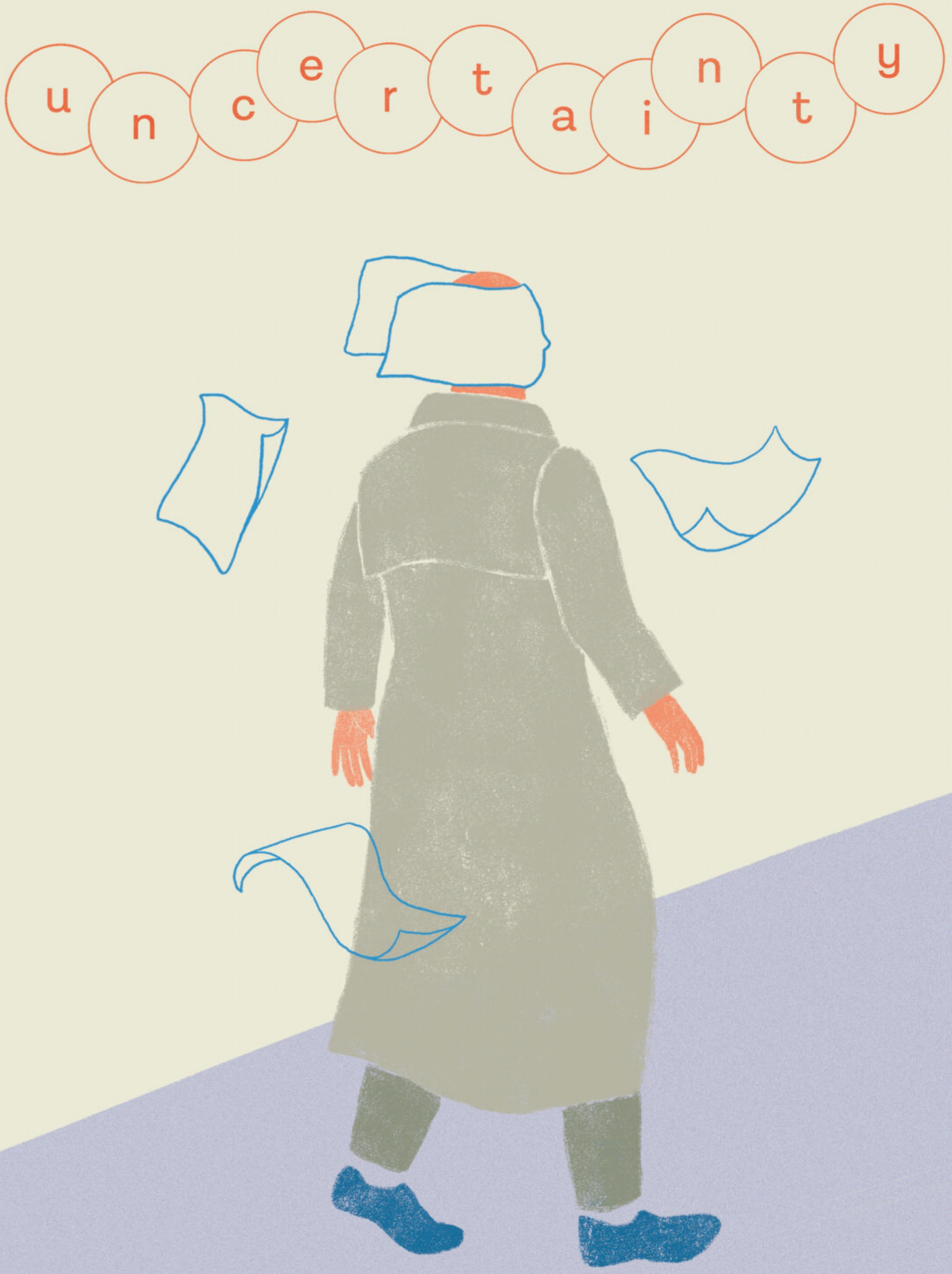
So far as game theory is concerned, saying "both players should always confess" creates a Nash Equilibrium – a state named after its discoverer, the mathematician John Nash, which describes any set of strategies able to ensure that no player regrets their strategy at the end of the game. Yet this particular Nash Equilibrium also enacts a minor tragedy. Two provably 'perfect' individual choices result in a collective loss. By taking self-interest for granted, mutual suffering is guaranteed.

What should we take from this? For me, it emphasises the importance of perspectives like Suits's when it comes to games and negotiations alike. Self-interest may be supreme so far as computation is concerned. But it's only by stepping away from the board that you can start to make sense of things: to ask *why* you're playing; *what* you really want; and *how* an unspoken agreement or a few honest words might alter the entire landscape of uncertainty.

“It is said to await certainty is to await eternity.”

Jonas Salk

Artwork: *Uncertainty*, by Arthur Hughes, 1878


uncertainty

UNCERTAINTY

/ʌnˈsəːtnti/

noun:

1. The state of being uncertain (‘times of uncertainty and danger’);

2. lack of certainty or doubt;

3. [usually uncertainties] Something that is uncertain or that causes one to feel uncertain (‘financial uncertainties’).

Origin:

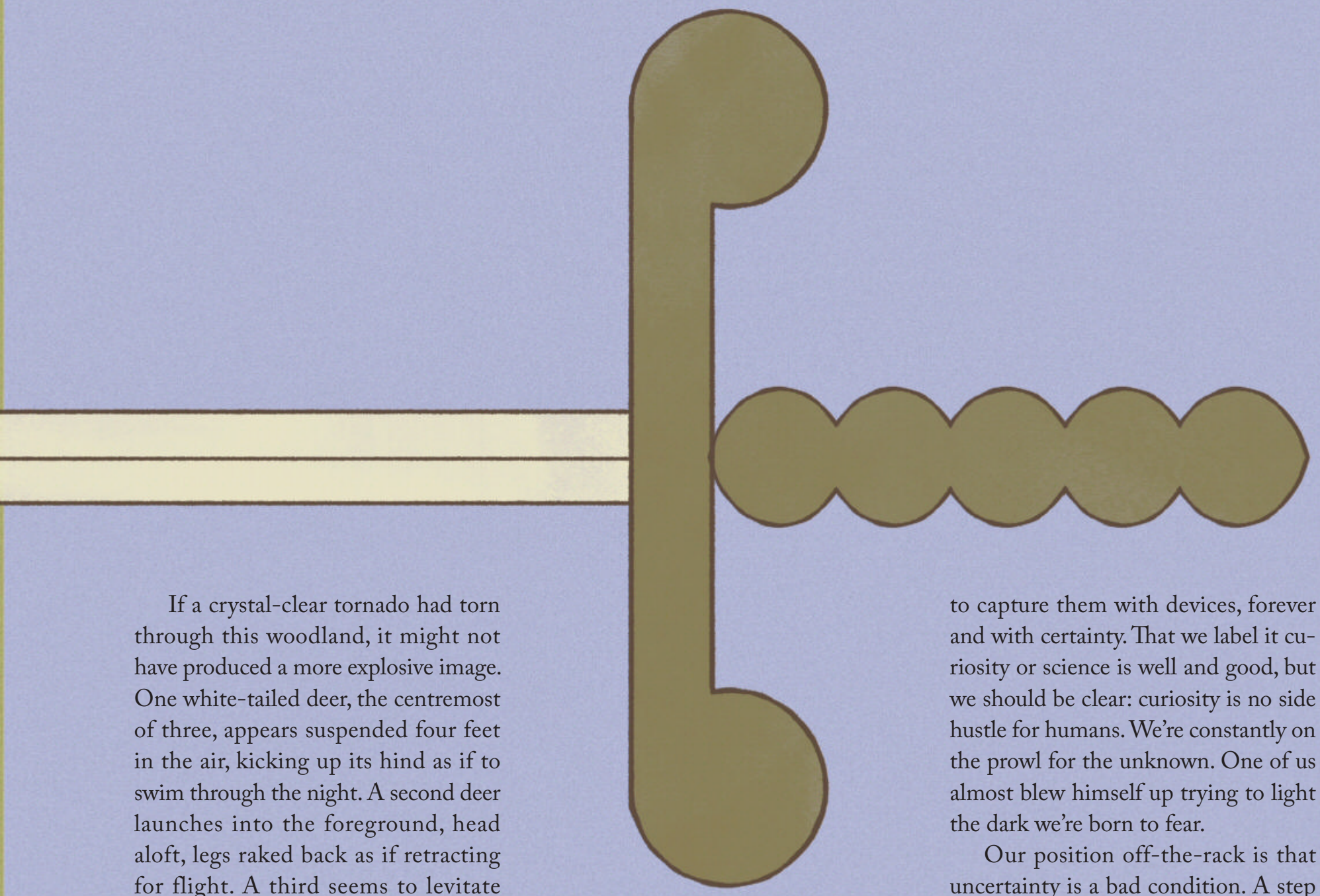
Middle English ‘*uncerteynte*’- c. 1300, “of indeterminate time or occurrence,” from *un-* “not” + *certain* (adj.). Meaning “not fully confident” is recorded from late 14c.

Source: Oxford English Dictionary and Collins Dictionary

The correct use of beasts

Illustration by Aida Novoa & Carlos Egan

By DBC Pierre



If a crystal-clear tornado had torn through this woodland, it might not have produced a more explosive image. One white-tailed deer, the centremost of three, appears suspended four feet in the air, kicking up its hind as if to swim through the night. A second deer launches into the foreground, head aloft, legs raked back as if retracting for flight. A third seems to levitate nose-down, perhaps after a mighty leap, or recoiling on all fours at once. This picture from the black of night at Whitefish Lake, Michigan, tells of an invisible blast.

But the blast hadn't been invisible: it was a flash from history's first remote camera-trap, tripped by the deer meeting a line of dark silk attached to the camera.

The photograph's precise date isn't known, but what's known is that in 1891, its creator, well-to-do lawyer, politician, and outdoorsman George Shiras, almost blew himself up with spirit-burning lamps and magnesium powder in this quest to finally capture the unobservable. That grandfather of the camera trap isn't a subject for us here, having tinkered for the usual human reasons. A subject for us to consider is what those reasons have become. The driving force behind the curiosity in the deer must surely have only been this: that we just couldn't be sure what went on outside in the dark. The dark is notoriously private; animals are notoriously private; ambulant things in the world just don't like being seen. We can guess what goes on and there are glimpses and clues; but we're persistently uncertain about the realpolitik of our environment. The planet's creatures tend to keep to themselves, and if we tame and befriend them we only defeat the nature we wanted to observe; we *change* them. The key was to observe them unawares and, ever since Shiras, to capture them with devices, forever and with certainty. That we label it curiosity or science is well and good, but we should be clear: curiosity is no side hustle for humans. We're constantly on the prowl for the unknown. One of us almost blew himself up trying to light the dark we're born to fear.

Our position off-the-rack is that uncertainty is a bad condition. A step away from the worst condition, equalling chaos, gateway to collapse. The dark is also a bad condition because it's supposedly full of beasts; but that only makes it a subtype of uncertainty, as the real issue is not knowing when the beasts come around. These images, then – deer, dark, beasts – are our setting for today, before we go on to pursue an even more disturbing situation than uncertainty. Since the upshot here is that the uncertainty we speak of arises outside ourselves, in the dark of our environment. It's not a part of us and we're even quite immune these days, having killed most of the beasts, and as for deer, no longer caring where they go provided it's not in front of our car; while inside our modern selves the arguably lesser quality of mental uncertainty is just a

gap between decisions, since we have the freedom to make the decisions and cure the uncertainty. Unlike the animals we have *free will*.

Then comes philosophy's two-headed beast: in the centuries before and after Shiras startled those deer, the old chestnut arguments of free will versus determinism crackled away in the background. For any of us who missed the episode, those arguments broadly proposed that we were either masters of our fates or else caught up in predictable cause and effect with our environments. And while we were out killing beasts, new ideas were budding in the determinist camp. Within six years of the flash-trap, Edward Thorndike advanced his *Law of Effect*, a notion of behavioural conditioning which stated: "Responses that produce a satisfying effect in a particular situation become more likely to happen again in that situation…" One of his experiments had a cat locked in a box to show that through satisfying and unsatisfying actions it would learn to escape. But note: Thorndike was a psychologist. His law was meant for humans. This was *behaviourism* blooming.

Human thoughts and feelings were no longer keys to ourselves as creatures. Barely two years before George Shiras mastered his flash, physicist Max Planck had joined the University of Berlin where he was impressed by a student named Max Meyer. Meyer went on to publish *The Fundamental Laws of Human Behaviour* a decade later, and twenty-six years after that, on March 6, 1937, the *Science News Letter* reported Planck's new stance on free will: "…that the laws of physics are iron and immutable in the outside world of observable phenomena, where no kind of will is exercised at all".

Forget our woodland, the crystal-clear tornado was gathering power around an idea: that freedom in humans is an illusion. That we are every bit as conditionable and conditioned as the beasts. And it happened that these new proponents, great thinkers no doubt every one of them, were mostly psychologists; a discipline that was scarcely thirty-seven years old at the time George Shiras set about mastering spirit lamps and magnesium.

Most seemed to have drawn their ideas from animals. Take psychologist John Broadus Watson, whose dissertation *Animal Education* earned him his PhD. He had been working with rats before going on to publish his behaviourist manifesto, the goal of which, he wrote, was: "the prediction and control of behaviour". "The behaviourist," he went on, "recognises no dividing line between man and brute." By 1948, when psychologist Burrhus Frederic Skinner published his novel *Walden Two,* it may have been clear to some that this tornado of an idea was bound

Forget our woodland, the crystal-clear tornado was gathering power around an idea: that freedom in humans is an illusion.

to make landfall on the minutes of our lives. *Walden Two* described a utopia where characters rejected the idea of free will, thriving instead under a program of behavioural engineering that started at birth. And from at least this point forward another element was being alloyed to the idea: that technology would be the means of control.

Cut to Monday May 1, 2017, and news was breaking that a social media company had told advertisers it could identify teens when they were feeling down. According to leaked internal documents, the corporation monitored users in real time to identify when teens were feeling 'stressed', 'defeated', 'overwhelmed', 'anxious', 'nervous', 'stupid', 'silly' and 'useless'. The same corporation had earlier been found to be experimenting with large numbers of users by seeking to alter their feelings without their consent.

After a hundred and thirty-one years in a straight rising line from an innocently conceived act of curiosity at Whitefish Lake, I ask: who are now the deer in focus?

And what becomes of our selves when the visible flashes are gone?

More concerning than the engineering so clearly underway is the engineering that allowed us to see this news and let it pass. The engineering we can see is not the one that should most dismay us; but rather the engineering that so successfully erased our dismay. As common thinkers or new philosophers we don't have to buy it. The notions are shaky and defeatable. Moreover, while results may be objective they rely on covert operations; the same is true of a successful poisoning. This lab-rat project went ahead without consent, making any evaluation of results immaterial. Since who can decide the engineering of a species?

This may be the greatest moral and legal question we will ever face. Surely the foremost right of a mind is to choose the way it will think. And if it chooses free will and free will is an illusion, then the greatest act of that mind is to choose its own illusion; under which idea the self-created license to control others covertly is an equal or greater illusion.

Plus let's pull rank: mainstream branches of both philosophy and physics admit the possibility of all reality being an illusion. Pop goes the notion's weasel right there.

As for uncertainty, here's a punchline: an idea's context and time lend it their weight. In this case one more rising line flew past Whitefish Lake into today.

The market.

Our behavioural engineers don't intend to build us a certain life.

They intend to make us a *certain market*.

From that outcome's point of view, uncertainty is suddenly the good old days.

We should cherish it.

And perhaps we can use it. Short-circuit their child-like world.

Where all we have to do to subvert is post about our pet giraffes. N

The corporation monitored users in real time to identify when teens were feeling 'stressed', 'defeated', 'overwhelmed', 'anxious', 'nervous', 'stupid', 'silly' and 'useless'.

NewPhilosopher

New Zealand soldiers heading off to World War I, Archives NZ

Taking risks

Interviewee: Sir David Spiegelhalter
Interviewer: Zan Boag

Sir David Spiegelhalter has been Winton Professor of the Public Understanding of Risk at the University of Cambridge since October 2007. His background is in medical statistics, with an emphasis on Bayesian methods: his MRC team developed the BUGS software which has become the primary platform for applying modern Bayesian analysis using simulation technology. He has worked on clinical trials and drug safety and consulted and taught in a number of pharmaceutical companies, and also collaborates on developing methods for health technology assessment applicable to organisations such as NICE. His interest in performance monitoring led to his being asked to lead the statistical team in the Bristol Royal Infirmary Inquiry, and he also gave evidence to the Shipman Inquiry. In his post he leads a small team which attempts to improve the way in which the quantitative aspects of risk and uncertainty are discussed in society. He works closely with the Millennium Mathematics Project in trying to bring risk and uncertainty into education, gives many presentations to schools and others, advises organisations on risk communication, and is a regular newspaper columnist on current risk issues. Sir Spiegelhalter is Chairman at The Winton Centre for Risk and Evidence Communication at the University of Cambridge, a Fellow at Churchill College Cambridge, and Associate Fellow of the Centre for Science and Policy. He was elected FRS in 2005 and awarded an OBE in 2006 for services to medical statistics. He received a knighthood in the Queen's Birthday Honours List in 2014 for services to statistics.

Zan Boag: You said that one regret in your life that you hadn't taken enough risks, that you'd been too cautious in your career and in your travels. You say, "I wish I'd done more adventurous things." What part do you think that your job played in being cautious and not taking more risks? Or is this simply a regret that most of us are likely to have as we get older?

Sir David Spiegelhalter: I don't think my job has hugely influenced my attitude to risk, it's much more that my personality has led me to my particular job, and it's also led me to my attitude to life. So they're correlated but only because I think they've got a common cause. Correlation is not causation as we all well know. And the common cause is me.

Do you think there's no possibility for change? We're stuck with ourselves, are we?

No, no, not completely. And, on second thoughts, I think maybe my work with risk has to some extent influenced how I approach my life, as I do try to get a feeling of magnitudes, particularly when faced with a situation

Photo: Sir David Spiegelhalter

which asks for 'thinking slow'. But in the end, how we feel about risk and threats is, I think, inevitably influenced by basic attitudes to caution and boldness. And I've tried to encourage my boldness by swallowing hard and going for it, jumping off high ledges into water and so on, sometimes walking on stage in front of a thousand people. You just have to do it even though your body is in a sense telling you not to.

But the more important thing is not, I think, your attitude when you are directly faced with a risk, it's whether you put yourself in that situation to start with, and whether you open yourself up to the opportunity of experience. And I think as I've got older, I realise I've not always done that in a way that I wish I had.

We're going to come back to a couple of different ideas about uncertainty, but I'd like to start with a definition of what you think uncertainty means.

This is tricky because risk and uncertainty can mean almost anything. For me, uncertainty is absolutely any situation that isn't completely specified and known and determined, which essentially means every aspect of life.

That's uncertainty; what about risk?

I don't like defining risk because it can be used in so many different ways, and who am I to say what it means? For me it's when you're faced with a situation where things may go well and may go badly, and so you're taking a risk. So it's the way that unavoidable uncertainty in the future plays out.

As you said earlier, everything is uncertain when it comes to the future. We think we have a high degree of control over what's going to happen…

And it's not just the future of course, we're hugely uncertain about what's going on at the moment, what's gone on in the past, and about how the world works. In fact, I'm less interested in uncertainty about the future than I am in uncertainty about the state of the world as it is and as it has been.

Yes, indeed. You mentioned in an interview that people overestimate the amount of control that they have in their lives. Do you think this is something that's quite common for all of us?

Well, I'm not a psychologist but from my colleagues and the reading

I've done, and just my personal experience, I feel people do overestimate the ability to control what happens. They search for meaning, they search for causes, they search for understanding about why things happen. I do think that it's actually quite difficult to grasp the fact that much of the time, maybe most of the time, we can't understand why things occur. They just happen.

That's certainly something that's come to the fore over the last few years. During this time, you and Anthony Masters drew on data to write a weekly Observer column that invariably centred on COVID statistics. What sort of response did you get from the general public about your columns?

Oh, enormously positive. And not just our columns, but throughout the two years of the pandemic I and other statisticians have seen it as their job to try to explain things to people. Not to say what should be done, not to say what policy is right or wrong, not to blame anybody for what they did, and not to predict what's going to happen. By avoiding all those things which so many other commentators were obsessed with, there was a very strong role just for trying to understand and explain what's going on. And this was deeply appreciated by audiences.

The media took some time to catch up on it. When I started doing media work on the pandemic it was just utterly predictable that I'd be asked, "What should we do? Who's to blame? What's going to happen?" And I'd say, "I'm not going to answer any of those questions, it's not my job." But I did try to explain what was going on, about the spread of the virus, positive tests, COVID deaths, trends in the data, and so on. I refused to engage in debates, speculate, or blame, and yet statisticians were very popular indeed. For me that's a very positive lesson, and I hope it is for the media too.

It was quite a novel idea because what tended to happen in the past was that statisticians would present this information, and then the information would be interpreted, or perhaps I should say misinterpreted, by the media, by politicians, and by interest groups.

That's why my book is called *The Art of Statistics*. It's all to do with what you can learn from data, its appropriate interpretation. Just getting the stuff and showing it to people is enormously important and very challenging in itself, but it's only part of the whole cycle. The people who are closest to the data know most about it and its limitations, so I feel they have a strong responsibility to help people judge what they can and can't learn from that data, and not to leave it to others who may have their own agendas. And so, again, I and others have spent a lot of the last two years countering the misinterpretation of data that so many people have done, with their own motivated reasoning. They have got a view of how society should work, and they choose and interpret data to support their argument. And that goes on not just in politics and the media but among a number of scientists as well, which I think is deeply unfortunate.

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I'll come back to your book because I'd like to ask you about that as well, but just staying on statistics and the way they

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can be manipulated and misinterpreted – they haven't had a great reputation. Perhaps Mark Twain can be blamed for popularising the phrase, "Lies, damn lies, and statistics," but I think this it is really because politicians and companies will often use statistics to obfuscate. As a result, many people distrust statistics and the information they receive – fake news has been selected as the word or term of the year by several dictionaries. So, how do people cut through the spin?

This is the most important issue, which I now dedicate most of my time to. Communication is a two-way process – there's two parties to this, there's the people doing the communicating and there's the people receiving – the audience. For the people doing the communication, there's an enormous responsibility to use what we would call 'trustworthy communication'. Those principles are well established – that you try to inform people rather than persuade them, and show a balance of evidence, both positive and negative. But not a false balance, not everything is equal. You are upfront about uncertainty and the quality of your underlying evidence, and also you try to pre-empt misinformation, you get in there and make clear what you *can't* say on the basis of the evidence.

We've even got a prize that we've given for trustworthy communication. So we know it can be done, but it needs care and thought, and for people to call out when it's not being done. And then to encourage audiences to take a sceptical view to the claims being made and that are being presented to them. But only sceptical, not cynical, because we need the numbers.

It has been all about numbers.

It has been all about numbers, and I've got to be honest, the experience of statisticians has been good. So it shows that the "Lies, damn lies," nonsense just isn't true, in that people do not have a totally cynical view of statistics. They want them and they appreciate them.

I'm not sure of the best way to improve the 'data literacy' in the population. I think a critical approach to data can be taught in schools, but in fact mostly this is just common sense. In particular, there are very general questions about, "How does this number make me feel? What's my emotional response to it? Why am I hearing this? What is someone trying to do to me by presenting?" And to realise that many times when numbers are used, they're being used to try to persuade you to change your emotional response – this is where the "Lies, damn lies," thing has got some truth in it. In my talks on this I show that I can make any number look large or small, frightening or reassuring, just by changing the framing and context. I know other people can do it and so I just want to show people how to spot the tricks.

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It's interesting, you talk about our emotional response to things – most people wouldn't expect to hear a statistician talking about an emotional response to numbers.

Oh, absolutely. Numbers convey and have got a lot of emotional impact.

You say that we can never take an objective view about evidence, that we always bring our personalities into it.

Yes, we should recognise we may have feelings about how the evidence

A tightrope walker above a clown, engraving after M. Laroon, 1688 , Wellcome

has turned out, and then actively try not to express them in our communication. Except to express enthusiasm for the knowledge gained, and for people engaging with it.

And you see yourself as an optimistic person. In what way does being optimistic or pessimistic affect our decision making or our interpretation of data?

If I quote a number, and it's got some range of uncertainty, then you might focus on the top or the bottom of the range. Alternatively, if say I am discussing climate change, I could say "The temperature change could be as high as three degrees," or say, "It's unlikely to be greater than three degrees." Okay, exactly the same number, exactly the same scientific basis, but the story told in a completely different way. We have got used to thinking of these two processes: a cooler, rational, slightly mechanistic, quantitative way of analysis, in contrast to a warmer, emotional, spontaneous response. For me, they're talking to each other the whole time, I don't just click from one to the other. As Daniel Kahneman and others would say, that at times of challenge, of importance, when we're really all faced with something that's quite difficult, we should try to think slowly. Just try to think slowly.

Well, that's the thing, in your recently published book you didn't call it The Science of Statistics, you called it The Art of Statistics.

That's right.

It's a different approach. Statistics is usually viewed as a science – that these figures simply are what they are. What kind of message are you trying to get across by calling your book The Art of Statistics?

I started with a quote from Nate Silver in *The Signal and the Noise* where he says, "The numbers do not speak for themselves, we imbue them with meaning." Statistical analysis is not some algorithmic process where you turn a handle and out comes an answer, and that's it. Obviously, there are some algorithms, you apply formulae, you do analyses, do all that all the time. You code and out comes an answer. But that's not the real answer, that's just the numbers that have been calculated based on what you've done, and those numbers cannot tell you what actually is happening. We have to use judgment and skill, context and experience, and understanding in order to interpret the stuff on the bottom of the screen. In my old days, at the bottom of the printout, but now it never sees paper.

COVID has demonstrated this enormously – you can ask, for example, a very simple thing, "How many COVID deaths?" And I can go on to a website and count COVID deaths. Well, what does that mean? It means different things in different countries. Is it people who died because of COVID? With COVID? Do we even know? So it's completely deluded to think these numbers are the end of it. Crucially, we have to know where they came from, and the context, in order to judge what we can learn. Data science is the whole process of learning from data, and not just some technical algorithmic process.

You say that we imbue numbers with meaning. Now I know in the 1980s you developed ways for AI to handle uncertainty – fast forward 35 years and AI is becoming more a part of our lives. Understanding that AI is developed by humans, do you foresee any issues with AI and data analysis over the coming years?

This is so important. To be clear, I'm not talking about 'general' AI, I'm just talking about algorithms that are applied to data and come out with some conclusion that then might be useful to people. But that's the sense in which pretty well all so-called AI is being used. And there's some wonderful stuff, even if it's not always called AI.

But we know that if you've used a biased data set, the output will just reflect what you put in. Of course, that is now deeply appreciated, and has brought calls for ethical use of AI, transparency, explainability and so on. All of which, I think, is excellent, as it might stop people imbuing algorithms with some mystical abilities just because they are called AI. Some of the work is remarkable, but it always requires critical faculties to use appropriately.

You were an expert witness into the inquiry into the serial killer doctor Harold Shipman. I know this was some time back. You noted that Shipman could have been caught much earlier if someone had been looking at the data, lives could have been saved perhaps. Now, over the past few years data and death have been at the forefront of people's minds because we've been grappling with COVID pandemic right around the world. How has data been used to save lives and is it possible to quantify how many?

The appropriate collection and analysis of data has saved countless lives. No medical treatment would be approved for use unless it was evaluated using statistical methods in randomised trials. Statins, vaccines, and so on have been evaluated using formal statistical techniques, regression analysis, P values, whatever. And statisticians have been at the very centre of all that work, and this has ended up saving vast numbers of lives. Let alone the work on determining the risks of smoking being run by statisticians and epidemiologists. So without that data collection, without that quantification, we'd be still in the age of medical anecdote, and it would be completely disastrous.

It's only through rigorous data collection and statistical analysis that you can produce the evidence that's convincing enough to make people like me take their statin every morning. I have no idea whether they will do me any good and I never will. However, if enough people take them, it saves huge numbers of lives. When I take my statin, I know it is lowering my cholesterol, but I have no idea that, if I don't get a heart attack, whether it was prevented by the statin. I just know, because of the statistical analysis alone, that on average this benefits people like me. So statistics save lives.

Over the course of your life many different things would have happened to you – you have had uncertainty throughout your life, just like anyone else. Now experience tells us that life is uncertain by its very nature and that we don't have much control over what the future holds, we discover this day in day out, but people still

But we know that if you've used a biased data set, the output will just reflect what you put in.

seem to have an aversion to uncertainty. Is there a way of overcoming this fear of not knowing what comes next?

Oh, that is such a good question. I feel I personally have overcome it to some extent, in that I embrace uncertainty in my life. I don't particularly seek to control what happens. Of course I don't like it when things go wrong, and I moan and shout, but in the end it happens. Now I don't know whether that's a function of just getting older, or the fact that I've studied this stuff for years, or my basic personality, or the shocks I've had in my life, and the fact that I have managed to live with them. So I have absolutely no idea why I feel this way but I'm glad I do – I think that an increased acknowledgement of the lack of control in our lives would be, in general, quite good.

Just to finish up, you wanted to say something about different types of uncertainty…

I struggle to understand uncertainty and the way to categorise it, although there's been many attempts to do so. But I first think of uncontrollable uncertainty about the future. It's sometimes called aleatory uncertainty, or chance – when you *can't* know what's going to happen. And then next you've got uncertainty about some state of the world. For example, if I flip a coin and cover it up, what's underneath my hand? That's epistemic uncertainty, it's our lack of knowledge about a potentially verifiable fact, what we *don't* know. But this is quite limited because it assumes you know what you don't know, in Rumsfeld's famous words.

So we need to go beyond that. And I think that Rumsfeld's "Unknown unknowns," is useful. What is outside your current ways of measuring and envisaging? It sounds a bit intellectual, but I like the phrase 'ontological uncertainty', meaning that the words and concepts you're using are inadequate to describe what you're trying to explore. This may mean possible futures you've never thought of, but also just the inadequacy of your measures. For example, we calculate statistics on wellbeing and ill health. Well, of course they're not measuring whatever people mean by wellbeing and ill health, they're just very limited metrics of it – there's a huge amount left over. What's left outside our conceptualisation and measurement, perhaps things we've never even thought of – that for me is the interesting area of uncertainty, because it acknowledges the limitations of everything we do with data. Which is suitably humbling.

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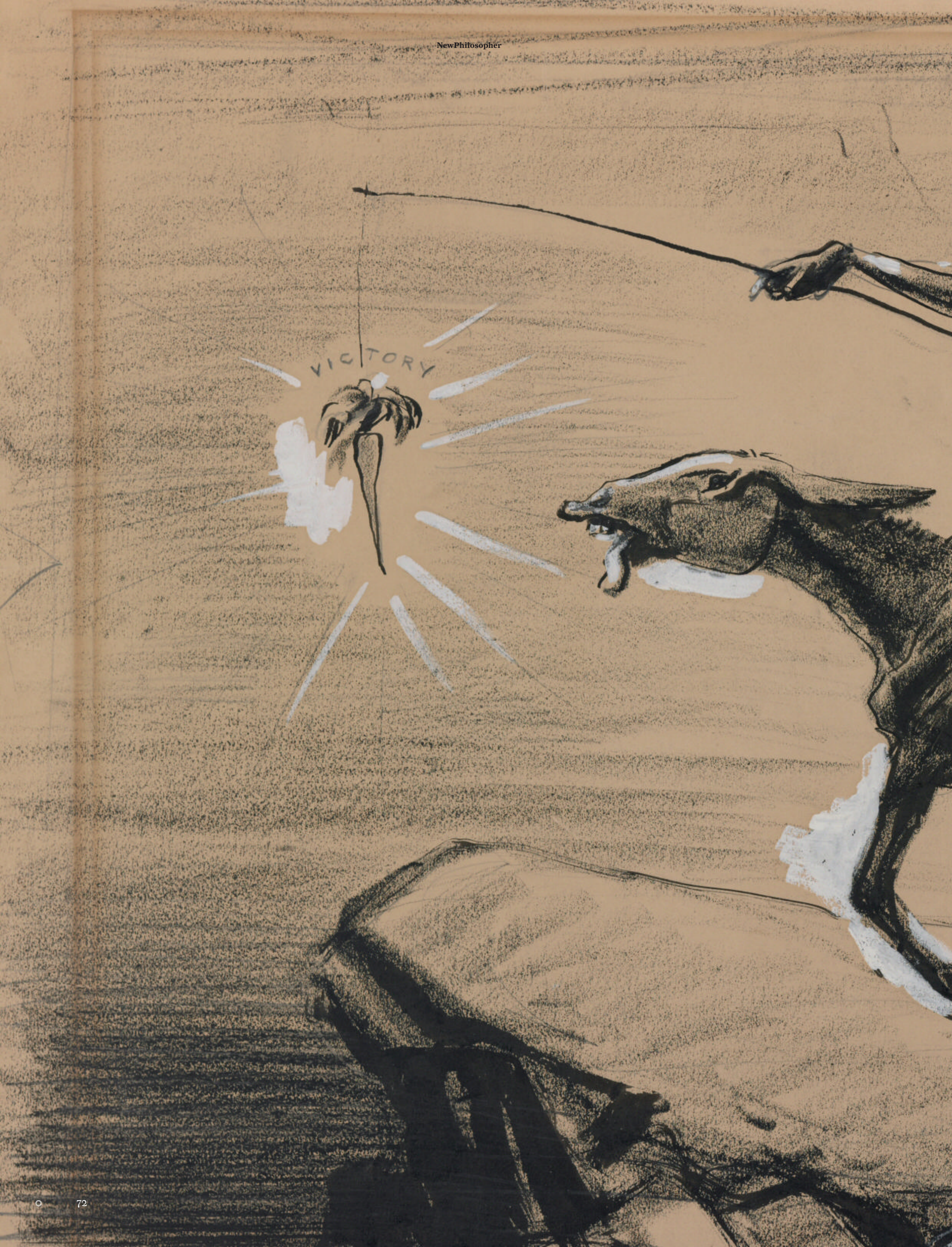
Artwork: *Industry*, by Arthur Durston, 1934


VICTORY

Europe 1916

by Nigel Warburton

When will you die?

When it comes to death, it is as if we are walking backwards through a minefield. There aren't that many mines in the field, but there are more of them the further you go into that field. At the furthest point they are so close together that hardly anyone gets beyond them, and then there are even more densely planted mines. Some people are fortunate enough to step directly on a mine that kills them instantly thus avoiding the nasty business of dying – moving straight from active life to death. Tough for those around them, but a painless exit for the individuals. The less fortunate are maimed to a greater or lesser degree and struggle on in pain backwards through the field, perhaps helped by their friends, and by doctors and nurses administering palliative care, until they step on another mine, or succumb to their injuries. Some die with dignity, others, less so. It's not possible to stop walking. We all trudge on backwards, mostly oblivious to how perilous our route is, occasionally shocked to see what happens to those around us, always missing those who have died before us. We never really know when we'll step on a mine, but if we think about it, we all realise that is inevitable.

One of the strange things about life is that we know with certainty that we will leave it, but for most people there's a high degree of uncertainty about when exactly that will be. That's an important aspect of the human condition. Cancer, a respiratory disease, heart problems, dementia, an accident, suicide, violence, or something else will get us all in the end. The longest anyone has lasted so far (or at least has been verified to have lived for so long) was Jeanne Louise Clament, a French woman who lived to 122. But if by some weird quirk of fate you too were to live to be 122 that doesn't mean that you'd definitely not live another month, or even a year. Perhaps you'd be the first person to reach 123. I doubt it. But it is possible. More likely, of course, is that if you're reasonably healthy you'll die in your 80s or perhaps 90s, hopefully without a long steep decline. Very sadly youth is no prophylactic against death. Babies, children, and young adults die, middle-aged people die, and older people die a lot.

You can decide to end your life, and if you have chosen your method of exit well, you can be fairly certain of when you will die. But even then, that might backfire. And those with terminal illnesses, or victims of severe injuries know they haven't got long. You can see the hourglass emptying without knowing exactly when that last grain of sand will fall through.

In some ways, it would have been better if we all had, for example, 70 years on the clock and could know that after that it would all be over. If we knew precisely when we were going to die and knew it sufficiently in advance, our time management might improve. It could be a good way of ending procrastination. As the English writer Dr Johnson observed, "when a man knows he is to be hanged in a fortnight, it concentrates his mind wonderfully". Think of what Boethius achieved as he waited for his execution by strangulation in a prison cell on charges of conspiracy to overthrow King Theodoric the Great: he wrote *The Consolation of Philosophy*, an early medieval bestseller in which Philosophy appears in Boethius's cell as a tall woman who implores him not

Illustration by Aida Novoa & Carlos Egan

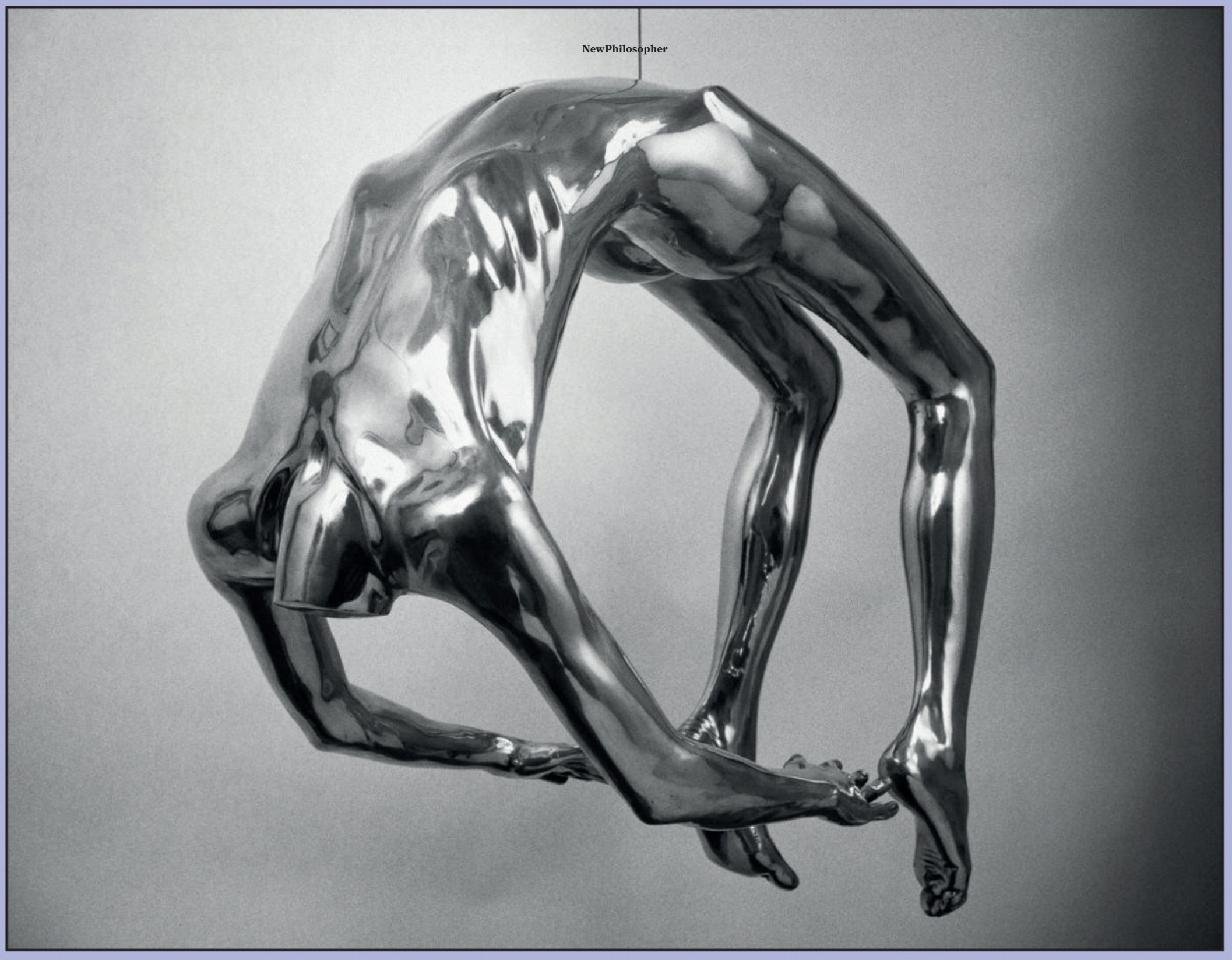
to forsake her even in these dark days. That book is still in print today. Boethius was exceptional, but there are other examples of those awaiting execution being galvanised to complete pieces of writing rather than succumbing to despair. To take just one example, in 1586, while imprisoned in the Tower of London awaiting execution for his part in a Catholic plot to assassinate Elizabeth the First (he was to be eviscerated, hanged, drawn, and quartered), Chidiock Tichborne, who was in his mid-20s, wrote his moving autobiographical elegy that included the lines:

> My prime of youth is but a frost of cares,
> My feast of joy is but a dish of pain,
> My crop of corn is but a field of tares,
> And all my good is but vain hope of gain.
> The day is past, and I yet I saw no sun,
> And now I live, and now my life is done.

Both Boethius and Tichborne knew they almost certainly had a very short time to live, and both faced a gruesome death. Both produced lasting works in that short space of time. I'm not sure that knowing in advance the hour of our death would help all of us to get more done, though. I'd like to see more empirical evidence in support of Dr Johnson's dictum on this. It's dangerous to argue from a few remarkable cases. To balance Boethius and Tichborne with hypothetical anecdotal evidence on the other side: knowing I was to be hanged in a fortnight would probably tip me into glum inaction rather than creative and productive fervour, and I suspect I'm not alone in this. That's because, as Seneca wisely observed, plenty of people would find life too short to achieve much no matter how long they were given, because they are, in a way, addicted to wasting time.

I'm inclined to see the uncertainty of knowledge about when we will die as a good thing, though. We can throw ourselves into long-term projects and activities that would seem absurd if we really knew how little time we had. If I knew I was going to die next month, would I still try to play that difficult Bach suite on the guitar? I doubt it. But if I didn't, I'd miss out on the fulfilment that comes from getting lost in the challenge of playing beautiful music, and the small but satisfying gains that an amateur musician can make. Would I start writing a book I knew I'd never complete? Again, probably not. But I'd miss out on the intellectual stimulation that the writing process can give. Knowledge of my imminent death would disturb every close relationship: each time I was with my family or close friends I'd be conscious that time was quickly running out and that very knowledge might well make those interactions too self-conscious with too much weight hanging on them, compared with how we would have been in ignorance that I was about to take another step backwards in that minefield for the last time.

I'm not sure that knowing in advance the hour of our death would help all of us to get more done.



“What a chimera, then, is man! What a novelty, what a monster, what a chaos, what a subject of contradiction, what a prodigy! A judge of all things, feeble worm of the earth, depositary of the truth, cloaca of uncertainty and error, the glory and the shame of the universe!”

Blaise Pascal

Artwork: *Arch of Hysteria*, Louise Bourgeois, 1993

What dreams may come

By William Shakespeare

Illustration by Aida Novoa & Carlos Egan

To be, or not to be, that is the question:
Whether 'tis nobler in the mind to suffer
The slings and arrows of outrageous fortune,
Or to take arms against a sea of troubles
And by opposing end them. To die — to sleep,
No more; and by a sleep to say we end
The heart-ache and the thousand natural shocks
That flesh is heir to: 'tis a consummation
Devoutly to be wish'd. To die, to sleep;
To sleep, perchance to dream — ay, there's the rub:
For in that sleep of death what dreams may come,
When we have shuffled off this mortal coil,
Must give us pause — there's the respect
That makes calamity of so long life.
For who would bear the whips and scorns of time,
Th'oppressor's wrong, the proud man's contumely,
The pangs of dispriz'd love, the law's delay,
The insolence of office, and the spurns
That patient merit of th'unworthy takes,
When he himself might his quietus make
With a bare bodkin? Who would fardels bear,
To grunt and sweat under a weary life,
But that the dread of something after death,
The undiscovere'd country, from whose bourn
No traveller returns, puzzles the will,
And makes us rather bear those ills we have
Than fly to others that we know not of?
Thus conscience doth make cowards of us all,
And thus the native hue of resolution
Is sicklied o'er with the pale cast of thought,
And enterprises of great pith and moment
With this regard their currents turn awry
And lose the name of action.

From *Hamlet*, Act 3, Scene 1

IN PRAISE OF UNCERTAINTY

by John Armstrong

The 19th century Scottish novelist, Robert Louis Stevenson, once wrote a charming little article in praise of idleness. The problem was, he was a great fan of hard work. But he sidestepped the apparent contradiction with a neat analogy: to talk of the excitements of travel isn't a denial of the love of home. To enthuse about loafing, dawdling, and lounging away an afternoon, isn't to ignore the obvious benefits of strenuous application. So too with uncertainty: of course we want and need security, so far as it can reasonably be found, in our lives. But there are ways in which we might need, and even relish, uncertainty as well.

The issue of how uncertainty can serve us well arises at the very start of philosophy in the persona of Socrates. The typical opening phase of a Socratic dialogue takes an interlocutor away from their current convictions and leaves them in a state of confusion. They've been confidently using key words like beauty, knowledge, or justice all their lives but now – under Socrates's gentle but pointed questioning – they come to feel confused and puzzled, they no longer feel sure of their beliefs.

It's tempting to think of these people as defeated: they felt certain and now they are anything but. Yet, for Socrates, this condition of coming to feel less certain, of realising that one doesn't know, is intellectually admirable. He's combating the normal inclination to associate intelligence with clear and definite answers. Instinctively, the cleverest person is the one who knows most: the pundit who can reel off statistics, the dinner party guests who quotes Karl Marx and St. Thomas Aquinas, the six-year-old who can spell 'scythe'. But, suggests Socrates, there's another – much deeper – mark of a bright mind: the capacity to reach – and recognise that one has reached – uncertainty. It's not the end of the discussion: one might wrestle profoundly and insightfully with a problem for a long time and not come to a final answer. Plato's *Theaetetus*, in which Socrates plays a starring role, grapples with the question, *What is knowledge?* It may be the single most insightful work ever written on the topic (and is probably the most influential) but it ends with uncertainty.

There are matters on which we should be unsure because the issues involved are so subtle, intricate, and elaborate that even our careful, clever

Woman with a Head of Roses, Salvador Dalí, 1930.

Uncertainty is noble if and when it is the proper result of intelligent investigation.

efforts to untangle them are baffled: uncertainty is the mind's homage to complexity. It's perhaps necessary to recall Stevenson at this point: Socrates, in his elevation of uncertainty to the place of a grand intellectual virtue, isn't implying that we should be uncertain whether chalk is cheese or if bears do their business in woods. Uncertainty is noble if and when it is the proper result of intelligent investigation. The Socratic motto "I know that I don't know" isn't an endorsement of contented ignorance; it's not another way of saying "don't know, don't want to know"). Instead, it's the concession of a mind yearning to understand but able to admit the defeat of its best efforts.

The benign capacity to be uncertain – and therefore to speculate and explore – is quite fragile. In a world mad for certainties, social legitimacy gets tied up with adherence: we belong where we share convictions. To admit to being unsure is to risk coming untethered from the group.

In 1817, a few days before Christmas, the poet John Keats (then in his early twenties) wrote a letter to his younger brothers, George and Thomas, in which he casually introduced the lovely phrase "negative capability". What he meant, he explained, was the ability to appreciate "uncertainties, mysteries, doubts". And he contrasts this with an "irritable grasping after facts and reasons".

We need to be careful parsing his meaning. To be directly opposed to genuine facts and valid reason is (bluntly) idiotic. But that's not what he's talking of: 'irritable grasping' implies a craving for anything that can be made to pass (even if only in one's own mind) as a fact or a reason. Keats is noting, I think, a tendency to want to run away from avowed speculation to pretend facts and spurious rationality. It's a deeply poignant insight. The world is full of people who have fanciful, strange notions, but they won't (or can't) see them for what they are: they feel they must insist they are true. They can't – for some reason – afford to be uncertain. They won't say, "I have no idea if this is really true but I'd like to elaborate a thought"; instead they insist that they are right.

The wonderful mid-century British psychoanalyst, Donald Winnicott used to delight in praising naughtiness and rule breaking. It is, he thought, often a sign of health. A child who is inwardly secure can afford to test the received norms of their society (all the genuine wisdoms of maturity will, of course, pass the test). Deepening Keats, he's suggesting that the capacity to be uncertain flourishes where there is an underlying emotional security. It's an idea to transform our perception of rigid partisans and those who conspicuously lack 'negative capability'. It's not that they are intellectually blighted but that, for them, it feels too risky – in some existentially threatening way – to simply wonder about an idea, or entertain a possibility as such or consider that they might be wrong or admit (finally) to confusion and doubt.

Our world is mad for certainty not so much because we are collectively stupid but because it feels (to so many people) too risky to recognise and be open about their confusions and uncertainties. Winnicott dreamed of a society in which people could feel that their intrinsic worth, in their own deepest estimate of themselves, could survive an admission of intellectual uncertainty.

The human longing for certainty is immensely unironic. There is perhaps no deeper theme in our lives than the

unknowability and unpredictability of the future. We are creatures thrown into a world that guarantees us nothing, yet in which we must plan and act and hope. We've pitted our unassailable convictions against the incomprehensible volatility of existence. The record of what people have been sure of almost defies the imagination: that we might be reincarnated, that religious ceremonies could affect the weather, that the descendants of a particular person were entitled to rule a nation, that Communism would bring universal peace and happiness, that T.S. Eliot was the greatest poet of all time, that progress was inevitable, that the young are always right, that the internet would be an instrument for good.

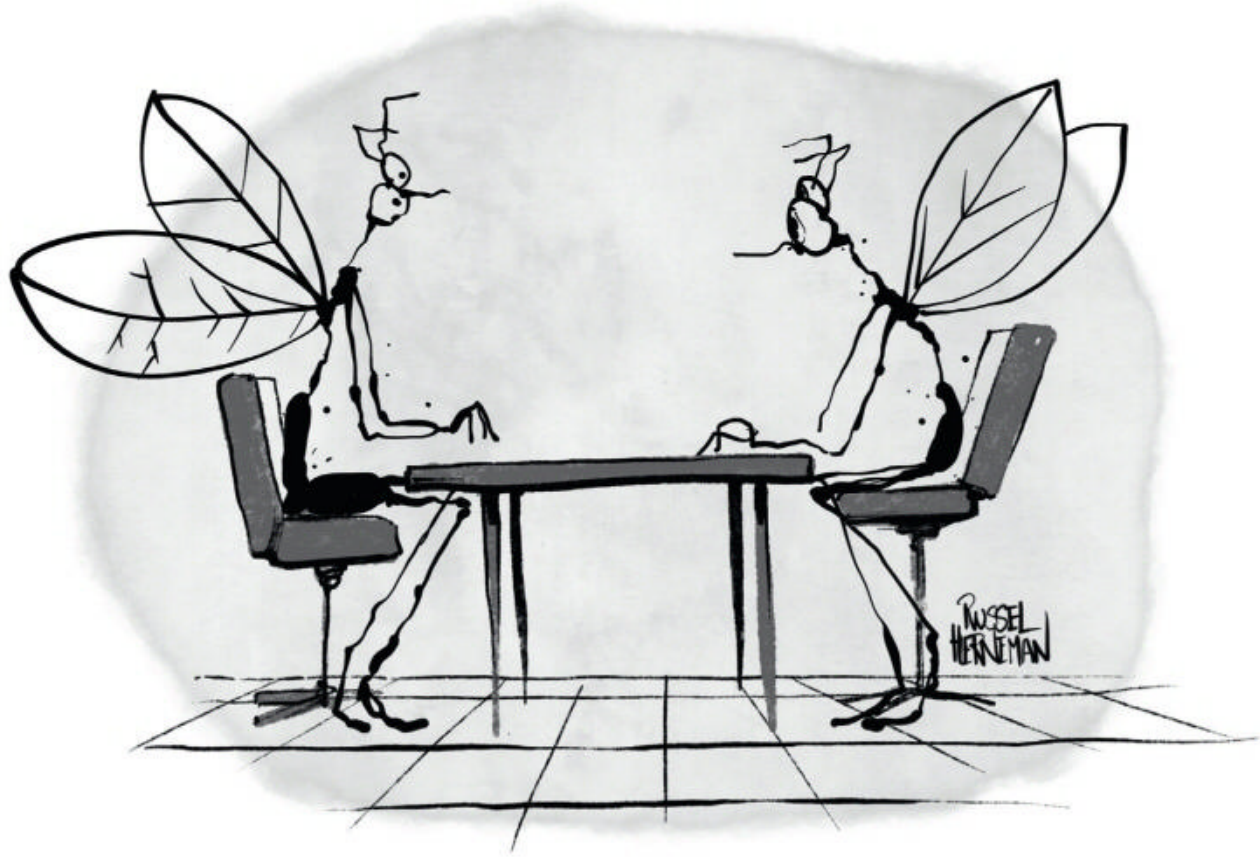
But we shouldn't be ashamed. These are, indeed, lovely (or at least well-intended) ideas: we've suffered because we couldn't collectively say that's what they were; we couldn't find the space between conviction and imagination; we couldn't be uncertain.

Being uncertain doesn't mean you see nothing in a proposition or that you hate it. You can feel uncertain about ideas that appeal, only you don't quite know if they really are justified. It doesn't mean that – if the right sort of evidence is presented tomorrow – you won't upgrade to believer status. What it does mean is that you can take a lot of interest in a line of thought, you can sense its allure, you can sympathise with it, without being convinced.

Uncertainty around ideas and creeds may be compared to flirtation. It's mutually delightful but, as yet, mutually non-committal. It's an intriguing exploration, not a declaration. It's enjoyable uncertainty a deux.

There are times and places where the search for certainty is necessary: but often people are using certainty to compensate for something else. They have to assert intellectual conviction in slightly crazy places to compensate for, and to mask, self-doubt around their own dignity.

The righteous and over-convinced, perhaps, are not to be outright condemned but rather to be understood and treated with the generous tenderness they secretly long for and need. ◘



"Where do you see yourself in five minutes?"

"The quest for certainty blocks the search for meaning. Uncertainty is the very condition to impel man to unfold his powers."

Erich Fromm

"Our doubts are traitors." — William Shakespeare

UNSURE

"Certainty generally is illusion, and repose is not the destiny of man."

Oliver Wendell Holmes, Jr

"A woman's guess is much more accurate than a man's certainty."

Rudyard Kipling

"I ever held worse than all certitude to know not what the worst ahead might be."

Algernon Charles Swinburne

"Who knows whether the gods will add to-morrow to the present hour?"

Horace

"If you tried to doubt everything you would not get as far as doubting anything."

Ludwig Wittgenstein

"When the mind is in a state of uncertainty the smallest impulse directs it to either side."

Terence

"Doubt is not a pleasant condition, but certainty is an absurd one."
Voltaire

"In this world nothing can be said to be certain, except death and taxes."
Benjamin Franklin

"If a man will begin with certainties, he shall end with doubts."
Francis Bacon

"If you don't like the weather, wait a minute."
Anonymous

"He is no wise man that will quit a certainty for an uncertainty."
Samuel Johnson

"Doubt is brother-devil to Despair."
John Boyle O'Reilly

Uncertainty

"Not to be absolutely certain is, I think, one of the essential things in rationality."
Bertrand Russell

"That's the thing about life: you can do what you do but in the end, some things remain stubbornly outside your control."
W. I. B. Beveridge

"It is certain because it is impossible."
Tertullian

"Ah, what a dusty answer gets the soul
When hot for certainties in this our life!"
George Meredith

by Mariana Alessandri

Relentless uncertainty

No matter what people say about pandemic doomscrolling or the abundance of negativity in the news, a good pessimist is still hard to find. One who, like the Roman Stoic Seneca, would tell you that life is, at bottom, a storm; one who, like the Greek Stoic Epictetus, was hyper-aware of death and highly invested in keeping it at the topmost of our minds; one who, like the Roman Emperor Marcus Aurelius, took for granted that the day brings with it a litany of ills, ranging from the inconvenient to the torturous. A good pessimist acknowledges life's imminent pain and its relentless uncertainty.

Stoics from Zeno of Citium (circa 300 BCE) to Marcus Aurelius (circa 170 CE) were certain of one thing: humans suffer and die. They did not avert their gaze from this fact – ever. And they came up with a whole philosophy to deal with it. If you believe that the worm of death gnaws at your core but – and here's the snag – you don't know when it will make a hole big enough to crawl through, then you'll need to do some planning. Prepare to be surprised, Stoics say, and you'll not be surprised. Prepare for uncertainty, same idea. But their particular brand of preparation made them Stoics and not Existentialists.

The Stoics' plan for certain death at an uncertain time, as well as for how to face our ever-growing wave of suffering, was to divide the world into two categories: what is in my control and what is out of my control. The Stoics ceded loads more control to the universe than our "there's-an-app-for-that" society does, including what profession we'll hold, how healthy and beautiful we'll be, and who will honour and esteem us. But they simultaneously won for us plebeians an unbelievable amount of inner control. For everything we can't control in the external world, they found something to subjugate in our psychic lives. Epictetus, for example, would hold that you don't get to decide whether you become a TV news anchor – that will depend on your level of God-given beauty, your voice, aptitude, and luck, all of which are out

Illustration by Aida Novoa & Carlos Egan

of your direct control (if it's not 100% in our control, he puts it out of our control). In the end, becoming a TV news anchor will depend in large part on whether or not you get disfigured in a car crash on your way to your audition. Don't put your hopes in becoming a TV news anchor, then, or even the bulk of your striving. Even if you win the position, it can be taken away from you tomorrow. But what you do get to decide is whether to become a good human being. Instead of calling any outer event 'good', believe the only good is moral good, and work hard to become it. Virtues are, for the Stoics, 100% in our control. Thus, they gain for us a new certainty in addition to death: moral progress. TV news anchor or not, we can form ourselves into patient souls, generous and temperate. The external world is (definitely) rife with car crashes, but the internal world is (potentially) abloom with virtues. The Stoics were smart pessimists: they had a good answer to life's misery, and it continues to make a whole lot of people feel better in times as uncertain as these. Since 2012, the Stoic revival has gotten traction in the US, UK, and Australia. So many of us want to hear that we're in control.

The 19th and 20th century Existentialists came up with a different, equally pessimistic, response to the certainty the Ancients alighted on. Like the Stoics, they saw suffering everywhere and wanted to help. Like prophets, they proclaimed that the end of the world is nigh, that we need to wake up today because tomorrow may never come, and that opting out is copping out. In 1843, Søren Kierkegaard dared us to look straight into the face of doom and not blink. He put us humans at the edge of a great abyss, looking down and getting wobbly. He described anxiety as "the dizziness of freedom," and said we are overwhelmed – and excited – by the idea that anything is possible. We intuit that, in addition to a sack of money appearing on our doorstep, 'anything is possible' also means we can ruin ourselves. Almost a hundred years later, Martin Heidegger said that the fundamental human condition was "being-towards-death": the unshakeable reality of our lives is that we die. But instead of living as though we're dying, he noticed that we spend most of our time in "flight", distracting ourselves with Very Important Things. We also cling to otherwise stories about how we control the inner landscape of our minds. Fifteen years after that, Jean-Paul Sartre likened the human condition to feeling nauseated, thanks to a worm he called "nothingness" that lies curled up in our core. Even before we make a decision, Sartre said, we know that by choosing A we will not choose B-Z. Entire books and movies have been dedicated to the what-ifs of life: what if I had chosen M, or C, or U? We have more freedom than we want, Sartre believed, so we retaliate by tying ourselves to spouses, jobs, kids, roles, email. The tethers stop the spinning for

In 1843 Søren Kierkegaard dared us to look straight into the face of doom and not blink.

a time, but eventually we start resenting our strings.

On the Existentialist story, the average human being is a coward, who, upon whiffing the end of life in the air, changes direction. We flee from the macabre right toward the mundane, and we dream of a world without turbulence. Sure, the pandemic had us living in full colour for a stretch, but the sooner we get back to black-and-white – low alert instead of high – the better. Rational humans don't choose to soak in suffering, loneliness, alienation, anxiety, and disappointment. We want something to cool our mind, take the edge off, help us regain our footing. Few Existentialists remain, especially compared to Stoics, and it's no coincidence. After all, how many of us today could agree with Miguel de Unamuno that it is better to exist in hell than to not exist?

Which story looks better (which is not to say truer): the Stoic one, in which what we do in the face of a certain storm at an uncertain time is to batten down the hatches, or the Existentialist story, in which we stumble around, perpetually uncertain but persistently unwilling to tell ourselves lies about battening down mythical hatches? Both schools see humans scampering about like so many white rabbits looking at their watches, and neither is particularly resonant with any bright-sided world. Stoicism asks us to picture the ills awaiting us, and Existentialism urges us to turn toward our anxiety. The difference between them is not a matter of pain tolerance but of uncertainty tolerance, and which philosophy you choose may come down to your feelings about control. Seneca was nauseated once, too. But he jumped out of his boat and risked his life swimming to dry land. The Stoic might ask: what kind of lunatic wants to live on choppy waters? To which the Existentialist might reply: what other waters does life offer?

"Well, it's not a problem for me to say both where you were and how fast you were going, Mr Heisenberg."

Interviewee: Stuart Vyse
Interviewer: Zan Boag

The illusion of control

Stuart Vyse is a behavioural scientist, teacher, and writer. He is a columnist with *Skeptical Inquirer* magazine and has written for the *Observer*, *The Atlantic*, and *Time*. He is the author of *Believing in Magic*, which won the William James Book Award; *Going Broke*; *Superstition*; and *The Uses of Delusion*. As an expert on superstition and irrational behaviour, he has been quoted in *The New York Times*, the *Washington Post*, the *Los Angeles Times*, and has appeared on CBS, CNN, and PBS NewsHour.

Zan Boag: How would you define certainty?

Stuart Vyse: As humans, certainty isn't something we have a lot of. Obviously, there are mathematical definitions of certainty and uncertainty, but psychologically, what matters is that you are able to predict and count on things happening in your world enough to function and to do so without anxiety or worry.

Is there a connection between certainty and superstition?

A very strong one, yes. The whole purpose of superstition is to combat uncertainty. If you know something is going to happen and you want it to happen, then there is no need for superstition. Superstition does not crop up in those cases. There are no superstitions about walking because mostly we can do that with some degree of confidence, but there are superstitions about things that we can't be confident about and yet we care about. Things we either want to happen or want not to happen. So, there's a very strong connection between uncertainty and superstition.

You're a behavioural scientist, so you investigate human behaviour to work out why people do what they do. What was it that drew you to investigate superstition and delusion in particular?

Well, it actually is based on a personal experience. I was in graduate school many years ago as a budding psychological scientist, and I read an article. This is the kind of thing that people who write boring journal articles dream of happening. As a result, Barry Schwartz should be very flattered. I read an article many years ago by psychologist Barry Schwartz about college students trying to earn points at a simple computer game when they didn't know what the contingencies were, how it worked. They were moving a little marker through a matrix, and they had just two keys that they could press. They were trying to figure out how to earn points and what they ended up doing was developing bizarre theories.

They came to believe that a specific pattern had to be followed in order to get points. And each person had a different theory as to how the game worked, because, in fact, there were many ways to get points. This immediately struck me as being analogous to superstition and just a really interesting and quirky thing that these well-educated college students were doing when placed in this context. So, several years later I did some similar experiments that were even more directed

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A Thirteen Club table, New York Public Library

at the question of uncertainty's effect on theories that people put together in a problem-solving situation like Schwartz's experiment. So, that's how I started. I was inspired by a specific article, and many years later, followed up and did research on it. Then after publishing a couple of papers, I realised that no one had written a book on superstition for quite a while, so I wrote my first book.

It's interesting that's what sparked interest for you. It made me think immediately about gambling, about the gambler's fallacy, how gamblers will create stories and superstitions around their gambling. Something with somewhat higher stakes is professional sports. Many players tend to have rituals and superstitions – baseball is a huge one, but also cricket, football, and tennis are all sports where players suffer from such behaviour. In tennis, Rafael Nadal stands out, he has a drawn-out routine before he serves, after each game, even when changing ends with the way he places his towels and water bottles. Although the routine or superstition seems ludicrous to those watching, is it simply an attempt by the player to control their environment and to gain focus?

Absolutely. I think in sport, superstition has a number of functions and obviously the most important thing for the athlete is winning. Most of the examples we hear about are from the better athletes. The few studies that have been done on the subject suggest that better athletes are more likely to have superstitions than athletes who are not as good. These superstitious athletes are trying to eke out some measure of control. There's a psychological phenomenon called the illusion of control, which involves simply doing something – anything – even when that something can't really have an effect on the outcome. Merely taking some action gives you the feeling that you've done one more thing, and that you have perhaps some additional control over the outcome.

It's also the case that athletes often have a waiting period. There is a period of time that has to be filled, be it in between serves or as you're waiting to come out onto the pitch. There is waiting time when you can't really do anything that might realistically improve your performance, but performing a ritual of some kind or engaging in a superstition that requires action provides a way to fill the time. And as you say, these superstitions also focus the athlete on the contest. So, I think athletes' superstitions have a number of functions, all of which can serve to maintain them over a long period of time.

There's that human tendency to create stories around events and try and explain away why things have happened. I know you assigned the Freakonomics podcast in your behavioural economics course, and also recently appeared in an episode of their podcast. I think it was titled, "What Do Broken-Hearted Knitters, Urinating Goalkeepers, and the C.I.A. Have in Common?" There's an interesting website by Tyler Vigen, he provides examples of a whole range of spurious correlations such as the CIA-knitters-goalkeepers one. What is it that causes us to make such connections? How can we sidestep the correlation-causation dilemma?

Well, I believe that's part of the reason why better athletes are more

There's a psychological phenomenon called the illusion of control, which involves simply doing something – anything – even when that something can't really have an effect on the outcome.

superstitious because their superstitions have appeared to work. There's a famous example that I've used in my books for many years, of Wade Boggs who was a baseball player in the US, who believed that eating chicken before each game would improve his hitting. And he was a very good hitter. He won the batting title several times and he's in the Hall of Fame. But he never *didn't* eat chicken. He always ate chicken. So, there was no way of knowing whether chicken had any effect on his hitting. All we know is that chicken was correlated with him, but we don't know whether it was really correlated with his hitting or not. But better players are going to assume that that's the case.

There are some cases in which the correlation is more detectable. In a few cases, players or teams were involved in a winning streak where, for example, the coach had worn a particular 'lucky' sweater that was thought to be responsible for the winning streak. Often, the minute the streak ends, the lucky item is thrown away.

So, there are some rare instances in which the correlation is detectable and acted upon. But often it's not that, it's simply that the person habitually performs their superstition. Also, there is a psychological benefit in the moment of doing a well-practiced superstition. Rafael Nadal by now, I'm sure, feels so much better whenever he does his ritual with the bottles and pulls at his shirt. Not doing his routine would probably create real anxiety. So, it's not just looking at the outcome, it's also the psychological benefit in the moment.

That brings me to your new book that's out now in the US and coming out later this year in the UK, in which you write that delusion can sometimes be useful, why it's not always rational to be rational. There are instances where irrational behaviour can be beneficial for us, aren't there?

Yes, there are. And superstition is one of the things that drew me to this topic. The research evidence that superstitions actually help performance is not strong, but it's obvious that they do have a psychological benefit. Superstition is not going to be an option for some people if their backgrounds were more rational, but it's clear that for the people who employ them, there is some benefit. And so, yes, this book is about many of these kinds of instances in which the action that seems rational and reasonable is not the best one for the individual.

There's another area that's been top of mind for you – and everyone else – for the last few years, where delusion hasn't been so useful. In a recent column, you wrote about jumping to superstitious conclusions, referring to those who were doubtful about the effectiveness of the COVID vaccine. And these people were certain that 'doing their own research' was the best course of action. But the quality of the research that such people do is highly questionable. You note that there is substantial evidence that people who endorse weird beliefs are not actually very good at research. Now, how common is it for people to seek research that confirms their already established beliefs? Or maybe, put another way, how rare is it for people to seek out research that challenges their beliefs?

It is remarkably rare. I think you have to be trained to look in the other direction and seek out disconfirming evidence. And it's a rare group of scientifically or

The research evidence that superstitions actually help performance is not strong, but it's obvious that they do have a psychological benefit.

rational-minded people who think that way and will deliberately look for disconfirming information. So that's a serious problem. And of course, what we have now is the ability to find confirming information very easily. Confirmation is a couple of keystrokes away, and the algorithms for social media platforms are set to provide it to you. The internet is tuned to your interests and your persuasions. That is the problem. In a less sculpted social context, where we randomly run into people with different viewpoints, we might actually benefit from hearing different points of view more often.

The issue is that a lot of people are online all the time, they are in this information bubble or tunnel, where they don't seem to be able to see outside. Is there a way of countering this?

I think the only way that is hopeful is to train people in sceptical reasoning, and that would have to be done early. I'm not teaching anymore, but when I was, I would sometimes engage my students in a classical debate where we would tune up a couple of topics that they were interested in. Then, I assigned them sides by a coin flip. They would not know going in what side they would have to argue. Setting up situations like that in schools would be very helpful. Many schools still do have debate clubs, but I think it should be worked right into the curriculum to get young people used to the idea that they need to look for the best evidence on both sides before forming an opinion.

It's a cornerstone of the law, isn't it.

Yes, absolutely. It is. The law is an adversarial system and that's why here in the US I believe that, of all of the branches of government, it is the law that is most based on reason, logic, and evidence. Sophistry can be found in the other branches, but our hope of survival in some sense is in the rule of law if we can hold onto it.

Combating superstition isn't an easy task. Back in 1880, the The Thirteen Club was created in an attempt to do this, and it attracted the support of several US presidents over the years. Robert Ingersoll said in a toast at the club in 1986, "We've had enough mediocrity, enough policy, enough superstition, enough prejudice, enough provincialism. And the time has come for the American citizen to say, hereafter, I will be represented by men who are worthy, not only of the great republic, but of the 19th century." How do you think Ingersoll would view contemporary American politics? Is it still filled with mediocrity, prejudice, and superstition?

Absolutely, yes, it is. And The Thirteen Club was a wonderful organisation. They used to deliberately have meals with 13 people around the table, but I do think that Ingersoll would be shocked that we have not made more progress. I hate to say it, but I think that we are going backwards now with a complete lack respect for truth. You may know that some years ago, the philosopher Harry Frankfurt wrote a book called *On Bullshit*.

It's a terrific book, and it still holds up. And what is disturbing in the present situation is that it's not just that bad arguments are being made and bad evidence is being used, it is actually a destruction of the value of logic and evidence. And we are, I think, in a situation where the goal of many people is simply to disrupt and not to really advocate anything other than a destruction of institutions and respect for expertise. Frankfurt's book was really prescient, and, if anything, things have gotten much worse since the book was written.

I think there was a line from it where he said that bullshitters simply aren't interested in the truth; that the truth is beside the point.

That's right. That's the most destructive thing. At the time he wrote

it, it was a new and somewhat controversial statement, but he said that bullshitting is worse than lying.

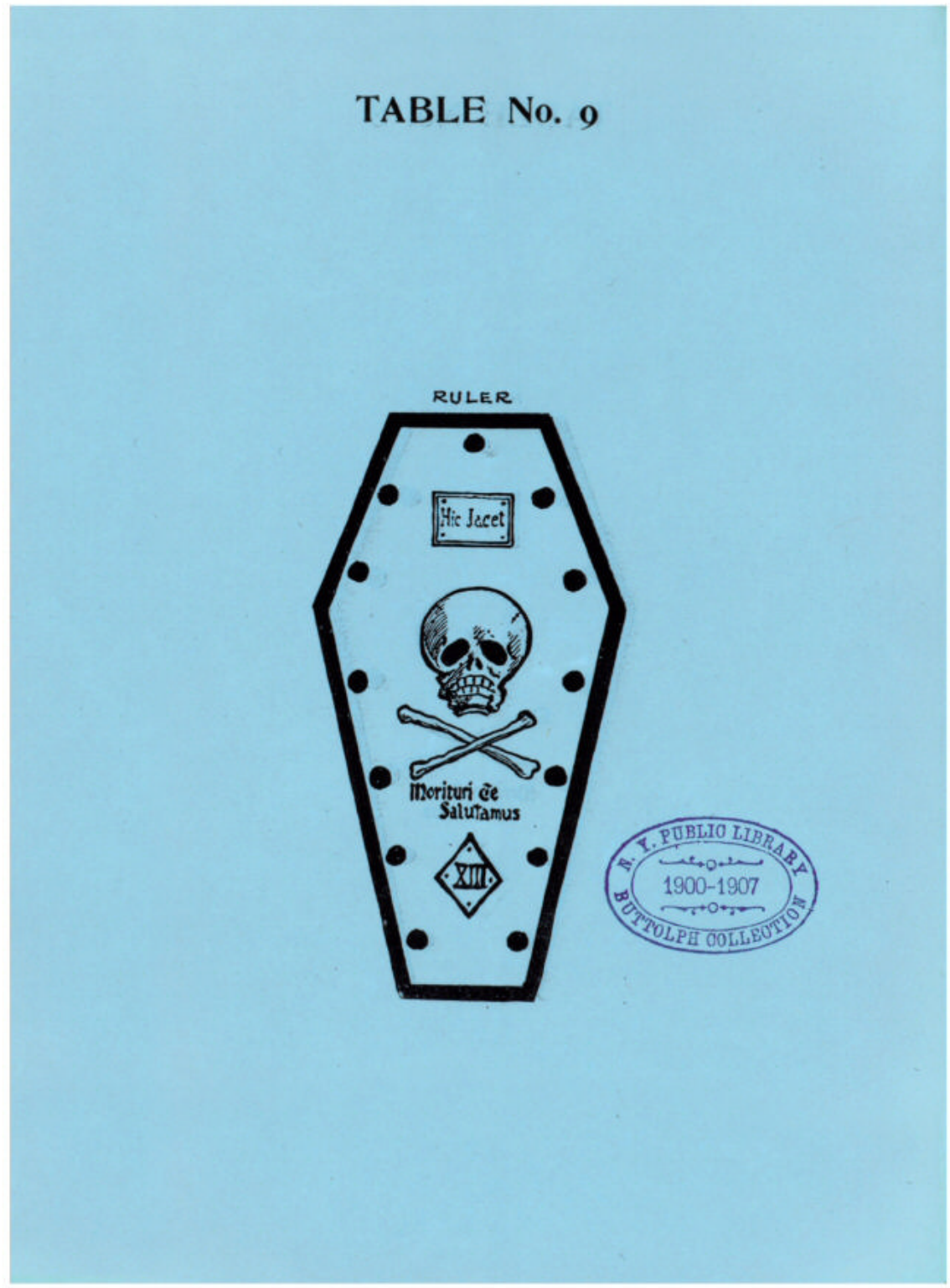
Although we used to think of bullshitting as just playing around, what Frankfurt meant, and I think what we've seen now, is that it's much worse to destroy the value of truth and to not care about the truth than to say a falsehood. The liar still cares about the truth. He or she knows what it is and deliberately says something else. So, we see now that Frankfurt was right about that.

That has come to the fore over the last few years, and you would have researched a range of odd beliefs, particularly when it comes to COVID and anti-vaxxers. What are some of the more deluded beliefs that you've encountered? What do you think is driving these beliefs? I know you just touched on that, that it's an attempt to disrupt, a disregard for the truth, but are there other things that are driving these beliefs?

One of the things that shows a clear dividing line is how people view information about how to deal with the virus and the vaccines. Most of this information and the administration of the vaccines has come through the government. In my country and in others as well, there is a real resistance from people who place great value on their freedom and independence and do not want to be told that they have to wear a mask or get vaccinated.

That has become a dividing line, but it's not entirely correlated with political parties. There are some people in both parties who don't want to be told what to do by a big entity. For some, it's a concern about government

Thirteen Club table no.9, New York Public Library



overreach, and for others it's distrust of Big Pharma. In this country the vaccines and the virus itself were highly politicised right out of the box. Some of the motivation, I believe, is simply to resist the other side and demonstrate your membership in your group.

Do you think this is partly related to a loss of trust, not just in government institutions, but in the media and in the information people are receiving?

Yes, absolutely. That's a big part of the problem. Many people have lost trust in authority and in authority figures. Everyone is an expert now. That's why "do your own research" is such a crazy phrase. Most people don't know how to do or evaluate research because they haven't been taught. They are not experts. We are faced with an undermining of the authority of experts and also a lack of trust in large groups like the government or pharmaceutical companies.

I dare say that these elements do play a part, but perhaps superstition is in a way an attempt to control one's life, an attempt to control the outcome. Do you think people tend to overestimate the amount of control they have in their lives?

Yes, I do, and there's good evidence of that. People who are not depressed believe they have a greater control over their lives than they really do, and there's some laboratory evidence to back that up. There's a famous study, the subtitle of which is "sadder but wiser" that showed that people who are moderately depressed actually had a more accurate view of how much control they had over the world. Whereas people who were not depressed, overestimated their degree of control.

There is a tendency, an optimistic tendency, among most people to think that they have more control over things than they really do. And generally speaking, it's a healthy thing.

In a way, it can be related to optimism?

Yes, absolutely. In my new book, I cover a number of cases in which unrealistic optimism can be beneficial. It's not helpful in every situation, but there are certain situations where being overly optimistic can be of great value.

Despite the fact that experience tells us that life is uncertain by its very nature, we have little control about how things will pan out, we see that day in, day out with what transpires in our daily life, in our plans for the future. But nonetheless, we seem to have an aversion to uncertainty. We love certainty. We want to know what's going to happen. How do we deal with this fear of not knowing what comes next and the tendency to try and control what's going to happen?

We deal with it through practice and experience and controlling your exposure to risky situations. For example, consider the realm of health. In the ancient world, people had no idea how things worked, how the body worked, what things were good or bad for them. We now have lots of good information about things that we can do to decrease our risk of sickness, avoid health problems, and extend our lives. If you think about it, there's quite a bit of money and effort spent aimed at those behavioural goals that will provide you with a long and healthy life.

As much as we may feel anxious and worried about the future, our lives

As much as we may feel anxious and worried about the future, our lives today are immeasurably more predictable and less risky than at earlier times.

Some people perform superstitions with full awareness that they're irrational and cannot have a direct effect.

today are immeasurably more predictable and less risky than at earlier times. Despite the pandemic and everything else, this is probably the best era to live in because we know so much about health and wellness and people can achieve a sense of certainty by doing the right things.

So, the stress of uncertainty has been diminished?

Yes, it has. It is hard to imagine how ancient people dealt with it, but if you stand back and look at the two situations, you realise how different life must have been and begin to understand that most of the things that we worry about are small by comparison. Obviously, we still have illness and cancer and accidents, but our life expectancies are much longer than they were in earlier times. And importantly, we know so much more about how to maintain health today.

Is there anything else you'd like to say about superstition and uncertainty?

I want to point out that sometimes we have a sense of what we're doing. A person might employ a superstition while being aware that it's a little silly. We are of two minds. Some people perform superstitions with full awareness that they're irrational and cannot have a direct effect. Not all people are this way. Some believers think superstitions really work. But many people who have been raised in a superstitious family will use a superstition, even though they know it doesn't make sense. They feel better doing it. And often they say, "I don't want to take a chance."

They don't want to take the chance that, if they omit the superstition, something bad will happen. One of the things that has drawn me to this work from the very beginning, continuing through this book on useful delusions, is the endearing ways that human beings exhibit paradoxical behaviour, often with full knowledge of the contradiction.

It brings to mind comments like "good luck", "fingers crossed", "break a leg". All of these expressions are used, I suppose, knowing that they're not going to have an effect on the outcome – people are simply showing that they care.

Yes. I think many of those examples have a social function. I don't believe many people knock on wood when there's no one else around. It's often done as a communication. Many of the team-level superstitions or rituals have a bonding function. They express a shared wish, a shared hope. And, yes, many people employ these social superstitions knowing they don't really have an effect. They just want to show their commitment to a common goal.

Uncertainty

The Philosopher

Plato

~428 – 348 BCE

Philosopher Kings

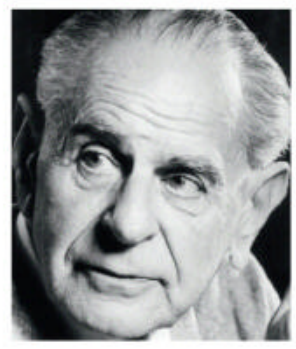
Until philosophers are Kings, or the kings and princes of this world have the spirit and power of philosophy, and political greatness and wisdom meet in one, and those commoner natures who pursue either to the exclusion of the other are compelled to stand aside, cities will never have rest from their evils, no, nor the human race, as I believe – and then only will this our State have a possibility of life and behold the light of day.

The Philosopher of Science

Karl Popper

1902 – 1994

To err is human



Knowledge consists in the search for truth – the search for objectively true, explanatory theories. It is not the search for certainty. To err is human. All human knowledge is fallible and therefore uncertain. It follows that we must distinguish sharply between truth and certainty. That to err is human means not only that we must constantly struggle against error, but also that, even when we have taken the greatest care, we cannot be completely certain that we have not made a mistake… To combat the mistake, the error, means therefore to search for objective truth and to do everything possible to discover and eliminate falsehoods. This is the task of scientific activity. Hence we can say: our aim as scientists is objective truth; more truth, more interesting truth, more intelligible truth. We cannot reasonably aim at certainty.

The Metaphysician

Duns Scotus

1266 – 1308

Most knowable

If all men by nature desire to know, then they desire most of all the greatest knowledge of science. So the Philosopher argues in chapter 2 of his first book of the work [*Metaphysics*]. And he immediately indicates what the greatest science is, namely the science which is about those things that are most knowable. But there are two senses in which things are said to be maximally knowable: either because they are the first of all things known and without them nothing else can be known; or because they are what are known most certainly. In either way, however, this science is about the most knowable. Therefore, this most of all is a science and, consequently, most desirable.

What is uncertainty? How does uncertainty shape our lives? Here are six thinkers' views on known and unknown unknowns.

The Political Theorist

Michael Oakeshott

1901 – 1990

Appearance of certainty

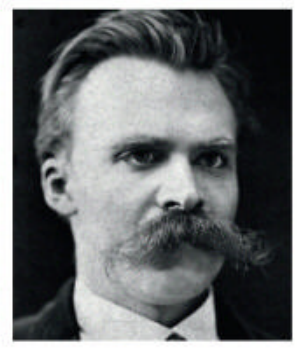
It is possible to write down technical knowledge in a book. Consequently, it does not surprise us that when an artist writes about his art, he writes only about the technique of his art. This is so, not because he is ignorant of what may be called the aesthetic element, or thinks it unimportant, but because what he has to say about that he has said already (if he is a painter) in his pictures, and he knows no other way of saying it. And the same is true when a religious man writes about his religion or a cook about cookery. And it may be observed that this character of being susceptible of precise formulation gives to technical knowledge at least the appearance of certainty: it appears to be possible to be certain about a technique.

The Philologist

Friedrich Nietzsche

1844 – 1900

The great majority



The great majority of people does not consider it contemptible to believe this or that and to live accordingly, without first having given themselves an account of the final and most certain reasons pro and con, and without even troubling themselves about such reasons afterward: the most gifted men and the noblest women still belong to this "great majority." But what is good-heartedness, refinement, or genius to me, when the person who has these virtues tolerates slack feelings in his faith and judgments and when he does not account the desire for certainty as his inmost craving and deepest distress – as that which separates the higher human beings from the lower.

The Politician

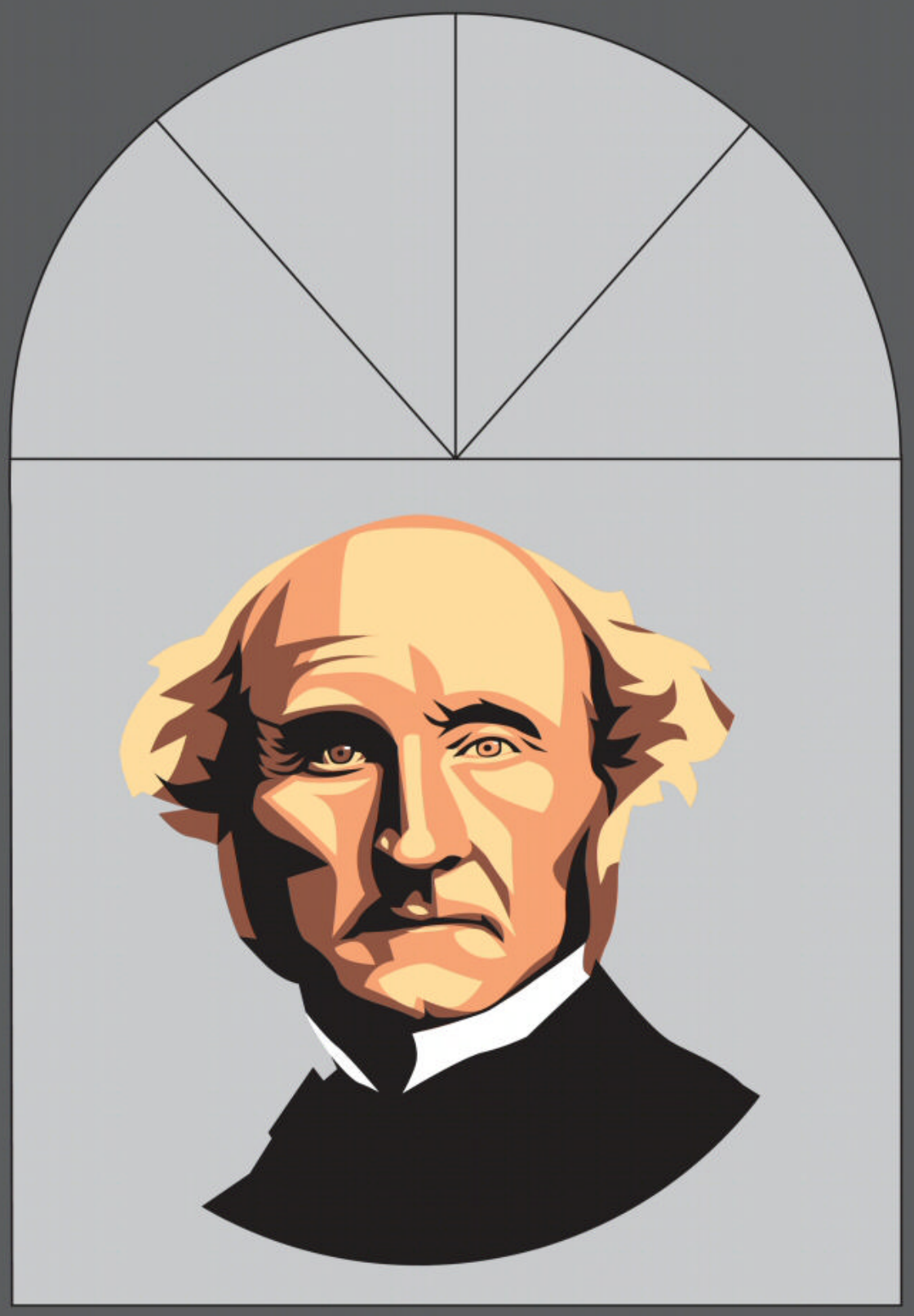
Donald Rumsfeld

1932 – 2021

Unknown unknowns

Reports that say that something hasn't happened are always interesting to me, because as we know, there are known knowns; there are things we know we know. We also know there are known unknowns; that is to say we know there are some things we do not know. But there are also unknown unknowns – the ones we don't know we don't know. And if one looks throughout the history of our country and other free countries, it is the latter category that tend to be the difficult ones. The absence of evidence is not evidence of absence, or vice versa.

Absolute



CERTAINTY

By John Stuart Mill

If all mankind minus one, were of one opinion, and only one person were of the contrary opinion, mankind would be no more justified in silencing that one person, than he, if he had the power, would be justified in silencing mankind.

But the peculiar evil of silencing the expression of an opinion is, that it is robbing the human race; posterity as well as the existing generation; those who dissent from the opinion, still more than those who hold it. If the opinion is right, they are deprived of the opportunity of exchanging error for truth: if wrong, they lose, what is almost as great a benefit, the clearer perception and livelier impression of truth, produced by its collision with error.

First: the opinion which it is attempted to suppress by authority may possibly be true. Those who desire to suppress it, of course deny its truth; but they are not infallible. They have no authority to decide the question for all mankind, and exclude every other person from the means of judging. To refuse a hearing to an opinion, because they are sure that it is false, is to assume that *their* certainty is the same thing as absolute certainty. All silencing of discussion is an assumption of infallibility.

Yet it is as evident in itself, as any amount of argument can make it, that ages are no more infallible than individuals; every age having held many opinions which subsequent ages have deemed not only false but absurd; and it is as certain that many opinions, now general, will be rejected by future ages, as it is that many, once general, are rejected by the present.

Because he has felt, that the only way in which a human being can make some approach to knowing the whole of a subject, is by hearing what can be said about it by persons of every variety of opinion, and studying all modes in which it can be looked at by every character of mind. No wise man ever acquired his wisdom in any mode but this; nor is it in the nature of human intellect to become wise in any other manner.

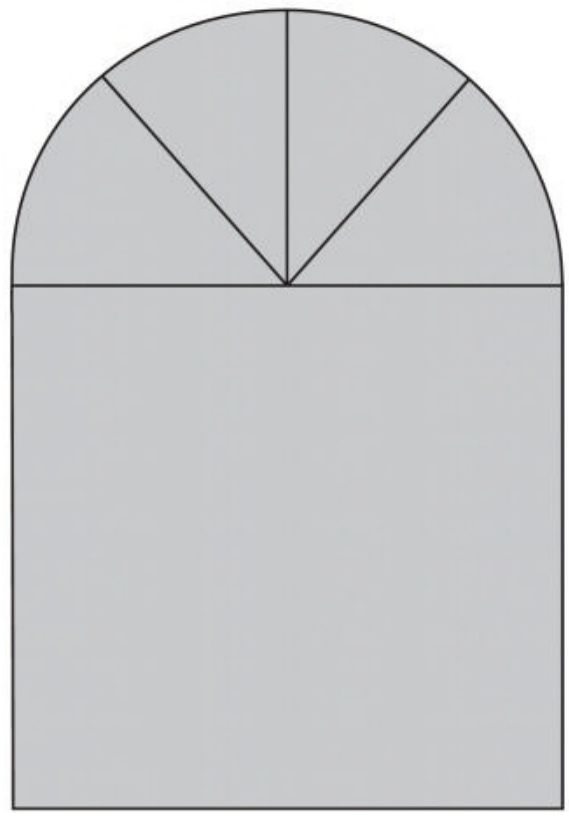
Strange that they should imagine that they are not assuming infallibility, when they acknowledge that there should be free discussion on all subjects which can possibly be doubtful, but think that some particular principle or doctrine should be forbidden to be questioned because it is so certain, that is, because they are certain that it is certain.

To call any proposition certain, while there is any one who would deny its certainty if permitted, but who is not permitted, is to assume that we ourselves, and those who agree with us, are the judges of certainty, and judges without hearing the other side.

Orthodox Christians who are tempted to think that those who stoned to death the first martyrs must have been worse men than they themselves are, ought to remember that one of those persecutors was Saint Paul.

For it is this – it is the opinions men entertain, and the feelings they cherish, respecting those who disown the beliefs they deem important, which makes this country not a place of mental freedom. For a long time past, the chief mischief of the legal penalties is that they strengthen the social stigma. It is that stigma which is really effective, and so effective is it, that the profession of opinions which are under the ban of society is much less common in England, than is, in many other countries, the avowal of those which incur risk of judicial punishment.

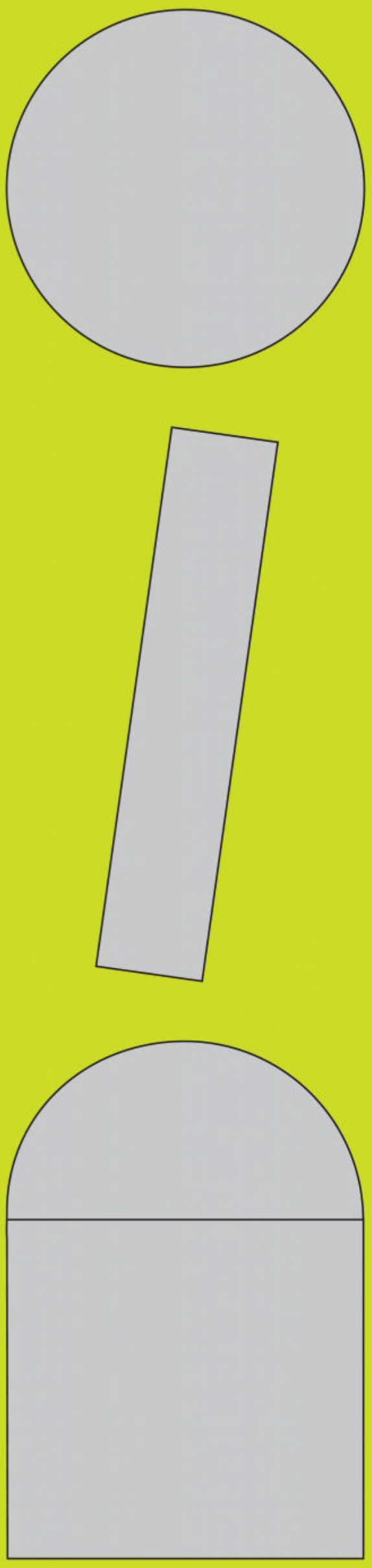
A state of things in which a large portion of the most active and inquiring intellects find it advisable to keep the general principles and grounds of their convictions within their own breasts, and attempt, in what they address to the public, to fit as much as they can of their own conclusions to premises which they have internally renounced, cannot send forth the open, fearless characters, and logical, consistent intellects who once adorned the thinking world. The sort of men who can be looked for under it, are either mere conformers to commonplace, or time-servers for truth, whose arguments on all great subjects are meant for their hearers, and are not those which have convinced themselves. Those who avoid this alternative, do so by narrowing their thoughts and interest to things which can be spoken of without venturing within the region of principles, that is, to small practical matters, which would come right of themselves, if but the minds of mankind were strengthened and enlarged, and which will never be made effectually right until then: while that which would strengthen and enlarge men's minds, free and daring speculation on the highest subjects, is abandoned.



But it is not the minds of heretics that are deteriorated most, by the ban placed on all inquiry which does not end in the orthodox conclusions. The greatest harm done is to those who are not heretics, and whose whole mental development is cramped, and their reason cowed, by the fear of heresy. Who can compute what the world loses in the multitude of promising intellects combined with timid characters, who dare not follow out any bold, vigorous, independent train of thought, lest it should land them in something which would admit of being considered irreligious or immoral?

Truth gains more even by the errors of one who, with due study and preparation, thinks for himself, than by the true opinions of those who only hold them because they do not suffer themselves to think.

There is a class of persons (happily not quite so numerous as formerly) who think it enough if a person assents undoubtingly to what they think true,



though he has no knowledge whatever of the grounds of the opinion, and could not make a tenable defence of it against the most superficial objections. Such persons, if they can once get their creed taught from authority, naturally think that no good, and some harm, comes of its being allowed to be questioned.

Whatever people believe, on subjects on which it is of the first importance to believe rightly, they ought to be able to defend against at least the common objections.

But much more of the meaning even of these would have been understood, and what was understood would have been far more deeply impressed on the mind, if the man had been accustomed to hear it argued pro and con by people who did understand it. The fatal tendency of mankind to leave off thinking about a thing when it is no longer doubtful, is the cause of half their errors. A contemporary author has well spoken of "the deep slumber of a decided opinion."

Popular opinions, on subjects not palpable to sense, are often true, but seldom or never the whole truth. They are a part of the truth; sometimes a greater, sometimes a smaller part, but exaggerated, distorted, and disjoined from the truths by which they ought to be accompanied and limited. Heretical opinions, on the other hand, are generally some of these suppressed and neglected truths, bursting the bonds which kept them down, and either seeking reconciliation with the truth contained in the common opinion, or fronting it as enemies, and setting themselves up, with similar exclusiveness, as the whole truth. … Such being the partial character of prevailing opinions, even when resting on a true foundation, every opinion which embodies somewhat of the portion of truth which the common opinion omits, ought to be considered precious, with whatever amount of error and confusion that truth may be blended.

We have now recognised the necessity to the mental well-being of mankind (on which all their other well-being depends) of freedom of opinion, and freedom of the expression of opinion, on four distinct grounds; which we will now briefly recapitulate.

First, if any opinion is compelled to silence, that opinion may, for aught we can certainly know, be true. To deny this is to assume our own infallibility.

Secondly, though the silenced opinion be an error, it may, and very commonly does, contain a portion of truth; and since the general or prevailing opinion on any subject is rarely or never the whole truth, it is only by the collision of adverse opinions that the remainder of the truth has any chance of being supplied.

Thirdly, even if the received opinion be not only true, but the whole truth; unless it is suffered to be, and actually is, vigorously and earnestly contested, it will, by most of those who receive it, be held in the manner of a prejudice, with little comprehension or feeling of its rational grounds. And not only this, but, fourthly, the meaning of the doctrine itself will be in danger of being lost, or enfeebled, and deprived of its vital effect on the character and conduct: the dogma becoming a mere formal profession, inefficacious for good, but cumbering the ground, and preventing the growth of any real and heartfelt conviction, from reason or personal experience. N

Excerpt from John Stuart Mill's *On Liberty* (1859).

The Scream, Edvard Munch, 1893

Friedrich Nietzsche, Edvard Munch, 1906

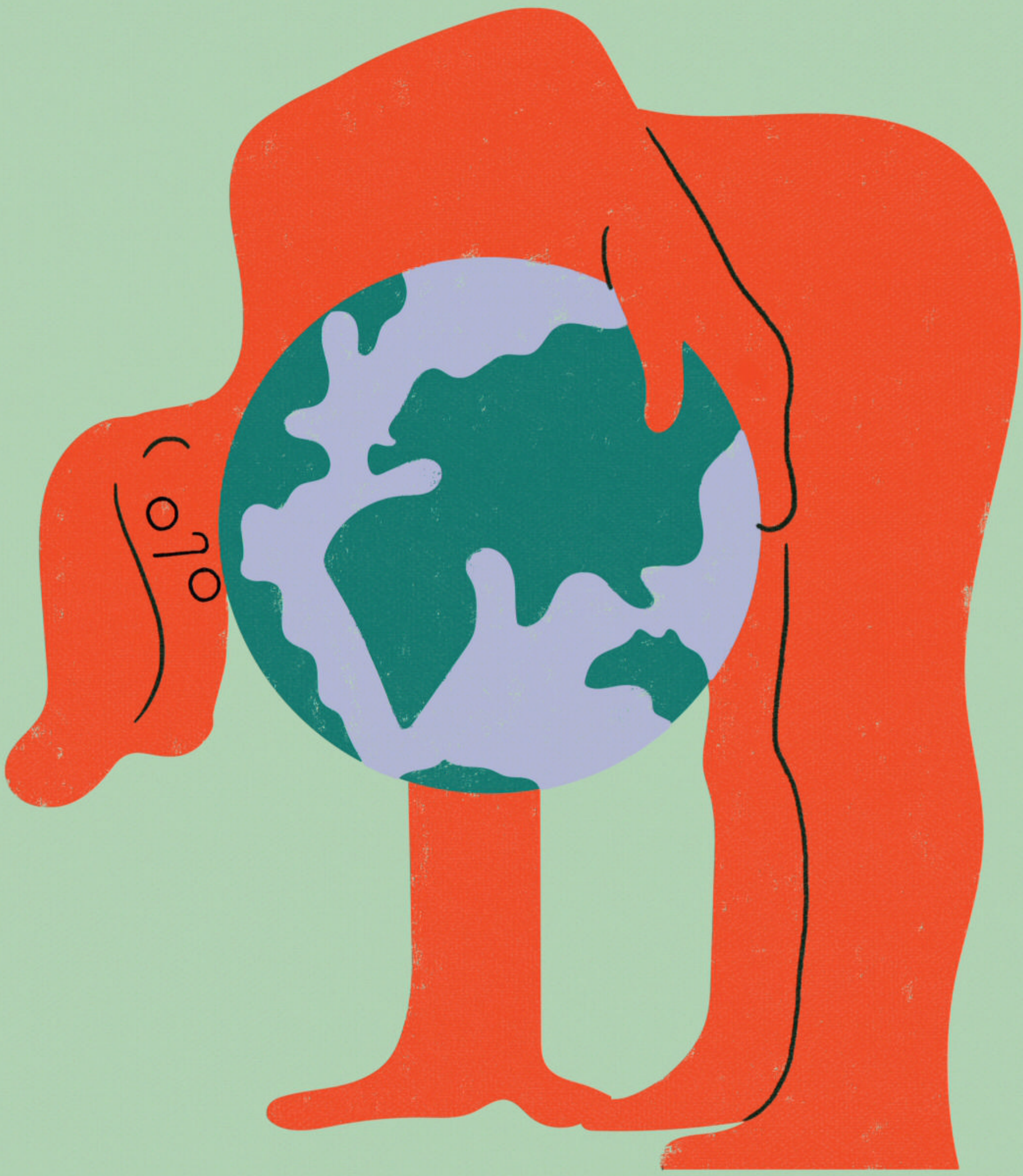


Illustration by Aida Novoa & Carlos Egan

By Stephen Law

Cast-iron certainty

Many of us desire certainty. We don't just want to believe, or even have fairly or very good grounds for believing. We want more still: cast-iron certainty.

Our craving for certainty can make us vulnerable to manipulation and even brainwashing. As scientist Kathleen Taylor points out, cults typically exploit the discomfort we feel when we experience uncertainty. By providing a simple set of certainties that cover and explain everything, and constantly reminding us of the chaos and uncertainties that lie outside, political and religious cultists can make their belief systems increasingly attractive, and increasingly difficult to leave.

As Taylor wrote in *The Guardian*, "The contrast between the complicated, fragmented muddle of the unbeliever's life and the pure simplicity of true belief is often emphasised by ideologues who understand the power of uncertainty to frighten and unnerve people who may already feel that their lives are out of control."Taylor considers this exploitation of uncertainty one of five key signs of brainwashing.

Certainty may be attractive, but what *exactly* are we after? Not just *psychological* certainty – *feeling certain* that something is true. Plenty of psychological certainties have turned out to be false. Many once felt quite certain that the Earth does not move. Fred might feel absolutely certain that his wife will never leave him. No matter the strength of convictions, these beliefs can still turn out to be false.

Epistemic certainty, on the other hand, is usually supposed to provide a *guarantee of truth*. If you're certain, in *this* sense, that your house is on fire, then you can't be mistaken. If it turns out you've been taken in by an elaborate hoax involving deceptive lighting and a smoke machine, then you were never really *certain* your house was on fire, even if you thought you were, and even if you *felt* quite certain.

How can we know when we have certainty? One popular suggestion is *indubitability*. A belief will be certain if it *can't be doubted.* Philosopher René Descartes famously tried to subject all his beliefs to doubt to discover if any were indubitable. He found that all his beliefs about the external world – including that he had a physical body – could be doubted. There might be a powerful evil demon intent on deceiving him about all these things. Descartes found he could even doubt simple mathematical truths, such as that 2 + 2 = 4, for that same evil demon could make him go wrong in even his simplest calculations. The one thing Descartes found he couldn't doubt was that he existed. In attempting to doubt that he exists, Descartes must be thinking, and so must exist. That he existed was something about which he could be certain.

But does indubitability guarantee certainty? Suppose a brain scientist puts a computer chip in my brain that makes it impossible for me to entertain certain thoughts and doubts. As a result, I'm unable to doubt that my underwear is red. That doesn't mean I'm epistemically certain my underwear is red. I could easily be mistaken. So, it seems that indubitability is no guarantee of either truth or certainty. If indubitability is to underwrite certainty, it's going to have to be a special sort of indubitability.

The ‘certainty’ attached to such an utterance is not a particularly secure kind of knowledge.

Might we get clearer about what certainty is by thinking about proof? People often associate certainty with proof: they suppose something is certain if it’s proved.

However, ‘proved’ is slippery term with a variety of meanings. In a court of law we seek, and often get, proof *beyond reasonable doubt*. But we can also set the bar far higher, demanding proof beyond *all possible doubt*. Set the bar that high, however, and very little is ever proved: certainly not the guilt of an accused person standing in the dock. No matter how strong the case presented by the prosecution, it’s always *possible* that there’s been some elaborate conspiracy to frame them. Still, as the evidence mounts, the accused’s guilt can become increasingly *probable* to the point where we can justifiably send them to prison for life. We can have proof beyond reasonable doubt – and reasonably suppose we *know* – even if we lack epistemic certainty.

One reason philosophers have been particularly interested in certainty is that they have sought an entirely secure foundation on which to build their beliefs. But what could provide such a foundation? Perhaps our immediate subjective experiences. Maybe I can’t be certain that there is a table currently before me (it’s possible I’m hallucinating), but I can be certain that I’m presented with the *subjective appearance* of a table. Similarly, perhaps I can’t be certain whether I have an injury to my foot, or even that I have a foot. Perhaps my foot has been amputated and I’m experiencing a phantom pain. Still, it seems I can be certain that I am experiencing a pain, even if it’s not in my foot. It appears that the subjective realm may allow us at least some cast-iron certainties.

Still, even supposing my subjective experiences can deliver some certainties, can those certainties provide an absolutely firm and unshakeable foundation for my belief in, say, an external world of trees, flowers, mountains, and stars? It seems not. After all, Descartes’ demon might be creating these appearances in me in order to deceive me about the external world.

The philosopher G.E. Moore thought the existence of the external world was certain, and offered the following ‘common sense’ proof. He held up his two hands, gesturing to one and then the other, and reasoned like so:

Here is one hand
And here is another
Therefore: I have two hands.

What’s true, perhaps, is that ordinarily, outside of a philosophical setting, Moore’s argument would probably be considered an adequate proof. But to many philosophers Moore’s argument looks hopelessly question-begging. Rather than *establishing* there’s an external world, Moore seems to *presuppose* that there is one. How can Moore be sure his two premises are true? How can Moore know he’s not being deceived into supposing he is waving two arms about by Descartes’ evil demon, or a supercomputer generating a Matrix-like virtual reality? After all, everything would seem exactly the same to Moore. It appears that not only can Moore not be certain of his conclusion; he can’t even know it.

We have seen that certainty is elusive – hard to pin down, and hard to achieve. But perhaps that’s because we are confused about what certainty is. Ludwig Wittgenstein responds to Moore’s proof by insisting that both Moore and the sceptic about the external world make the same mistake. Both think that ‘Here is a hand’, used in the way Moore uses it, is used to make a claim. On Wittgenstein’s view it is not used to make a claim about the existence of a hand; rather, it’s used to fix what the word ‘hand’ *means*. No one ever holds up a hand in broad daylight and says ‘Here is a hand’ unless they are, say, explaining the meaning of the word ’hand’ to foreign-language students. Consequently, says Wittgenstein, “If I wanted to doubt whether or not this was my hand, how could I avoid doubting whether the word ‘hand’ has any meaning?”

The ‘certainty’ attached to such an utterance is not a particularly secure kind of knowledge about how things stand in the world. Rather, in such circumstances, the sentence ‘This is a hand’ is used to articulate part of the framework of meaning within which any such doubt might be raised. As Wittgenstein famously wrote: “The questions that we raise and our doubts depend on the fact that some propositions are exempt from doubt, are as it were like hinges on which [those] doubts turn.”

Epistemic certainty is siren-like yet elusive. I have to admit that not only am I not sure how to achieve epistemic certainty, I’m not even sure what it is.

NewPhilosopher

“A common fallacy in much of the adverse criticism to which science is subjected today is that it claims certainty, infallibility, and complete emotional objectivity. It would be more nearly true to say that it is based upon wonder, adventure, and hope.”

Sir Cyril Norman Hinshelwood

Artwork: *Maison Pour Érotomane*, Salvador Dalí, circa 1932

Interviewee: Kate Sweeny
Interviewer: Zan Boag

Waiting for news

Kate Sweeny is Professor of Psychology and Teresa and Byron Pollitt Endowed Term Chair at the University of California, Riverside, where she runs the Life Events Lab. She received her B.S. in 2002 from Furman University and her M.S. in 2003 and Ph.D. in 2008 from the University of Florida where she studied social psychology with Dr James Shepperd. Her research examines two broad questions. First, how do people cope with acute and stressful moments of uncertainty, like the wait for important news? Second, how should doctors talk to their patients? To answer these questions, Professor Sweeny has studied law graduates awaiting news about the bar exam, patients awaiting biopsy results, researchers awaiting manuscript decisions, surgeons talking to their adult patients, and asthma specialists talking to paediatric patients, among many other topics and studies.

Zan Boag: To start with, could you define what 'uncertainty' means to you?

Kate Sweeny: Uncertainty is a very broad concept, capturing any case in which a person is aware that some information is unknown or ambiguous. In my research, I've focused on a specific but familiar form of uncertainty: waiting for some kind of uncertain news.

You've researched uncertainty in some depth – with your Life Events Lab at the University of California, Riverside, exploring how people cope with uncertainty and waiting. What prompted you to begin your research on this topic?

I've actually studied this topic even longer, starting at the beginning of graduate school in 2002. I was first interested in how people manage their expectations, either embracing optimism or bracing for the worst. Shortly after I started the Life Events Lab at UCR, I broadened my scope to investigate not just expectations but the full experience of uncertainty during waiting periods. From another angle, my interest in this topic stems from my own struggles with worry and waiting, combined with my surprise upon learning that we knew very little about that experience before I started my research.

Why do you think humans have such an aversion to uncertainty?

We can't know for sure, but uncertainty is essentially risky for survival; if we don't know what's coming, we can more easily be caught unaware or off-guard. From an evolutionary perspective, then, it's very useful to notice uncertainty and find it uncomfortable or even distressing, thus motivating us to resolve it whenever possible.

Uncertainty could be viewed as a fear of the future, and fear is one of the greatest human motivators. What part does fear play when people are dealing with uncertainty?

Fear and anxiety are close cousins, and worry is also in that emotional family. Fear tends to arise in response

Illustration by Aida Novoa & Carlos Egan

to a specific threat – you hear a suspicious sound at night, or you see something out of the corner of your eye. Anxiety and worry, on the other hand, tend to arise when we don't know what threat might be coming, or when we fear a threat but don't know whether that fear will come to pass.

Life is uncertain by its very nature, which is clear by what has transpired over the past few years. Why do humans fool themselves into thinking that they have any control over the future?

Sometimes we do! We may not be able to control the big things – politics, climate, war – but we have lots of control over smaller decisions and experiences in our lives. That said, another reason we are motivated to feel in control is that, like uncertainty, a lack of control over our fate is similarly dangerous for our survival. Our evolutionary ancestors who found it distressing to lack control likely lived longer because of it, being motivated to grab hold of opportunities for control wherever available.

Note that when we're in a waiting period like the ones I study, we get a double dose of discomfort: uncertainty plus lack of control.

You experienced uncertainty as a child growing up, as you dealt with your mother's illness. How did this impact your early years and your view of what was and wasn't in your control?

To be honest, my parents protected me from much of the uncertainty surrounding my mom's cancer diagnosis and treatment. However, I did face significant uncertainty repeatedly during my childhood, as my dad's job moved around quite a bit. I had lived in four states by the time I got to high school. Each time, there was quite a bit of uncertainty about when/if/where we would move, and each time we moved my life was completely upended. I don't know for sure how that uncertainty affected me, but I may have learned not to hold on to anything too tightly, and not to worry about things you can't control.

Your research shows that optimism helps when waiting for important news, if only because it lowers anxiety. Why is it good to be optimistic?

Optimism feels good! It can also be motivating, prompting us to persist even when the going gets tough. However, it's important to note that optimism has a downside when we're waiting for news: if we're unrealistically optimistic, we might be shattered by unexpected bad news. It's best to be optimistic most of the time but brace

"I didn't know that I didn't know that."

for the worst as you approach a moment of truth.

Your studies have shown that flow, or complete immersion in one activity, can help reduce the pain of waiting for a diagnosis, exam result, or a lengthy period of quarantine. Can you run through your research into 'flow'? What sort of 'flow activities' are most effective for relieving uncertainty?

We've published a number of findings about flow and waiting, but all have essentially the same message: flow makes waiting easier. It gets you out of your head, thus quieting worrisome thoughts, and it makes time seem to fly by. Your worries and uncertainty will be there when you come out of flow, but at least it gives you a much-needed break from the stress.

Fortunately, really any flow activity can have the benefits I just described, and everyone's flow activities are different. Think about something you do that makes you completely lose track of time, such that you can't start doing that thing if you need to leave the house in 15 minutes. That's probably a flow activity for you. You're looking for activities that challenge you but not too much and that let you see when you're making progress.

Many people who are diagnosed with an illness must live with the uncertainty of 'not knowing' – sometimes for decades. Can people habituate to the uncertainty of 'not knowing', in the sense that they learn to live with it? And can we learn something from such people?

I find open-ended uncertainty, as many cancer survivors face when they're in remission, to be an interesting and complex form of waiting. I do think people habituate to uncertainty to an extent – in fact, we even habituate to uncertainty over short periods of time, such that people waiting for the outcome of an academic or professional exam report a lot less worry in the middle of the waiting period than at the start and end. It's like a buzzing sound that's always there but eventually fades into the background until you don't really notice it anymore. Unfortunately, that buzzing sound tends to get a lot louder when we come face to face with our uncertainty, like when a cancer survivor goes in for an annual scan or blood test.

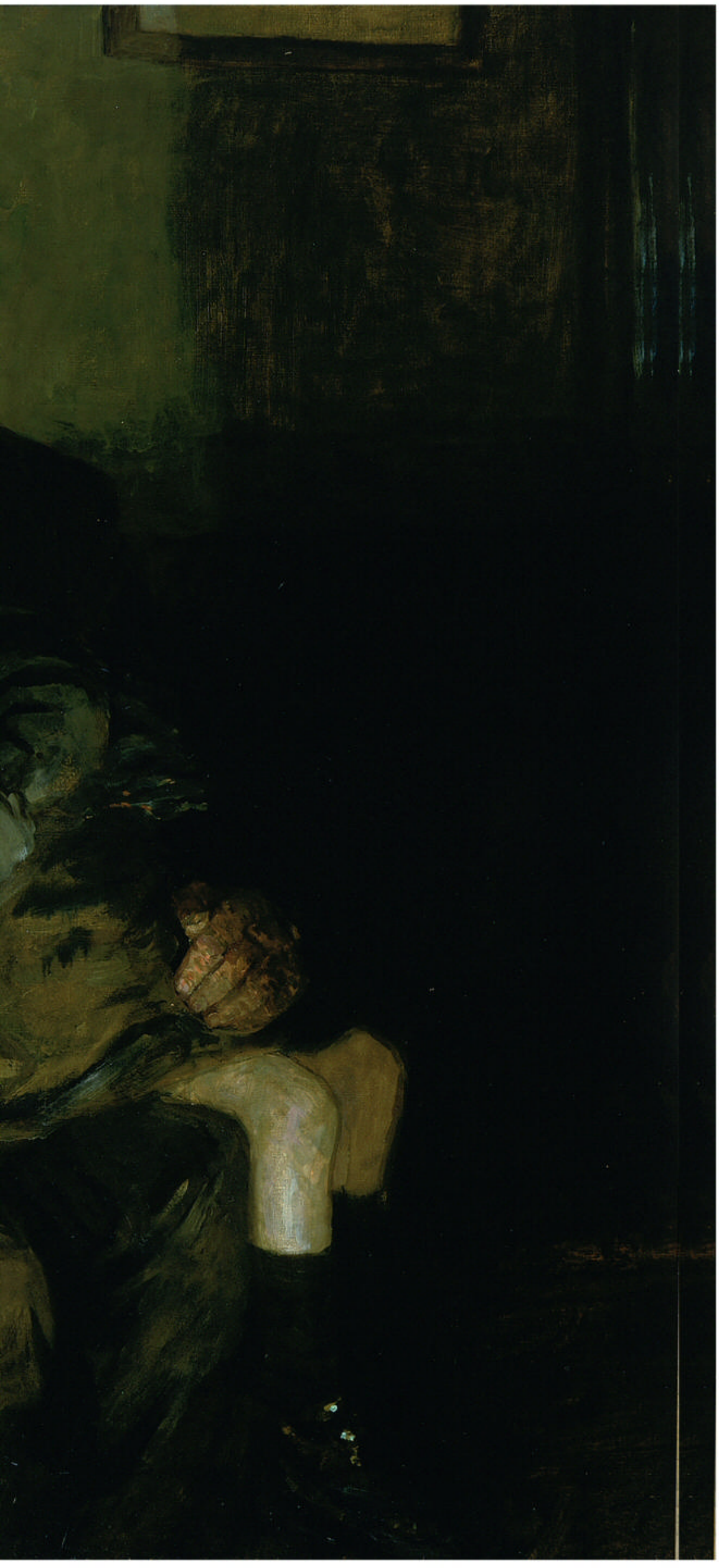
How can we learn to embrace uncertainty a little more in our lives?

We sure can't get rid of it! I'm not sure we can ever get comfortable with uncertainty, for the reasons I described above. That said, I do think we can come to have a healthier relationship with waiting and worry. Worry is actually very helpful to us – it draws our attention to potential threats and motivates us to do something to prevent or prepare for them. If we notice ourselves worrying about some uncertainty, it's useful to ask yourself, "is there something more I could be doing to ensure a good outcome or at least prepare for a bad outcome?" If you've done everything you can, then it's time to get into flow or maybe practice mindfulness to relieve ourselves of toxic worry.

One last question: Buddhist thinking teaches the impermanence of all things. Why is uncertainty the centre of so much eastern philosophical thought and can those in the west learn something from such thinking?

I greatly appreciate the Buddhist perspective on impermanence and uncertainty. As hard as uncertainty might be for all of us, it's a great deal harder if we believe that any change is bad, that we won't survive a shift in our experience. Embracing the fact that change is inevitable can reduce the fear that often accompanies uncertainty.

Flow makes waiting easier. It gets you out of your head, thus quieting worrisome thoughts, and it makes time seem to fly by.



Waiting for the Doctor, Robert Lee MacCameron, 1910

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On certainty

By Ludwig Wittgenstein

1. If you do know that *here is one hand*, we'll grant you all the rest.

When one says that such and such a proposition can't be proved, of course that does not mean that it can't be derived from other propositions; any proposition can be derived from other ones. But they may be no more certain than it is itself. (On this a curious remark by H.Newman.)

2. From its *seeming* to me – or to everyone – to be so, it doesn't follow that it *is so*.

What we can ask is whether it can make sense to doubt it.

3. If e.g. someone says "I don't know if there's a hand here" he might be told "Look closer". – This possibility of satisfying oneself is part of the language-game. Is one of its essential features.

4. "I know that I am a human being." In order to see how unclear the sense of this proposition is, consider its negation. At most it might be taken to mean "I know I have the organs of a human". (E.g. a brain which, after all, no one has ever yet seen.) But what about such a proposition as "I know I have a brain"? Can I doubt it? Grounds for *doubt* are lacking! Everything speaks in its favour, nothing against it. Nevertheless it is imaginable that my skull should turn out empty when it was operated on.

5. Whether a proposition can turn out false after all depends on what I make count as determinants for that proposition.

6. Now, can one enumerate what one knows (like Moore)? Straight off like that, I believe not. – For otherwise the expression "I know" gets misused. And through this misuse a queer and extremely important mental state seems to be revealed.

7. My life shows that I know or am certain that there is a chair over there, or a door, and so on. – I tell a friend e.g. "Take that chair over there", "Shut the door", etc. etc.

8. The difference between the concept of 'knowing' and the concept of 'being certain' isn't of any great importance at all, except where "I know" is meant to mean: I *can't* be wrong. In a law-court, for example, "I am certain" could replace "I know" in every piece of testimony. We might even imagine its being forbidden to say "I know" there. [A passage in "Wilhelm Meister", where "You know" or "You knew" is used in the sense "You were certain", the facts being different from what he knew.]

9. Now do I, in the course of my life, make sure I know that here is a hand – my own hand, that is?

10. I know that a sick man is lying here? Nonsense! I am sitting at his bedside, I am looking attentively into his face. – So I don't know, then, that

Stamp printed by Austria, shows Ludwig Wittgenstein, 1989

there is a sick man lying here? Neither the question nor the assertion makes sense. Any more than the assertion "I am here", which I might yet use at any moment, if suitable occasion presented itself. – Then is "2x2=4" nonsense in the same way, and not a proposition of arithmetic, apart from particular occasions? "2x2=4" is a true proposition of arithmetic – not "on particular occasions" nor "always" – but the spoken or written sentence "2x2=4" in Chinese might have a different meaning or be out and out nonsense, and from this is seen that it is only in use that the proposition has its sense. And "I know that there's a sick man lying here", used in an *unsuitable* situation, seems not to be nonsense but rather seems matter-of-course, only because one can fairly easily imagine a situation to fit it, and one thinks that the words "I know that..." are always in place where there is no doubt, and hence even where the expression of doubt would unintelligible.

11. We just do not see how very specialised the use of "I know" is.

12. – For "I know" seems to describe a state of affairs which guarantees what is known, guarantees it as a fact. One always forgets the expression "I thought I knew".

13. For it is not as though the proposition "It is so" could be inferred from someone else's utterance: "I know it is so". Nor from the utterance together with its not being a lie. – But can't I infer "It is so" from my own utterance "I know etc."? Yes; and also "There is a hand there" follows from the proposition "He knows that there's a hand there". But from his utterance "I know..." it does not follow that he does know it.

14. That he does know remains to be shown.

15. It needs to be *shown* that no mistake was possible. Giving the assurance "I know" doesn't suffice. For it is after all only an assurance that I can't be making a mistake, and it needs to be *objectively* established that I am not making a mistake about *that*.

16. "If I know something, then I also know that I know it, etc." amounts to: "I know that" means "I am incapable of being wrong about that." But whether I am so must admit of being established objectively.

17. Suppose now I say "I'm incapable of being wrong about this: that is a book" while I point to an object. What would a mistake here be like? And have I any *clear* idea of it?

18. "I know" often means: I have the proper grounds for my statement. So if the other person is acquainted with the language-game, he would admit that I know. The other, if he is acquainted with the language-game, must be able to imagine *how* one may know something of the kind.

19. The statement "I know that here is a hand" may then be continued: "for it's *my* hand that I'm looking at." Then a reasonable man will not doubt that I know. – Nor will the idealist; rather he will say that he was not dealing with the practical doubt which is being dismissed, but there is a further doubt *behind* that one. – That this is an *illusion* has to be shown in a different way.

20. "Doubting the existence of the external world" does not mean for example doubting the existence of a planet, which later observations proved to exist. – Or does Moore want to say that knowing that here is his hand is different in kind from knowing the existence of the planet Saturn? Otherwise it would be possible to point out the discovery of the planet Saturn to the doubters and say that its existence has been proved, and hence the existence of the external world as well.

21. Moore's view really comes down to this: the concept 'know' is analogous to the concepts 'believe', 'surmise', 'doubt', 'be convinced' in

Now do I, in the course of my life, make sure I know that here is a hand – my own hand, that is?

that the statement "I know..." can't be a mistake. And if that *is* so, then there can be an inference from such an utterance to the truth of an assertion. And here the form "I thought I knew" is being overlooked. – But if this latter is inadmissible, then a mistake in the *assertion* must be logically impossible too. And anyone who is acquainted with the language-game must realise this – an assurance from a reliable man that he *knows* cannot contribute anything.

22. It would surely be remarkable if we had to believe the reliable person who says "I can't be wrong"; or who says "I am not wrong".

23. If I don't know whether someone has two hands (say, whether they have been amputated or not) I shall believe his assurance that he has two hands, if he is trustworthy. And if he says he *knows* it, that can only signify to me that he has been able to make sure, and hence that his arms are e.g. not still concealed by coverings and bandages, etc. etc. My believing the trustworthy man stems from my admitting that it is possible for him to make sure. But someone who says that perhaps there are no physical objects makes no such admission.

24. The idealist's question would be something like: "What right have I not to doubt the existence of my hands?" (And to that the answer can't be: I *know* that they exist.) But someone who asks such a question is overlooking the fact that a doubt about existence only works in a language-game. Hence, that we should first have to ask: what would such a doubt be like?, and don't understand this straight off.

25. One may be wrong even about "there being a hand here". Only in particular circumstances is it impossible. – "Even in a calculation one can be wrong – only in certain circumstances one can't."

26. But can it be seen from a *rule* what circumstances logically exclude a mistake in the employment of rules of calculation?

What use is a rule to us here? Mightn't we (in turn) go wrong in applying it?

27. If, however, one wanted to give something like a rule here, then it would contain the expression "in normal circumstances". And we recognize normal circumstances but cannot precisely describe them. At most, we can describe a range of abnormal ones.

28. What is 'learning a rule'? – *This*.

What is 'making a mistake in applying it'? – *This*. And what is pointed to here is something indeterminate.

29. Practice in the use of the rule also shows what is a mistake in its employment.

30. When someone has made sure of something, he says: "Yes, the calculation is right", but he did not infer that from his condition of certainty. One does not infer how things are from one's own certainty. Certainty is *as it were* a tone of voice in which one declares how things are, but one does not infer from the tone of voice that one is justified.

31. The propositions which one comes back to again and again as if bewitched – these I should like to expunge from philosophical language.

From Ludwig Wittgenstein's *On Certainty*, 1969.

The idealist's question would be something like: "What right have I not to doubt the existence of my hands?"

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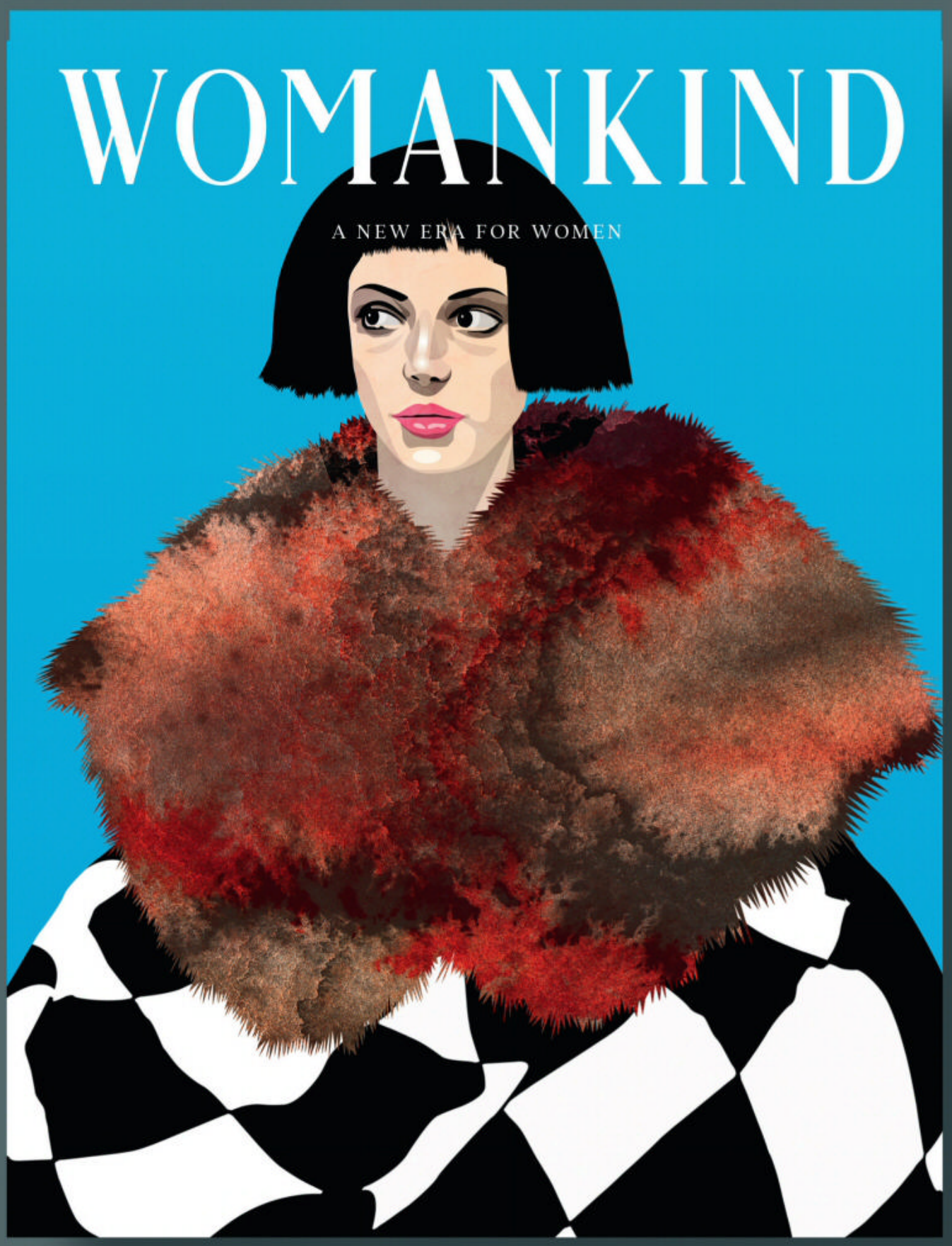

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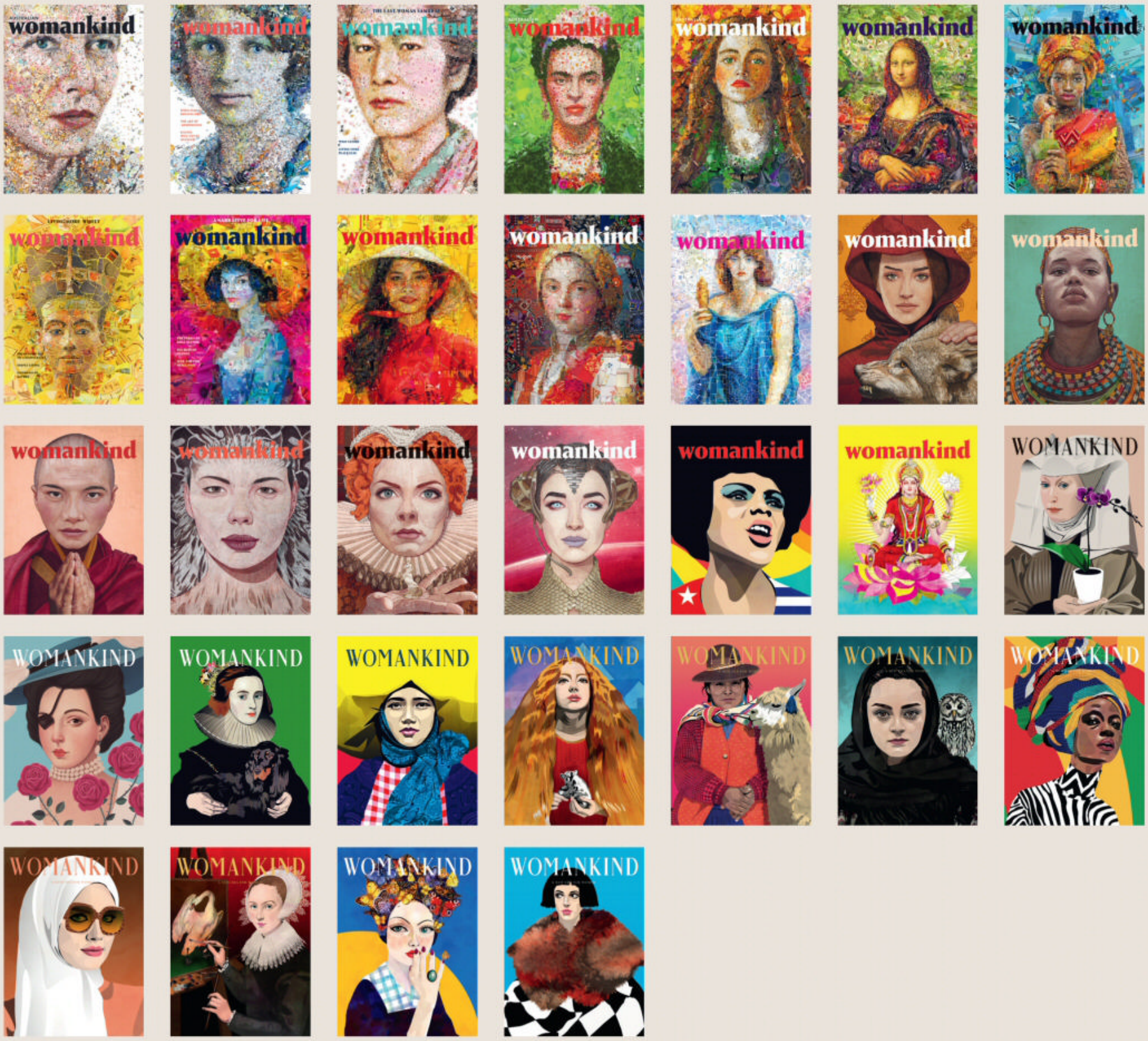

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Our library

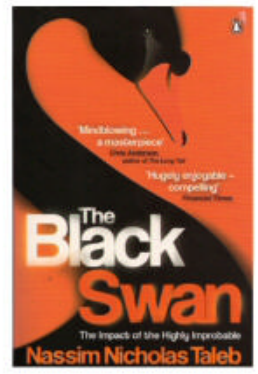
Superstition: A Very Short Introduction

Stuart Vyse

Systems of belief

The concept of superstition has been with us for millennia, and yet today it has no agreed-upon meaning. We tend to know it when we see it – as in the examples above. If it carries a single enduring connotation, it is one of disapproval. From almost the very beginning, it was not a compliment to call someone superstitious. Throughout its long history, superstition has been a transactional concept with no fixed meaning of its own except in contrast to some other, more accepted worldview. As governments and systems of belief have changed, so have the targets of the label. As a result, the history of superstition is largely the story of a word and the different ways it has been applied.

The Black Swan

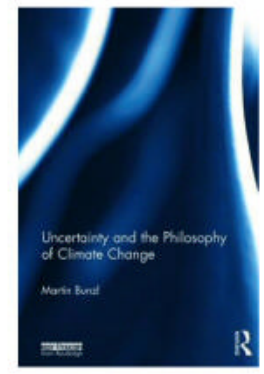


Nassim Nicholas Taleb

Unassailable belief

Before the discovery of Australia, people in the old world were convinced that all swans were white, an unassailable belief as it seemed completely confirmed by empirical evidence. The sighting of the first black swan might have been an interesting surprise for a few ornithologists (and others extremely concerned with the colouring of birds), but that is not where the significance of the story lies. It illustrates a severe limitation to our learning from observations or experience and the fragility of our knowledge. One single observation can invalidate a general statement derived from millennia of confirmatory sightings of millions of white swans. All you need is one single (and, I am told, quite ugly) black bird.

Climate Change and Uncertainty



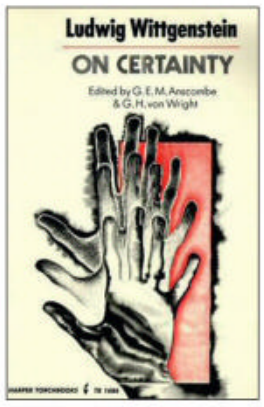
Martin Bunzi

Future generations

The standard literature on future generations assumes there will be future beings but that their identities and numbers are a function of our actions today. But what if my actions now cause total human extinction in the future? Then whom have I wronged? Constructing a philosophical framework to support the intuition that here too I have done wrong, even though I wronged no one, is the challenge I take on in Chapter Three. The to do so lays the groundwork for a much more controversial project, to make sense of the idea that our actions can also wrong nonhumans.

Food for thought from the *New Philosopher* library. We discover books that can change the way you view the world.

On Certainty



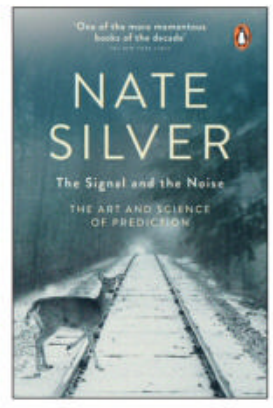
Ludwig Wittgenstein

Writing Chinese

I should like to say: "If I am wrong about this, I have no guarantee that anything I say is true." But others won't say that about me, nor will I say it about other people.

For months I have lived at address A, I have read the name of the street and the number of the house countless times, have received countless letters here and given countless people the address. If I am wrong about it, the mistake is hardly less than if I were (wrongly) to believe I was writing Chinese and not German.

The Signal and the Noise

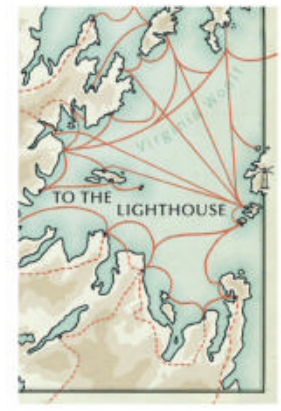


Nate Silver

Work ethic

Forecasting reflected the new Protestant worldliness rather than the otherworldliness of the Holy Roman Empire. Making a forecast typically implied planning under conditions of uncertainty. It suggested having prudence, wisdom, and industriousness, more like the way we now use the word foresight. The theological implications of this idea are complicated. But they were less so for those hoping to make a gainful existence in the terrestrial world. These qualities were strongly associated with the Protestant work ethic, which Max Weber saw as bringing about capitalism and the Industrial Revolution. This notion of forecasting was very much tied in to the notion of progress. All that information in all those books ought to have helped us to plan our lives and profitably predict the world's course.

To the Lighthouse



Virginia Woolf

Clear planets

What after all is one night? A short space, especially when the darkness dims so soon, and so soon a bird sings, a cock crows, or a faint green quickens, like a turning leaf, in the hollows of the wave. Night, however, succeeds to night. The winter holds a pack of them in store and deals them equally, evenly, with indefatigable fingers. They lengthen; they darken. Some of them hold aloft clear planets, plates of brightness. The autumn trees, ravaged as they are, take on the flesh of tattered flags kindling in the doom of cool cathedral caves where gold letters on marble pages describe death in battle and how bones bleach and burn far away in Indian sands.

Documentaries

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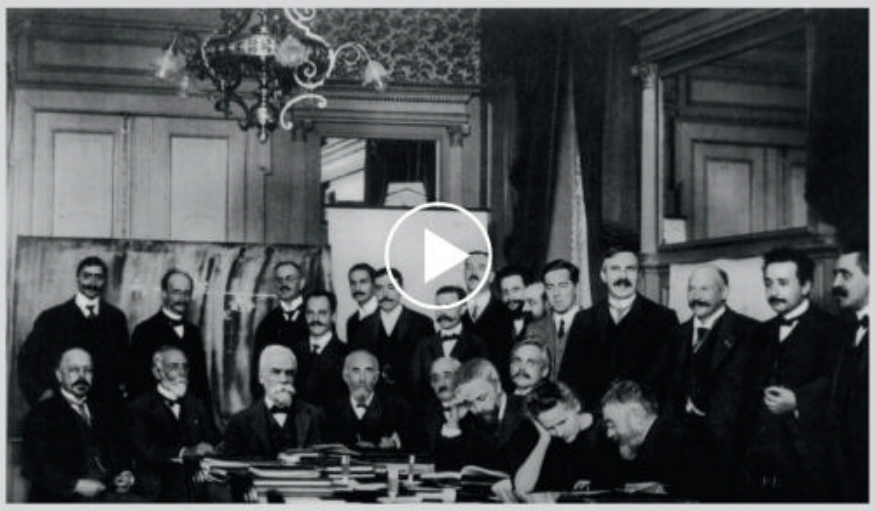
Uncertain Principles



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In 1927, Albert Einstein boarded a train to the Solvay Conference in Basel. His journey would take him to the centre of an argument between the greatest collections of scientists assembled this century. By the end of this conference, Einstein had lost a battle and our understanding of the world changed forever. Quantum mechanics, in a nutshell, is the physics of the very small," says John Gribben, author of *In Search of Schrödinger's Cat*. "Anything as small as an atom, or smaller than that, works in the rules of quantum physics."

The Age of Uncertainty

newphilosopher.com/videos/the-age-of-uncertainty/

The Age of Uncertainty is a 1977 television series about economics, history, and politics, written and presented by Harvard economist John Kenneth Galbraith. Galbraith acknowledges the successes of the market system in economics but associated it with instability, inefficiency, and social inequity. He advocates government policies and interventions to remedy these perceived faults. The content of the series was determined by Galbraith, with the presentation style directed by his colleagues in the BBC. Galbraith began by writing a series of essays from which the scripts were derived and from these a book by the same name, emerged which in many places goes beyond the material covered in the relevant episode.

Around the web

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The waiting game

Everyone – mums, therapists, rock stars – says the waiting is the hardest part. It turns out there is a way to "wait well", researchers say. People who feel anxious or pessimistic or who ruminate while awaiting news fare better than others when it finally arrives, the researchers say. They're more prepared for bad news and more excited about good news.
wsj.com/articles/how-to-manage-a-long-wait-for-news-1495465180

Devil's dictionary

CARTESIAN, adj. Relating to Descartes, a famous philosopher, author of the celebrated dictum, *Cogito ergo sum* – whereby he was pleased to suppose he demonstrated the reality of human existence. The dictum might be improved, however, thus: *Cogito cogito ergo cogito sum* – "I think that I think, therefore I think that I am"; as close an approach to certainty as any philosopher has yet made.

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The Chinese edition of New Philosopher has launched and is available in print, digital, epub, mobile: https://item.jd.com/13062738.html

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Known and unknown

Into the Great Unknown goes Donald H. Rumsfeld, who died in 2021 at 88, and whose tenure as George W. Bush's defence secretary was a monument to both ramrod certainty and tactical equivocation, the twin dogmas of Washington. "There are known knowns – there are things we know we know," Rumsfeld said in 2002, when asked for evidence about Saddam Hussein's weapons of mass destruction. "We also know there are known unknowns – that is to say, we know there are some things we do not know. But there are also unknown unknowns, the ones we don't know we don't know."
washingtonpost.com/lifestyle/style/rumsfeld-dead-words-known-unknowns/2021/07/01/831175c2-d9df-11eb-bb9e-70fda8c37057_story.html

Fear the tsunami?

Hilo Bay faces the Pacific Ocean on the northeastern shore of the island of Hawaii. It is 4,000 kilometres due south of the Aleutian Islands, home to one of the most active subduction fault lines in the world. From 1900 to 1964, a tsunami with a run-up of at least a meter (inundating all areas below one meter above sea level) struck Hilo an average of once every five years. On May 7, 1986, warning of a magnitude 7.7 Aleutian earthquake reached the Pacific Tsunami Warning Center in Honolulu. Based on this seismological information the state of Hawaii decided to evacuate Hilo and the surrounding area.
amnh.org/explore/videos/earth-and-climate/tsunami-science-reducing-the-risk/fear-the-future-tsunami

Video: Surrealism
People use the term surrealism to describe things that are too strange to be real but Surrealism was more than just melting clocks. It was about philosophy, politics, art, and revolution. This documentary from the Dalí museum is a condensed history of the Surrealist Movement, providing a foundation for artists learning to make their own surreal art.

newphilosopher.com/videos/surrealism-a-short-history/

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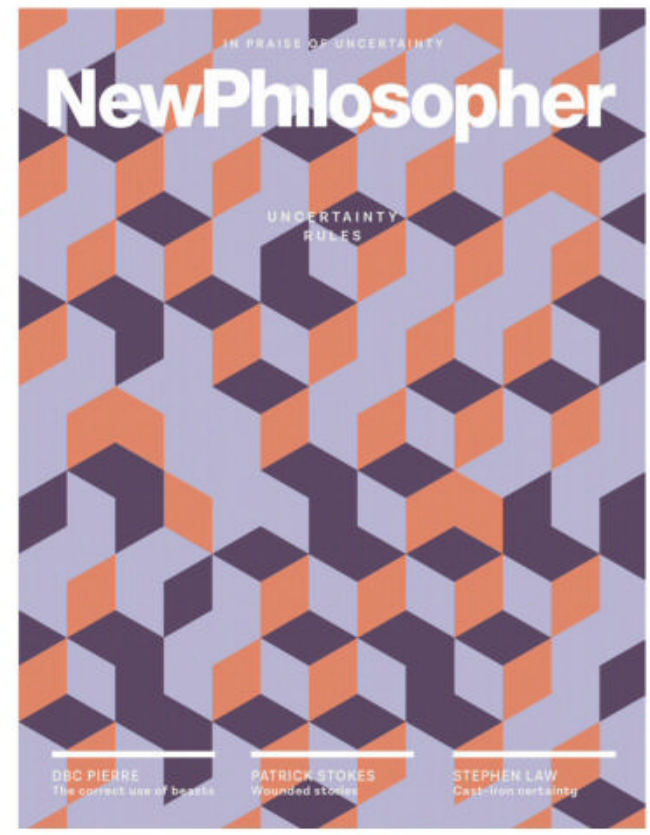

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what our readers say

Extraordinary quality in all respects. *New Philosopher* is one of the best things happening.

I love how the writing reminds me that the types of issues we are grappling with today are similar to those grappled with by philosophers for centuries.

I love this magazine. I no longer like reading or listening to mainstream news as it is not good for my health. Your magazine is good for me. It has real discussions that relate to life rather than reporting sensational 'news'. Thank you.

Your magazine is so helpful and is churning up the little grey cells, as Hercule Poirot says.

Thank you, this world is brighter, more aware, and more meaningful with your publication in it.

Great magazine. Finally one worth reading. Will recommend it to all friends.

This is a quality magazine: intelligent, lucid, lively, and beautifully designed and produced. Recommended.

A magazine that treats its audience as intellectual equals, that doesn't insult their intelligence by simplifying or manipulating concepts or appealing to crass consumerism as do most of the publications out there. Thank you for creating something that brings philosophy out of the sandstone universities and into our lounge rooms without losing its depth or critical stance.

You have introduced such a wonderfully fresh & intelligent publication to expand our minds & hearts. Thank you.

I feel like I have been waiting for this magazine all my life.

In a country so defined by the shallow, crass propaganda and cognitive conformity coming out of its papers, magazines, and broadcasters, yours is a true gem. You have no idea how happy I was to discover it, it has given me hope for the future of this country's media landscape and for the public discussion emanating from it.

I have to say that it is a tremendous relief and joy that your publication has arrived. Your magazine provides oxygen to the soul. I have often felt alone, alienated or old-fashioned in my outlook but now know I have a soulmate to reflect and take action with, namely *New Philosopher*.

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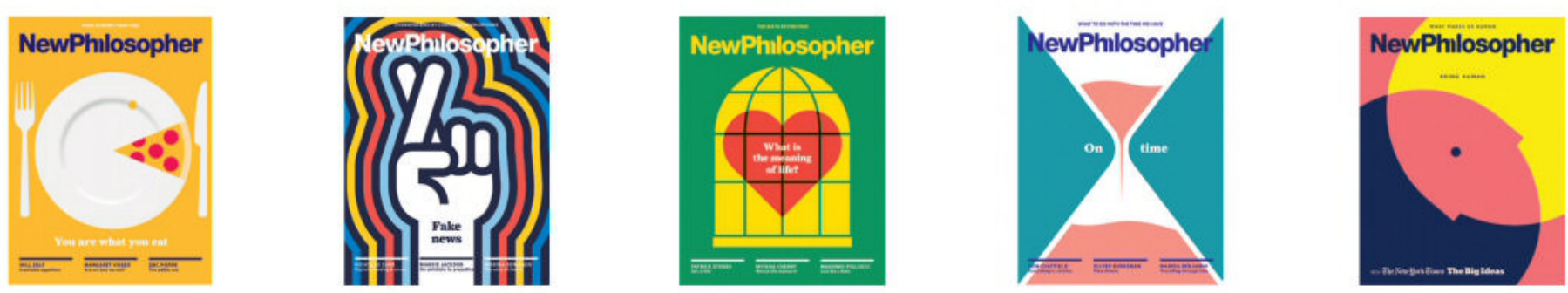

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13 questions:

Haaris Naqvi

In conversation with Zan Boag

Haaris Naqvi is an Editorial Director at Bloomsbury.

Illustration by Aida Novoa & Carlos Egan

What is your demon?
Fear of settling for just getting by.

What is the most important part of your education?
Whenever I began to value uninhibited enthusiasm and curiosity – the kind that leads you to worry and obsess – as much as intelligence and industry.

Which thinker has had the greatest influence on your life?
For that first serious, independent, excitement for reading, writing, and ideas: Dostoevsky, no question.

What do you doubt most?
Clean-cut efficiency.

If you could change one thing about the world, what would that be?
Less nationalism.

What illusion do you suffer from?
There's enough time.

What would you never do, no matter the price?
Eat mouldy cheese.

If you could choose, what would you have for your last meal?
A lamb curry cooked by my mother.

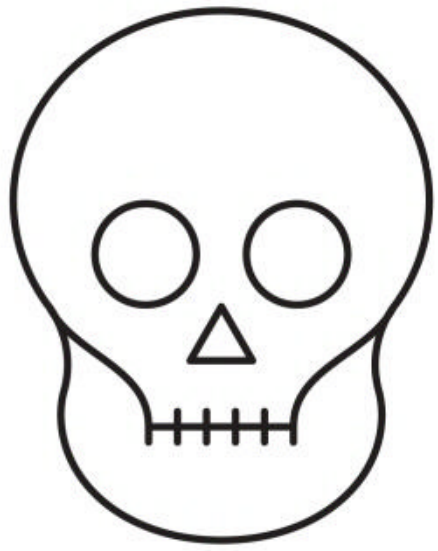
The question you'd most like to ask others?
What's your story?

What is your motto?
Lighten up.

What is a good death?
Not having to worry, overly, about the circumstances you leave behind.

What do people accuse you of?
Too many jokes.

What is the meaning of life?
I've thought about it, and it's hard to be sure.



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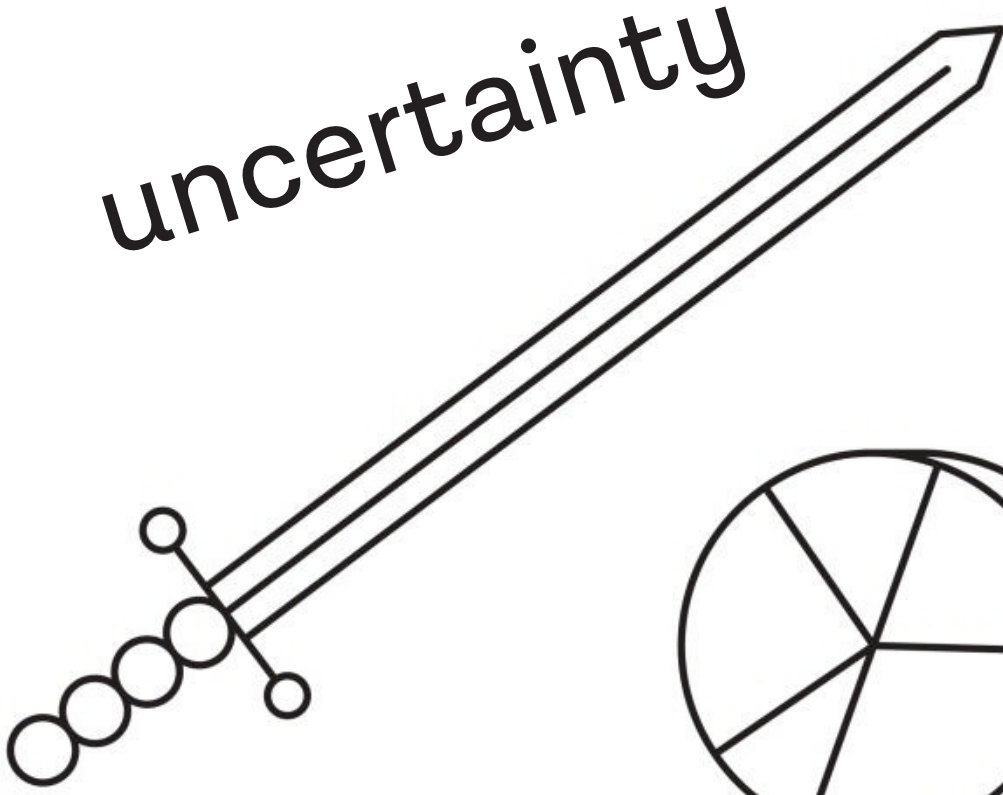

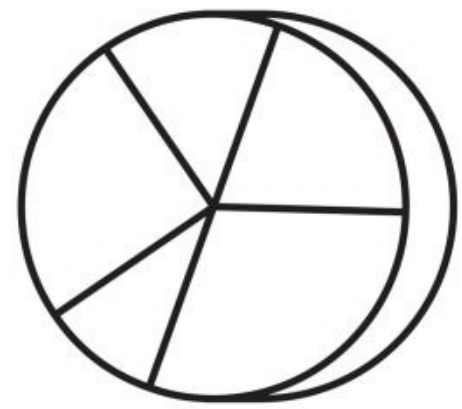

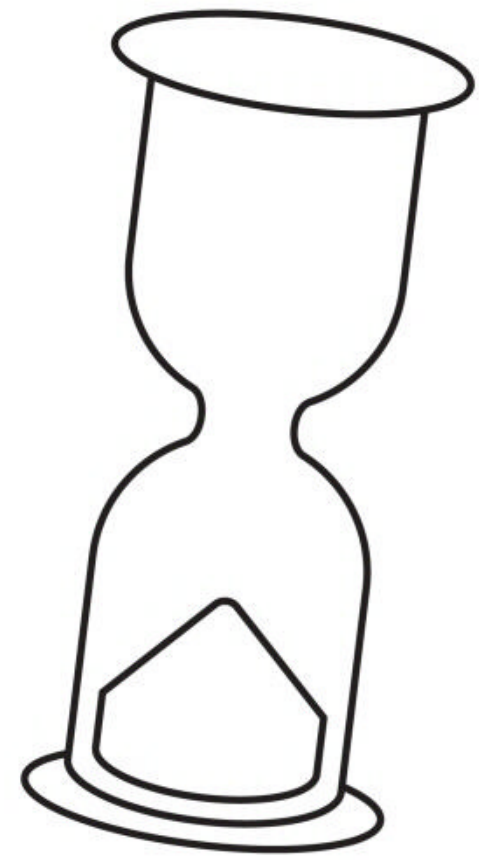

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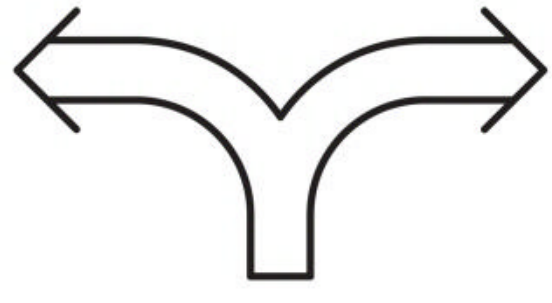

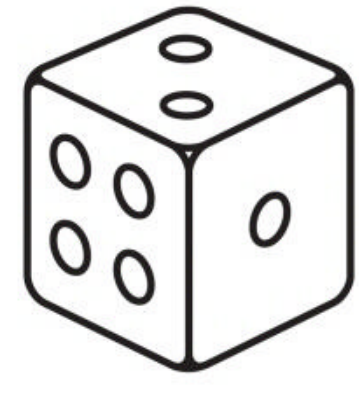

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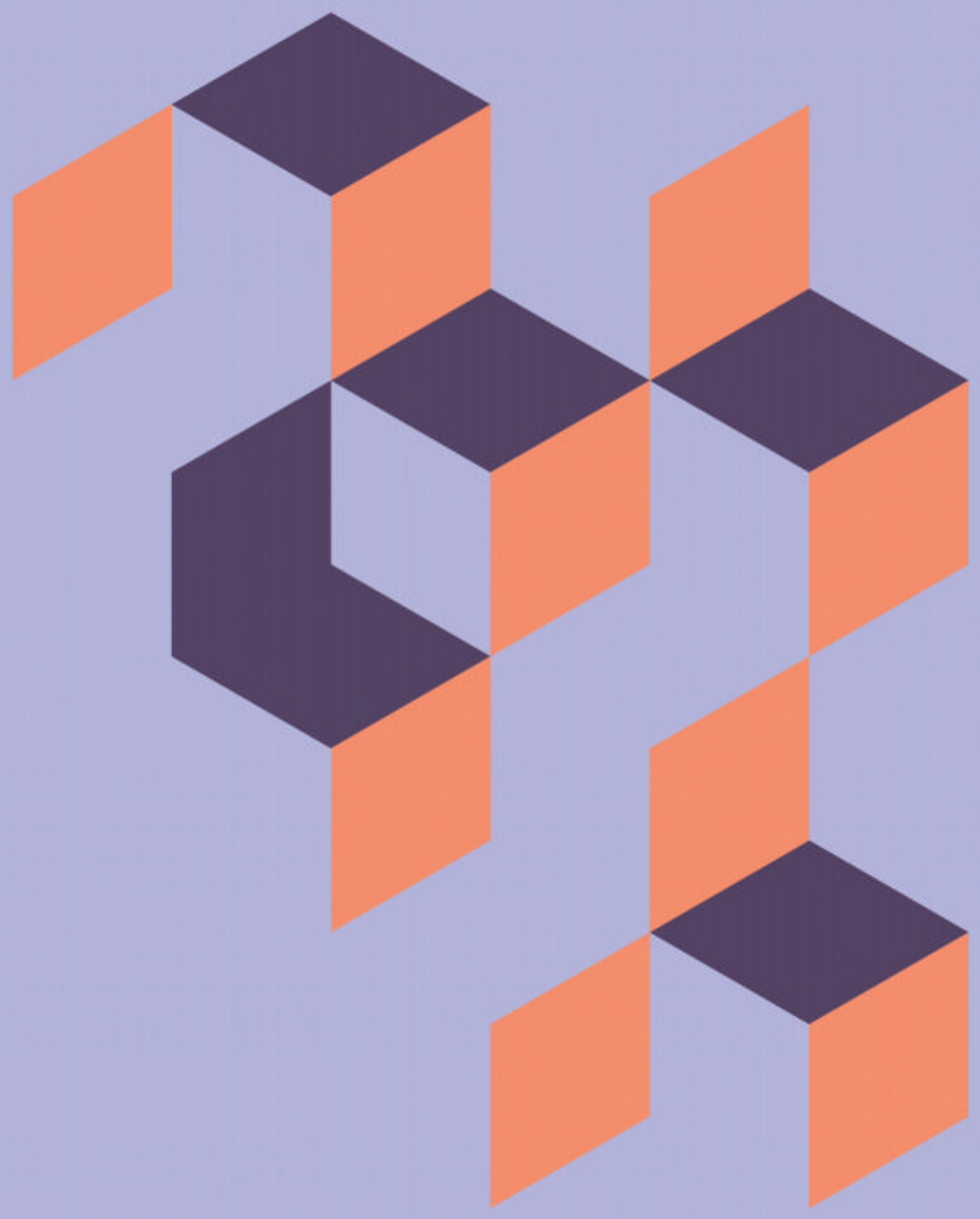

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